CIRCULAR STRATEGIES FOR INTERNAL MANUFACTURING SCRAP MANAGEMENT

A Case Study of Circular Opportunities in ABB Smart Power.

FATMA ZOHRA AMIRA DAHMAS

Department of Engineering Sciences
Course: Degree project in environmental engineering
Course code: MTK343
Credits: 30 hp
Program: Master's program in Environmental Engineering for Sustainable Development

Supervisor: Yahya Jani, Malin Estelli
Examiner: Erik Dahlquist
Customer: ABB Smart Power
Date: 2026-05-20
Email: fds24002@student.mdu.se

ABSTRACT

This thesis investigates the integration of circular economy principles into industrial manufacturing systems to improve the management of internal production scrap within ABB Smart Power. The study focuses on AF4–6 contactors, where defective newly manufactured units are currently treated as scrap and sent to external recycling without functional recovery. The research evaluates how circular recovery strategies, including dismantling, selective repair, component reuse, and eco-design approaches, can reduce environmental impacts, preserve embedded product value, and improve resource efficiency.

An AS-IS baseline assessment was first conducted to quantify the environmental and operational implications of internal production scrap. Production data from 2025 indicated that approximately 236,500 contactors were manufactured, with 485 rejected units, the rejected units represented approximately 1.78 tonnes of annual material loss and significant embedded environmental burdens associated with manufacturing activities.

The environmental impacts of the rejected contactors were assessed through Life Cycle Assessment (LCA) using SimaPro under cradle-to-gate system boundaries and based on verified Environmental Product Declarations (EPDs) for representative contactor models. The baseline analysis revealed measurable environmental impacts equivalent to approximately 13.5 t CO_2 eq, 53 MWh of primary energy demand, and 216 m^3 of freshwater consumption annually.

Building upon the baseline analysis, a TO-BE circular manufacturing framework was developed to prioritize higher-value recovery strategies before recycling. The proposed framework integrates defect-based decision-making, dismantling workflows, selective repair operations, component recovery, and eco-design principles to improve product recoverability and extend product value retention within the manufacturing system.

Comparative LCA results demonstrated that repair-oriented strategies significantly outperform dismantling and direct scrapping scenarios from an environmental perspective. Across all analyzed contactors (AF116-146, AF190-205, and AF265-370), the repair scenario consistently achieved the lowest environmental impacts by preserving the majority of functional components and avoiding complete remanufacturing processes. The dismantling scenario also reduced environmental impacts compared to the AS-IS baseline but remained environmentally more intensive than repair due to additional sorting, transportation, and recycling operations.

From a sustainability perspective, this research contributes directly to the United Nations Sustainable Development Goals (SDGs), particularly SDG 9 (Industry, Innovation and Infrastructure), SDG 12 (Responsible Consumption and Production), and SDG 13 (Climate Action). By shifting from linear scrap-oriented practices toward value-retention-based

circular strategies, the proposed framework supports the development of more resource-efficient, low-carbon, and sustainable industrial manufacturing systems.

Keywords: Circular Economy; Life Cycle Assessment (LCA); Circular Manufacturing; Internal Production Scrap; Repair and Dismantling; Sustainable Manufacturing; Eco-Design; Value Retention Strategies; Data-Driven Circularity.

PREFACE

This thesis was carried out as part of the Master's programme in Environmental and Sustainable Engineering and represents the final stage of my academic studies. The research was conducted in collaboration with ABB Smart Power, where I was given the valuable opportunity to carry out my Master's thesis within the factory environment. The study focuses on improving the management of internal manufacturing scrap through the application of circular economy principles and Life Cycle Assessment (LCA).

I would like to express my sincere gratitude to ABB Smart Power for granting me the opportunity to conduct this research within their industrial environment. Access to real production processes, technical expertise, and operational data has been essential for the successful completion of this work and has greatly enriched my learning experience.

I would also like to extend my deepest appreciation to my academic supervisor, Yahya Jani, for his continuous guidance, constructive feedback, and academic support throughout this project.

Special thanks are also directed to my industrial supervisor, Malin Estelli, for her practical guidance, availability, and support during the industrial phase of this thesis. Her insights and professional advice have contributed significantly to the development and practical relevance of this work.

On a personal level, I would like to express my heartfelt gratitude to my husband for his unwavering support, encouragement, and assistance throughout this demanding academic journey. His support has been a fundamental pillar in helping me complete this work.

A special thank you is also extended to my son Amir, whose presence has been a constant source of motivation and inspiration. I am deeply grateful to my family for their patience, encouragement, and continuous belief in me throughout my studies.

This thesis represents an important milestone in my professional development within the field of sustainable manufacturing and circular economy engineering.

Västerås, Sverige, May 2026

Amira Dahmas

Summary

The transition toward circular manufacturing systems represents a critical pathway for improving resource efficiency and reducing environmental impacts within industrial production environments. Internal production scrap generated during quality control processes constitutes a significant loss of materials, manufacturing energy, and embedded economic value, particularly in the production of complex electromechanical products. This thesis investigates how circular economy strategies supported by Life Cycle Assessment (LCA) can improve the management of internal manufacturing scrap within industrial manufacturing systems.

The research was conducted in collaboration with ABB Smart Power and focuses on the production of AF4–6 contactors within an industrial manufacturing environment. These contactors consist of multiple interconnected components, including metallic conductive elements, coils, engineering polymers, and mechanical switching systems. In the current AS-IS production model, defective newly manufactured contactors identified during quality control are classified as scrap and directed to external recycling without functional recovery of components or products.

An AS-IS baseline analysis was conducted to quantify the magnitude of internal production scrap and evaluate its environmental and economic implications. Production data from 2025 indicated that approximately 236,500 contactors were manufactured, of which 485 units were rejected during internal quality control processes, corresponding to an internal scrap rate of approximately 0.205%. Although relatively low from an operational perspective, the rejected units represented approximately 1.78 tonnes of annual material loss and considerable embedded manufacturing value.

The environmental impacts of internal scrap were quantified using verified Environmental Product Declarations (EPDs) and Life Cycle Assessment modelling in SimaPro under cradle-to-gate system boundaries. Three representative contactor models were analyzed: AF116-146-30, AF190-205, and AF265-370-30. The baseline assessment estimated that internal production scrap generated environmental impacts equivalent to approximately 13.5 tonnes CO_2 equivalent, 53 MWh of primary energy demand, and 216 m^3 of freshwater consumption annually. Sensitivity analysis was also performed to evaluate the influence of product-size distribution uncertainty.

In addition to environmental impacts, the economic implications of internal scrap were evaluated through a simplified techno-economic assessment. The findings indicate that direct scrapping of defective newly manufactured contactors represents a substantial loss of embedded product value, particularly considering the high manufacturing complexity and material content of the products.

Based on the AS-IS analysis, a TO-BE circular manufacturing framework was developed to enable higher-value recovery pathways before recycling. The proposed framework integrates:

- defect-based decision-making;
- selective repair strategies;
- controlled dismantling operations;
- component recovery;
- eco-design principles;
- circular workflow management.

The framework prioritizes repair and component reuse before recycling in order to maximize value retention and minimize unnecessary material and energy losses.
Comparative Life Cycle Assessment results demonstrated that repair-oriented strategies significantly outperform dismantling and direct scrapping scenarios from an environmental perspective. Across all analyzed contactor models, the repair scenario consistently generated the lowest environmental impacts by preserving the majority of functional components and avoiding complete remanufacturing processes. Dismantling strategies also reduced environmental burdens compared to the AS-IS baseline but remained more impactful than repair due to additional sorting, transportation, and recycling operations.

The study further highlights the importance of structured manufacturing and recovery-related data in supporting circular decision-making processes. The findings indicate that future circular manufacturing systems will increasingly depend on digital traceability, intelligent recovery workflows, and data-driven decision systems capable of optimizing repairability, dismantling efficiency, and resource recovery operations.

Overall, this research contributes to the development of practical and scalable circular manufacturing strategies applicable to complex assembled industrial products. The proposed framework provides a structured pathway for transitioning from conventional linear scrap-management practices toward value-retention-based circular manufacturing systems capable of improving environmental sustainability, operational efficiency, and long-term industrial resource resilience.

Table of Contents

1 INTRODUCTION 11

1.1 Background 12

1.2 Purpose/Aim 14

1.3 Research questions 14

1.4 Delimitation 15

2 THEORETICAL FRAMEWORK 16

2.1 Circular Economy and Sustainable Manufacturing 16

2.2 Circular Economy in Electrical and Electronic Equipment (EEE / WEEE) 16

2.3 Value Retention Strategies for Manufacturing Scrap 17

2.4 Circular Manufacturing and Industrial Waste Management 18

2.5 Manufacturing Scrap Management 19

2.6 EcoDesign and Design for Circularity 19

2.7 Life Cycle Assessment (LCA) and Environmental Evaluation 20

2.8 Digital Technologies Supporting Circular Decision-Making 20

2.9 Summary of Key Findings from Previous Studies 21

3 METHOD 22

3.1 Research Design 22

3.2 Case Study Context 24

3.2.1 Material Composition of Selected AF Contactor Sizes 26

3.3 Data Collection 27

3.3.1 Semi-Structured Interviews 27

3.3.2 Production and Operational Data 28

3.4 Data Analysis 29

3.4.1 Qualitative Analysis 29

3.4.2 Quantitative Analysis 29

3.4.3 Scenario Development 30

3.4.4 Integration with Environmental Modelling 30

3.5 Life Cycle Assessment (LCA) 30
3.5.1 Goal and Scope Definition 31
3.5.2 Functional Unit 31
3.5.3 System Boundaries 31
3.5.4 Data Sources 32
3.5.5 Life Cycle Inventory (LCI) 33
3.5.6 Impact Assessment Method (LCIA) 33
3.5.7 Scenario Definition 33
3.5.8 Methodological Limitations 34

4 RESULTS 35
4.1 Industrial Process Description 35
4.1.1 Manufacturing Process Overview 35
4.2 AS-IS Baseline: Internal Production Scrap Assessment 37
4.2.1 System boundary and scope 37
4.2.2 Production performance 37
4.2.3 Scrap rate analysis 38
4.2.4 Physical material loss 39
4.2.5 Economic impact 40
4.2.6 Environmental Impact Assessment of the AS-IS Baseline 40
4.2.7 Summary of the As Is Baseline 44
4.3 Objective of the TO-BE Scenario 45
4.3.1 Circular Strategy Framework 46
4.3.2 Dismantling and Component Recovery 47
4.3.3 Repair and Product Life Extension 53
4.3.4 Design for Repair and Circularity 56
4.3.5 Environmental Results — TO-BE Scenarios 58
4.3.6 Integrated Circular Workflow 63
4.3.7 Expected Economic and Operational Benefits 65
4.3.8 Summary of the TO-BE Scenario 65
4.4 Environmental Results and Comparative Assessment 66
4.4.1 Comparative Environmental Assessment of Contactors 66
4.4.2 Comparative Interpretation and Single-Score Analysis 66

4.5 Circularity and Industrial Implications 67

5 DISCUSSION 68

5.1 Interpretation of the Results 68

5.2 Implications for Circular Manufacturing 69

5.3 Environmental Implications (LCA Perspective) 70

5.4 Methodological Reflection and Limitations 70

5.5 Relation to Literature and Generalization 71

6 CONCLUSION 72

7 FUTURE RESEARCH DIRECTIONS 73

8 DECLARATION OF AI USE 75

9 REFERENCES 75

List of Tables

Tabell 1 Production and Scrap Overview 36
Tabell 2 Production Performance Indicators (2025) 37
Tabell 3 Scrap Rate Calculation 38
Tabell 4 Product Weight of Representative AF Contactor Models 39
Tabell 5 Summary of Material Loss Indicators 40
Tabell 6 Manufacturing-phase Global Warming Potential (GWP) per Contactor Model 41
Tabell 7 Manufacturing-phase energy demand per Contactor Model 42
Tabell 8 Manufacturing-phase Freshwater Consumption per Contactor Model 42
Tabell 9Scenario-based environmental impact of rejected units (AS-IS baseline) 44
Tabell 10 Defect code regarding the dismantling process 49
Tabell 11 Labour cost estimation for dismantling operations 50
Tabell 12 Annual dismantling workload estimation 51
Tabell 13 Single Score Results for AF116-146 59
Tabell 14 Single Score Results for AF190-205 60
Tabell 15 Single Score Results for AF256 61
Tabell 16 Environmental differences between the evaluated scenarios. 63
Tabell 17 Comparative interpretation between scenarios. 66
Tabell 18 Comparative single-score results across evaluated scenarios. 67
Tabell 19 Circular Manufacturing Roadmap for Future Industrial Implementation 74

List of Figurs

Figur 1 AF contactors line 4-6 sizes 25
Figur 2 Materiels compositions of contactors sizes 4-6 27
Figur 3 Whole contactor Scrapped 35
Figur 4 AS-IS Manufacturing and Scrap Management Flow 36
Figur 5 Transition from linear to circular management of internal production scrap 46
Figur 6Labelling process with colours distinction in the contactors. 49
Figur 7 Labour cost estimation for dismantling operations 51
Figur 8 Traceability for the recovery components. 53
Figur 9 Diagram of the repair scenario 56
Figur 10 Ecodesign and Circular recovery integration Framework 57
Figur 11 The chart bar of single score results on the AF116-146. 59
Figur 12 The chart bar of single score results on the AF190-205 60
Figur 13 The chart bar of single score results on the AF256 61
Figur 14 Integrated Circular Workflow for defective newly manufactured contactors. 64

List of Abreviations.

AI	Artificial Intelligence
AS-IS	Current state (baseline scenario)
CED	Cumulative Energy Demand
CO_2 eq	Carbon dioxide equivalent
EEE	Electrical and Electronic Equipment
EF 3.1	Environmental Footprint method 3.1
EN 15804+A2	European standard for construction and product EPDs
EPD	Environmental Product Declaration
GWP	Global Warming Potential
IoT	Internet of Things
LCA	Life Cycle Assessment
LCI	Life Cycle Inventory
LCIA	Life Cycle Impact Assessment
mPt	Millipoint (single-score unit, EF 3.1)
RFID	Radio-Frequency Identification
SDG	Sustainable Development Goal (United Nations)
SEK	Swedish Krona
TO-BE	Proposed future state (circular scenarios)
WEEE	Waste Electrical and Electronic Equipment

1 INTRODUCTION

The transition toward sustainable industrial systems has become an essential objective in response to accelerating environmental pressures, resource scarcity, and global climate commitments. Manufacturing industries are major contributors to greenhouse gas emissions, resource depletion, and industrial waste generation due to energy-intensive material extraction and production processes. As environmental regulations and sustainability expectations continue to intensify, manufacturing operations must evolve from conventional efficiency-driven approaches toward more resource-efficient and circular production models.

The circular economy offers a robust framework to support this transition. Unlike the traditional linear "take–make–dispose" model, circular strategies aim to maximize value retention by preserving products, components, and materials for as long as possible. Through approaches such as repair, reuse, remanufacturing, and material recovery, manufacturers can significantly reduce environmental impacts while improving resource efficiency and operational sustainability. These strategies are particularly relevant in complex manufacturing environments where even small quantities of rejected products may represent considerable embodied material, energy, and environmental losses.

Despite growing interest in circularity across academic and industrial domains, its implementation within internal manufacturing processes remains limited. In many production systems, products or components that fail quality control tests are treated as scrap and sent for external recycling without evaluating their potential for repair, component recovery, or reintegration into the production system. Although recycling enables partial material recovery, it does not preserve the significant functional and manufacturing value already embedded in the product. Consequently, even highly optimized manufacturing systems continue to operate according to a predominantly linear logic, resulting in avoidable material losses and unnecessary environmental burdens.

This issue is particularly significant in the production of complex electromechanical equipment composed of metals, engineering polymers, coils, and switching mechanisms, where rejected products may contain fully functional components and substantial embedded manufacturing value. Existing literature predominantly focuses on end-of-life waste management and recycling systems, particularly Waste Electrical and Electronic Equipment (WEEE), while comparatively limited attention has been given to internal production scrap generated directly during manufacturing processes. In particular, the integration of repair, structured dismantling, and component reuse strategies into internal scrap management remains insufficiently explored.

Implementing such circular strategies requires approaches that combine operational feasibility, structured decision-making systems, and quantitative environmental assessment methods capable of evaluating alternative recovery pathways. Within this context, Life Cycle Assessment (LCA) provides an important methodological framework for identifying environmental hotspots and assessing the environmental benefits associated with circular manufacturing strategies.

This thesis addresses these challenges through an industrial case study conducted at ABB Smart Power, focusing on the production of AF4–6 contactors. These industrial electromechanical devices are composed of metallic conductors, polymer housings, coils, and switching mechanisms that embody significant material, energy, and manufacturing investments. Their rejection during manufacturing quality control therefore generates considerable environmental losses, particularly when defective newly manufactured contactors are scrapped and replaced by entirely newly manufactured units.

To investigate these challenges, the research develops an AS-IS baseline model to quantify the magnitude and impacts of internal production scrap and proposes a TO-BE circular framework integrating dismantling, selective repair, defect-based recovery workflows, and design-for-repair principles. Particular attention is given to component-level repair strategies, where only defective components are replaced while the remaining functional structure of the contactor is preserved. This approach enables the retention of embedded manufacturing value while reducing the environmental burdens associated with complete product replacement.

The environmental implications of these circular strategies are evaluated through Life Cycle Assessment using cradle-to-gate system boundaries in SimaPro. The comparative analysis assesses the environmental performance of conventional scrapping practices against alternative circular recovery scenarios for three representative contactor models: AF116-146, AF190-205, and AF265-370.

From a broader sustainability perspective, this research directly supports the United Nations Sustainable Development Goals (SDGs), particularly SDG 9 (Industry, Innovation and Infrastructure), SDG 12 (Responsible Consumption and Production), and SDG 13 (Climate Action). By promoting value retention strategies and reducing unnecessary material and energy losses, the study contributes to the development of more resource-efficient, low-carbon, and sustainable industrial manufacturing systems.

Overall, this research contributes to bridging the gap between theoretical circular economy principles and their practical implementation within industrial manufacturing systems. By combining operational workflow development, circular recovery strategies, and quantitative environmental assessment within a real industrial environment, the study provides a practical and potentially scalable framework for improving circularity, reducing environmental impacts, and enhancing resource efficiency in industrial manufacturing operations.

1.1 Background

Global industrial development has significantly improved living standards and economic productivity. However, it has also contributed to increasing environmental pressures through intensive resource extraction, high energy consumption, and the growing generation of waste. In recent decades, concerns about climate change, resource depletion, and environmental degradation have led governments, industries, and researchers to reconsider how products are designed, manufactured, used, and disposed (Acerbi et al., 2021).

Traditionally, industrial production systems have followed a linear economic model, often described as *take–make–dispose*. In this model, raw materials are extracted, transformed into products, consumed, and eventually discarded as waste at the end of their useful life. While this model enabled rapid industrial growth, it has proven environmentally unsustainable due to continuous extraction of virgin resources and the accumulation of waste streams (Bovea et al., 2018).

In response, the concept of the circular economy (CE) has emerged as a fundamental paradigm for achieving sustainable production and consumption. Circular economy approaches aim to close material loops by maintaining the value of products, components, and materials for as long as possible through strategies such as reuse, repair, remanufacturing, refurbishment, and recycling. These *value-retention strategies* are

increasingly regarded as essential for reducing environmental impacts while improving resource efficiency in industrial systems (Pan et al., 2022; Psarommatis et al., 2025).

The need for circular strategies is particularly relevant in the electrical and electronic equipment (EEE) sector, which involves complex assemblies composed of valuable metals, polymers, and electronic components. Although much attention has focused on post-consumer Waste Electrical and Electronic Equipment (WEEE), recent studies have stressed the importance of upstream interventions to improve material efficiency and minimize waste generation during production (Pérez-Belis et al., 2015; Chu et al., 2024).

Increasingly, researchers emphasize circular manufacturing, which integrates circular principles directly into industrial production systems. During manufacturing, some products are rejected due to quality failures, testing deviations, or component defects. These internal rejects—known as manufacturing scrap—represent a significant loss of materials, energy, and economic value. Despite considerable recovery potential, such products are often sent directly to recycling without evaluating alternative value-retention options (Acerbi et al., 2021).

Evidence from recent studies indicates that repair, reuse, refurbishment, and remanufacturing can substantially reduce environmental impacts when compared with replacing rejected products by new ones. Extending product lifetimes preserves the embedded materials and energy invested in manufacturing and improves both environmental and economic performance (Bovea et al., 2018; Pini et al., 2019).

At the same time, advances in digital technologies—including digital twins, product passports, and data-driven monitoring systems—offer new opportunities to understand product performance, material composition, and defect patterns throughout the life cycle. These tools can facilitate more informed decisions regarding repair, dismantling, and reuse (Pehlken et al., 2024).

In parallel, quantitative environmental assessment tools such as Life Cycle Assessment (LCA) are essential for measuring the implications of alternative recovery strategies. LCA helps identify environmental hotspots and supports evidence-based decision-making by comparing linear and circular pathways (Psarommatis et al., 2025).

Despite these developments, the practical implementation of circular strategies in manufacturing remains limited. Many companies still lack structured workflows, integrated data systems, and decision-support tools capable of managing internal product rejects effectively within a circular framework (Psarommatis et al., 2025). This gap underlines the need for methodologies that unite technical feasibility, environmental evaluation, and organizational decision support in real industrial contexts.

Understanding how circular strategies can be meaningfully integrated into industrial production processes is therefore vital for advancing sustainable manufacturing. Managing internal manufacturing scrap through dismantling, repair, and component recovery can significantly minimize waste generation and improve resource efficiency.

However, despite extensive research on WEEE and end-of-life circularity, internal manufacturing scrap remains largely overlooked. Few studies have simultaneously addressed operational feasibility, component-level value retention, and LCA-based quantification within an integrated decision framework. Bridging this gap forms the scientific foundation of the present research.

1.2 Purpose/Aim

The purpose of this study is to investigate how circular economy strategies can improve the management of internal manufacturing scrap generated during industrial production and to evaluate the associated environmental and sustainability implications. The research is conducted as an industrial case study within ABB Smart Power, focusing on the production of AF4–6 contactors, where rejected units generated during assembly and quality control processes represent a measurable source of material, economic, and environmental loss.

The study aims to identify, develop, and systematically assess circular value-retention strategies such as repair, dismantling, component recovery, and design-for-repair approaches in order to reduce material losses, extend product lifetimes, and improve resource efficiency within the manufacturing system. Particular attention is given to the transition from conventional recycling-dominated practices toward higher-value circular strategies capable of preserving the functional and manufacturing value already embedded in scrapped manufactured contactors.

In addition, the research seeks to develop a structured decision-making framework enabling rejected products to be classified and managed according to their recovery potential, technical feasibility, and operational relevance. The proposed framework integrates defect-based recovery pathways intended to support more intelligent and adaptive scrap management processes within industrial manufacturing environments.

From an environmental assessment perspective, the study integrates Life Cycle Assessment (LCA) using cradle-to-gate system boundaries in SimaPro to quantify the environmental burdens associated with current scrap management practices and to evaluate the potential environmental benefits achievable through circular recovery strategies. The comparative assessment between the AS-IS baseline and the proposed TO-BE circular scenarios aims to establish a quantitative basis for evaluating reductions in environmental impacts related to climate change, energy demand, resource consumption, and water use.

The study further aims to analyse how product size, material composition, and recovery pathways influence environmental performance and circularity potential across different contactor models. Special focus is placed on component-level repair strategies, where only defective components are replaced while the remaining functional structure of the contactor is preserved, thereby minimizing the environmental burdens associated with complete product replacement.

1.3 Research questions

Building on the purpose and scope of this thesis, the research aims to address how circular economy and sustainability principles can be practically applied to improve internal manufacturing scrap management in the AF 4–6 contactor production line at ABB Smart Power.
The study is guided by the following research questions:

RQ 1. What circular economy strategies—such as repair, dismantling, and component recovery—can be integrated into the existing manufacturing process to enhance resource efficiency and reduce the environmental impacts associated with internal production scrap?

RQ 2. To what extent are these circular strategies technically and operationally feasible within the current production environment, and how can they be structured through a decision-making framework that supports value-retention and traceability?

RQ 3. How can Life Cycle Assessment (LCA) be applied to quantify and compare the environmental performance of the current linear waste-management system (AS-IS) with proposed circular recovery scenarios (TO-BE), and what are the expected contributions to climate-mitigation and sustainability goals?

1.4 Delimitation

Several delimitations define the scope of the research:

- The study focuses only on internal manufacturing scrap and excludes post-consumer waste streams and customer returns;
- The analysis is limited to AF4–6 contactors and does not represent all ABB product families;
- Circular strategies are evaluated through scenario modelling rather than full industrial implementation;
- Operational and economic data are presented in aggregated form due to ABB confidentiality requirements.

By maintaining these clearly defined boundaries, the study provides a focused evaluation of how practical circular manufacturing strategies can reduce environmental burdens and improve resource efficiency within industrial production systems while contributing to SDG 9, SDG 12, and SDG 13.

2 THEORETICAL FRAMEWORK

2.1 Circular Economy and Sustainable Manufacturing

The concept of the circular economy (CE) has gained increasing attention as a response to the environmental challenges associated with traditional linear production models. Conventional industrial systems typically follow a "take–make–dispose" approach, where natural resources are extracted, transformed into products, used, and eventually discarded as waste (Reike et al., 2018; Prieto-Sandoval et al., 2018). This linear model leads to increasing resource depletion, environmental pollution, and growing waste generation.

The circular economy proposes an alternative model that aims to maintain the value of products, materials, and resources for as long as possible within the economic system. Instead of focusing solely on waste management, circular economy principles emphasize resource efficiency, product life extension, and value retention (Reike et al., 2018; Messmann et al., 2019). Strategies such as repair, reuse, remanufacturing, and recycling are often highlighted as key mechanisms to reduce environmental impacts and improve resource productivity (Kissling et al., 2013).

Within manufacturing industries, circular economy principles are increasingly being integrated into production systems to reduce material losses and improve operational sustainability. However, implementing circular economy strategies within industrial contexts remains challenging due to technological, organizational, and economic barriers (Acerbi et al., 2021).
In manufacturing environments, circular economy principles are increasingly applied to address inefficiencies associated with internal production losses. Internal manufacturing scrap represents a significant source of material and energy loss, particularly in industries producing complex technical products. Improving the circular management of manufacturing scrap is therefore considered an important pathway toward achieving sustainable manufacturing systems.

2.2 Circular Economy in Electrical and Electronic Equipment (EEE / WEEE)

The electrical and electronic equipment (EEE) sector represents one of the fastest growing sources of waste globally. Waste Electrical and Electronic Equipment (WEEE) contains a complex mixture of valuable materials such as copper, aluminum, and rare metals, but also hazardous substances that require appropriate handling (Boldoczki et al., 2020).

Due to the rapid growth of electronic products and increasing consumption patterns, improving the management of electronic waste has become a major sustainability challenge. Circular economy strategies in the electronics sector aim to extend product lifetimes, improve recycling systems, and increase resource recovery (Parajuly et al., 2017; Corsini et al., 2020).

a record 62 million tonnes of WEEE were generated worldwide in 2022, an increase of 82% since 2010, projected to reach 82 million tonnes by 2030(Iraldo et al., 2017).. The global e-waste stream contained an estimated 31 million tonnes of embedded metals (copper, aluminium, gold, rare earths), alongside 17 million tonnes of plastics. The recoverable metal value was estimated at USD 91 billion, yet only 22.3% was formally recycled, leaving USD 62 billion worth of resources unaccounted annually.

However, most research on electronic waste focuses on post-consumer waste streams, while relatively limited attention has been given to internal manufacturing waste generated during production processes. Manufacturing rejects and defective units often contain valuable components and materials that may still hold functional value (Acerbi et al., 2021).
In addition to post-consumer electronic waste, manufacturing processes within the EEE sector also generate internal rejects that contain valuable materials and components. These internal waste streams represent an important but often overlooked opportunity for implementing circular economy strategies within industrial production environments.

2.3 Value Retention Strategies for Manufacturing Scrap

Within circular economy research, value retention strategies aim to preserve the economic and material value embedded in products, components, and materials for as long as possible (Reike et al., 2018). These strategies play an important role in reducing the environmental impacts associated with manufacturing new products and extracting additional raw materials.

In manufacturing environments, internal production scrap often includes rejected products removed from the production line due to defects or quality deviations. Despite these defects, many of these products may still contain functional components that could potentially be recovered and reused.

One important strategy for recovering value from manufacturing scrap is dismantling, which involves separating products into their individual components in order to recover usable parts or materials (Messmann et al., 2019). Through dismantling processes, manufacturers may be able to identify components that remain functional and suitable for reuse in new production processes.

Another relevant strategy is repair, which focuses on correcting localized defects in products rather than discarding the entire unit. Studies comparing repair and replacement scenarios demonstrate that extending product lifetimes through repair can significantly reduce environmental impacts compared with producing new products (Bovea et al., 2020; Ardente et al., 2012).

These strategies are particularly relevant for complex industrial products such as electrical equipment, where products consist of multiple components and materials with different levels of value.

The economic scale of the global WEEE sector underlines the importance of value retention strategies. According to the Global E-Waste Monitor 2024, approximately 62 million tonnes of electrical and electronic equipment waste were generated worldwide in 2022, representing an increase of over 82% since 2010.
The recoverable materials embedded in this waste stream including iron, copper, gold, and other critical metals, carry an estimated global value of USD 91 billion. Despite this, only 22.3% of global WEEE was formally collected and recycled, resulting in an annual loss of approximately USD 62 billion in recoverable material value. These figures confirm that the application of value retention strategies to electrical and electronic products, including both post-consumer devices and internal manufacturing rejects, represents not only an environmental priority but also a significant economic opportunity. (Iraldo et al., 2017).

Value retention strategies are commonly structured within a hierarchical framework that prioritizes options preserving the highest level of product functionality. Strategies such as repair and reuse typically retain more value than recycling, as they preserve both materials and embedded manufacturing effort. This hierarchical perspective supports decision-making processes aimed at maximizing resource efficiency and minimizing environmental impacts.

2.4 Circular Manufacturing and Industrial Waste Management

Circular economy principles are increasingly being applied not only to product lifecycles but also to manufacturing systems themselves. Manufacturing processes frequently generate internal waste streams, including defective components, rejected products, and production scrap. These materials represent a loss of embedded resources such as raw materials, energy, and labor.

Traditional industrial waste management practices often focus on recycling or disposing of production waste. However, circular manufacturing approaches aim to reduce waste generation at the source and recover value from materials that would otherwise be discarded (Messmann et al., 2019).

Improving the management of manufacturing scrap requires better understanding of production processes, product composition, and failure mechanisms. Without such knowledge, manufacturers often choose conservative strategies such as recycling rather than exploring higher-value recovery options (Kissling et al., 2013).

In industrial environments producing complex electrical components, internal manufacturing scrap often represents a hidden environmental burden. The cumulative impact of rejected units can be substantial, particularly when products consist of multiple materials and energy-intensive manufacturing processes.

2.5 Manufacturing Scrap Management

Manufacturing scrap refers to materials or products rejected during the production process due to defects, quality issues, or deviations from technical specifications. In many industrial environments, rejected products are directly sent to recycling streams without further evaluation of their recovery potential.

However, rejected products often contain valuable materials and components that have already undergone several production stages. Discarding these products therefore represents not only a loss of raw materials but also a loss of the energy and resources invested in their manufacturing (Messmann et al., 2019).

Effective scrap management strategies aim to reduce the generation of production rejects while improving the recovery of value from defective products. This may involve analysing failure modes, improving quality control processes, and identifying opportunities to recover components or materials from rejected units (Acerbi et al., 2021).

From an environmental perspective, rejected products contain embedded impacts accumulated throughout earlier manufacturing stages. These impacts include material extraction, component manufacturing, and assembly energy consumption. Discarding rejected products without evaluating recovery options therefore results in the loss of previously invested environmental resources.

2.6 EcoDesign and Design for Circularity

EcoDesign integrates environmental considerations into product development in order to reduce environmental impacts across the product life cycle (Ardente & Mathieux, 2014). Design decisions made during the early stages of product development can determine a large proportion of the environmental impacts associated with a product.

At the European Union level, EcoDesign principles are formally established through the Ecodesign for Sustainable Products Regulation (ESPR), Regulation (EU) 2024/1781, which entered into force on 18 July 2024 and replaces the earlier Ecodesign Directive 2009/125/EC. As a cornerstone of both the EU Circular Economy Action Plan (2020) and the European Green Deal, the ESPR extends ecodesign requirements to virtually all physical products placed on the EU market, mandating improvements in product durability, reparability, reusability, and recyclability. The regulation also introduces Digital Product Passports (DPPs) to improve product traceability and sustainability transparency throughout the product life cycle."

Several design strategies can support circularity, including design for repair, design for disassembly, and modular product architecture. These strategies facilitate component recovery and material reuse, enabling products to remain in use for longer periods (Bovea et al., 2020).

In industrial contexts, EcoDesign principles can support the development of products that are easier to diagnose, repair, and dismantle. Such design strategies can therefore contribute to improving the circular management of manufacturing scrap.

Product architecture plays a critical role in enabling circularity. Modular designs and accessible component layouts facilitate dismantling and repair operations, thereby improving the feasibility of circular recovery strategies within manufacturing systems.

2.7 Life Cycle Assessment (LCA) and Environmental Evaluation

Life Cycle Assessment (LCA) is a widely used methodology for evaluating environmental impacts associated with products, processes, or services throughout their lifecycle. The method is standardized under the ISO 14040 and ISO 14044 frameworks.

LCA enables the identification of environmental hotspots within product systems and allows researchers to compare different product management strategies such as repair, reuse, recycling, or replacement (Bovea et al., 2020).

In circular economy research, LCA is frequently used to assess whether extending product lifetimes leads to environmental benefits compared with producing new products (Ardente et al., 2012).

In this study, a cradle-to-gate perspective is adopted, focusing on environmental impacts embedded in products during the manufacturing stage. This approach allows the analysis to highlight the environmental implications of internal manufacturing scrap.
In industrial applications, Environmental Product Declarations (EPDs) are frequently used as reliable sources of standardized life cycle data. In this study, EPD data are used to represent baseline environmental impacts associated with rejected products. Additional Life Cycle Assessment modelling is conducted using SimaPro software supported by datasets from the ecoinvent database to evaluate alternative circular scenarios.

2.8 Digital Technologies Supporting Circular Decision-Making

Digital technologies are increasingly recognized as important enablers for implementing circular economy strategies in industrial systems. Technologies such as data analytics, digital twins, and data-driven decision-support systems can support improved monitoring and management of product lifecycles (Acerbi et al., 2021).

These technologies enable manufacturers to collect and analyse large volumes of production data, which can improve understanding of product failures and support better decision-making regarding repair, reuse, or recycling options (Psarommatis et al., 2025).

Digital tools may therefore support more efficient implementation of circular strategies by reducing uncertainty in decision-making and improving traceability across manufacturing systems.

2.9 Summary of Key Findings from Previous Studies

The literature reviewed in this chapter highlights several important insights regarding the implementation of circular economy strategies within industrial systems. First, previous research demonstrates that circular economy approaches aim to reduce environmental impacts by maintaining the value of products, components, and materials for as long as possible through strategies such as repair, reuse, remanufacturing, and recycling (Reike et al., 2018; Messmann et al., 2019).

Second, studies focusing on the electrical and electronic equipment sector emphasize the growing environmental challenges associated with Waste Electrical and Electronic Equipment (WEEE). Research shows that electronic products often contain valuable materials that can be recovered through improved circular management practices (Boldoczki et al., 2020; Corsini et al., 2020). However, most existing studies focus on post-consumer waste streams, while relatively limited attention has been given to internal manufacturing waste generated during production processes. The environmental significance of this sector is reinforced by its substantial resource consumption. The EEE manufacturing industry is one of the most resource-intensive sectors globally: it accounts for approximately 20% of worldwide copper consumption (around 4 million tonnes per year), consumes an estimated 300 tonnes of gold and over 5,000 tonnes of silver annually through electronic components and contact surfaces, and requires large volumes of steel and engineering polymers for housings and insulation systems (UNEP, 2013; European Commission, 2020). From an energy perspective, the electronics manufacturing sector is estimated to consume over 1,000 TWh of electricity per year globally, making it a significant contributor to industrial energy demand (IEA, 2022). Water consumption is also considerable, as manufacturing and surface-treatment processes for electrical components require high-purity water, with some facilities consuming hundreds of litres per unit produced. These resource flows reinforce the urgency of applying value retention strategies not only at the end-of-life stage but also within manufacturing processes, where rejected units represent a direct loss of embedded energy, water, and critical materials.

Third, the literature highlights the importance of value retention strategies, such as dismantling, repair, and component recovery, for improving resource efficiency and reducing environmental impacts. These strategies enable manufacturers to recover functional components from defective products and prevent the loss of embedded materials and energy (Bovea et al., 2020).

In addition, research on EcoDesign and Life Cycle Assessment (LCA) demonstrates that design decisions and lifecycle thinking play a crucial role in improving product sustainability. EcoDesign strategies such as design for repair and design for disassembly can facilitate circular product management, while LCA provides a methodological framework for

evaluating the environmental benefits of different circular strategies (Ardente & Mathieux, 2014).

Finally, recent studies highlight the potential of digital technologies, including data analytics and artificial intelligence, to support circular decision-making within manufacturing systems. These technologies can improve product diagnostics, traceability, and resource management across production processes (Acerbi et al., 2021; Psarommatis et al., 2025).

Despite these advances, the literature reveals a significant gap regarding the practical implementation of circular strategies for internal manufacturing scrap within industrial production environments. In particular, limited research has examined how rejected products generated during manufacturing processes can be systematically assessed and managed using circular approaches such as dismantling, repair, and component recovery. This gap highlights the need for further investigation into how circular economy principles can be applied to improve the management of manufacturing scrap in industrial contexts such as the production of electrical equipment.
Furthermore, limited research has combined industrial case studies, quantitative environmental evaluation, and circular decision workflows within real manufacturing environments. This limitation highlights the importance of developing integrated frameworks capable of supporting practical circular implementation in industrial production systems.

3 METHOD

3.1 Research Design

This research adopts an exploratory industrial case study design to investigate how circular economy strategies can improve the management of internal manufacturing scrap within a real manufacturing environment. Case study methodology is widely used in sustainability and industrial engineering research when analysing complex industrial systems within their operational, technological, and environmental contexts. This approach enables a detailed understanding of production workflows, quality control procedures, scrap management practices, and decision-making processes implemented at factory level.

The study focuses on the production of AF contactors (sizes 4–6) at ABB Smart Power, where rejected units are generated during assembly, testing, and quality control operations. These products were selected because they represent a realistic and data-rich example of electromechanical manufacturing involving multi-material assemblies, metallic conductors, polymer housings, coils, and switching mechanisms requiring high manufacturing precision. The AF contactor series therefore provides an appropriate industrial context for evaluating

material losses, circular recovery potential, and environmental impacts associated with internal production scrap.

This methodological approach was considered particularly appropriate because the management of internal manufacturing scrap involves both measurable environmental impacts and operational decision-making processes that cannot be fully captured through a single analytical method.

- The qualitative component included semi-structured interviews, facility observations, and discussions with stakeholders from production, quality, maintenance, engineering, sustainability, and business operations departments. This approach enabled a better understanding of current scrap management practices, operational constraints, recovery opportunities, and decision-making processes within the manufacturing environment.
- The quantitative component was based on the analysis of production data obtained from ABB internal monitoring systems, including Power BI dashboards, and scrap reports. These data were used to evaluate scrap quantities, material losses, and the environmental implications associated with rejected products.

The combination of qualitative industrial observations and quantitative environmental modelling enables a more comprehensive understanding of both operational feasibility and environmental performance within the proposed circular framework. This methodological triangulation strengthens the validity, reliability, and industrial relevance of the research by connecting theoretical circular economy principles with practical manufacturing conditions.

The analytical framework follows a scenario-based comparative structure between:

- The AS-IS scenario represents the current industrial scrap management workflow implemented within the ABB production environment, where rejected contactors generated during manufacturing and quality control processes are classified as manufacturing scrap and directly sent to external recycling partners without any circular strategies like dismantling, repair, or component-level recovery.
- In contrast, the TO-BE framework introduces circular value-retention strategies including dismantling, repair, component recovery, and design-for-repair approaches intended to preserve functional value, reduce material losses, and improve resource efficiency within the manufacturing system.

Each scenario is evaluated in terms of:

Resource efficiency; operational feasibility; value-retention potential; and environmental performance.

Life Cycle Assessment (LCA) is integrated into the research design as the primary quantitative environmental assessment method. Environmental modelling is conducted using SimaPro software with a cradle-to-gate boundary was selected because the study focuses specifically on defective newly manufactured contactors rejected before market entry. Since the products never reach the customer or use phase, downstream stages such as product use and end-of-life treatment were considered outside the scope of this assessment.

Although several environmental impact categories are generated during the LCA modelling, the study primarily focuses on:

- Global Warming Potential (GWP); freshwater use; and cumulative energy demand.

As these indicators represent the most relevant environmental dimensions associated with manufacturing-intensive industrial systems, particularly regarding material production, energy consumption, and resource use.

The LCA modelling is supported by ecoinvent datasets and Environmental Product Declarations (EPDs) to ensure methodological consistency with ISO 14040 and ISO 14044 standards.

3.2 Case Study Context

The empirical research was conducted within the ABB Smart Power manufacturing facility, which produces electrical switching devices used in industrial and commercial power distribution systems. Among these products, AF contactors represent critical electromechanical components designed to control electrical circuits and manage power distribution in a wide range of industrial applications.

Due to their functional importance, technological complexity, and high production volumes, AF contactors provide an appropriate industrial context for evaluating circular manufacturing strategies and environmental performance within a real production environment. Their multi-material composition and precision-based assembly processes make them particularly relevant for analysing material efficiency, internal manufacturing scrap, and opportunities for component-level recovery.

AF contactors are composed of multiple components manufactured from metals, polymers, and electromechanical elements. These components are produced through manufacturing processes such as injection melding, metal forming, machining, assembly, and functional testing, all of which contribute to significant embodied material, energy, and environmental impacts. Efficient management of rejected products therefore becomes important not only from an operational perspective, but also from a sustainability and resource-efficiency standpoint.

For analytical consistency, this study focuses on AF contactor sizes 4, 5, and 6 as illustrated in Figure 1, specifically:

- AF116-146; AF190-205; AF265-370



Figur 1 AF contactors line 4-6 sizes

These products are manufactured through two complementary production systems within the facility:

1. The fully automated production line.
2. The semi-automated assembly and testing line.

Both production systems involve multiple assembly, inspection, and quality control stages. During these operations, products may be rejected due to manufacturing deviations, failed functional tests, assembly defects, or quality non-conformities. Under the current AS-IS workflow, these rejected contactors are classified as manufacturing scrap and directly sent to external recycling partners without repair, dismantling, or component-level recovery operations.

From a circular economy perspective, this current scrap management approach results in the loss of significant embedded material, manufacturing, and functional value. Based on the 2025 operational data for the AF4–6 production lines, a total of 485 units were rejected out of 236,500 produced, corresponding to a scrap rate of approximately 0.21%. These rejected contactors represent an estimated annual material loss of approximately 1.78 tonnes and an embodied climate impact of approximately 13.5 tonnes CO_2 equivalent. In terms of primary energy, the manufacturing of the scrapped units is estimated to have consumed approximately 53 MWh, while the associated freshwater footprint amounts to approximately 216 m^3 per year. The annual economic loss attributable to internal manufacturing scrap is estimated at approximately 100,000 SEK, accounting for raw materials, energy, labor, and machine operation already invested in the rejected products, despite the fact that many rejected units still contain reusable or functional components. These rejected products

therefore represent not only material losses, but also the loss of energy, labor, manufacturing effort, and environmental value already invested during upstream production stages.

A detailed understanding of the causes, frequency, and component characteristics of rejected contactors is therefore essential for identifying feasible circular recovery opportunities within the manufacturing system. This analysis supports the development of alternative recovery pathways such as repair, dismantling, and component reuse strategies intended to improve value retention, reduce material losses, and minimize environmental burdens associated with internal production scrap.

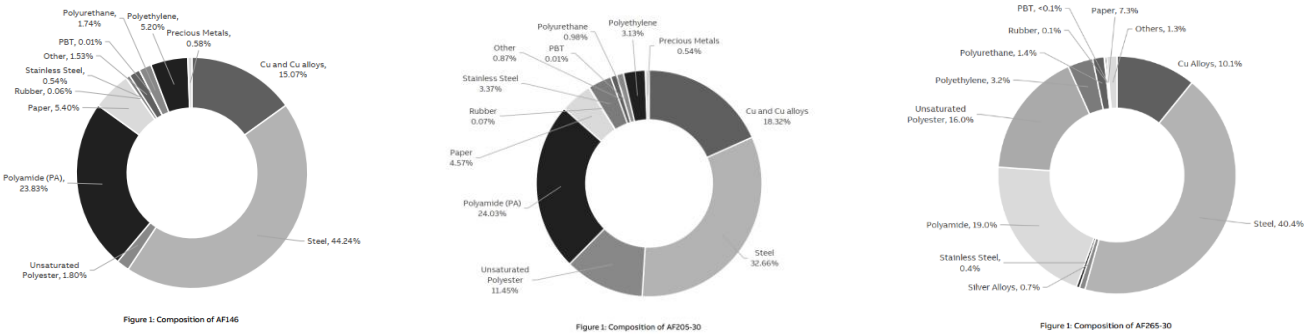
3.2.1 Material Composition of Selected AF Contactor Sizes

The AF contactors analysed in this study (sizes 4–6) consist of multiple material categories, including metallic, polymeric, and auxiliary materials. Although material quantities vary between product sizes, the general product architecture remains relatively consistent across the AF contactor family.

- **Metals:** Steel components provide structural rigidity and mechanical resistance, while copper and copper alloys are used in conductive parts responsible for electrical transmission and switching functions. Small quantities of precious metals are also incorporated into contact surfaces to improve conductivity, durability, and wear resistance.
- **Polymers:** Engineering thermoplastics, particularly polyamide (PA), are used in housings, insulators, and internal protective structures to ensure mechanical stability and electrical insulation.
- **Auxiliary materials:** Additional materials include elastomers used for sealing functions as well as packaging materials such as paper and cardboard used during handling and transportation. Depending on the product configuration, certain models may also contain small electronic elements or printed circuit inserts.

Although the material proportions differ between AF116-146, AF190-205, and AF265-370 contactors, the presence of energy-intensive materials such as steel, copper, and engineering plastics significantly influences the environmental profile of the products. These materials are associated with substantial environmental burdens related to raw material extraction, processing, and manufacturing activities.

Understanding the material composition is also essential for evaluating circular recovery opportunities within the manufacturing system. High-impact materials such as copper, steel, and engineering polymers represent significant value-retention potential when recovered through repair, dismantling, or component reuse strategies. Their preservation can contribute to reducing raw material demand, minimizing manufacturing-related environmental impacts, and improving overall resource efficiency.



Figur 2 Materiels compositions of contactors sizes 4-6

Figur 2 show us the different pourcentage of materiels compositions of the contactors size4-6, but you will find it more detailed in the EPDs in Appendix2.

3.3 Data Collection

Data collection for this study followed a multi-source empirical approach combining qualitative insights with quantitative production and environmental data. The objective was to obtain a comprehensive understanding of current scrap management practices, operational constraints, and circular recovery opportunities within the ABB Smart Power manufacturing environment.

Data collection activities were conducted between January 2025 and December 2025, enabling the integration of operational observations, stakeholder perspectives, and production records over an extended industrial production period.

3.3.1 Semi-Structured Interviews

Qualitative data were collected through semi-structured interviews with key stakeholders involved in the production, quality control, sustainability, and management of AF contactors at ABB Smart Power.

A total of 10 interviews were conducted with representatives from multiple functional departments, including production, maintenance, sustainability and communication management, research and development (R&D), quality management, business operations, and Life Cycle Assessment (LCA) and EcoDesign specialists. The inclusion of multiple stakeholder groups was intended to capture diverse operational perspectives and reduce potential bias associated with single-department observations.

The interviews aimed at:

• Current scrap management practices; rejection causes and defect classification systems; decision-making procedures regarding rejected products; operational constraints affecting circular recovery implementation; and opportunities for implementing circular strategies such as repair, dismantling, and component reuse.

Semi-structured interviews were selected because they provide sufficient flexibility for investigating complex industrial systems while maintaining alignment with predefined research objectives. This approach was considered particularly appropriate given the operational complexity of manufacturing scrap management, where important process knowledge, informal practices, and recovery-related decision-making are often embedded within stakeholder experience and not fully documented through formal operational records.

3.3.2 Production and Operational Data

- Quantitative production and operational data were obtained from internal ABB monitoring systems in order to establish an empirical understanding of manufacturing scrap generation and operational performance within the AF contactor production environment.
- The datasets were collected from multiple internal sources, including: Power BI dashboards; scrap reports; manufacturing records; and structured Excel datasets.

This data provided detailed information regarding:

• Production volumes; Rejected units; scrap rates; rejection categories; and operational performance indicators for both the fully automated production line and the semi-automated assembly and testing line.

The use of multiple internal data sources enabled cross-verification of production information and contributed to improving data reliability and consistency throughout the analysis.

The quantitative datasets were primarily used to:

• Quantify manufacturing scrap generation; estimate material losses associated with rejected contactors; identify recurring rejection patterns; and establish the AS-IS baseline representing the current internal scrap management system.

The AS-IS baseline served as the empirical reference scenario for evaluating alternative TO-BE circular recovery strategies. Within the current operational workflow, rejected contactors are classified as manufacturing scrap and directly sent to external recycling partners without repair, dismantling, or component-level recovery operations.

- ### 3.3.3 Environmental Data Sources

Environmental data were obtained from verified Environmental Product Declarations (EPDs), ecoinvent datasets, and Life Cycle Assessment databases integrated within SimaPro.

The environmental assessment was developed in accordance with ISO 14040 and ISO 14044 standards using cradle-to-gate system boundaries.

The environmental modelling primarily focused on:

- Global Warming Potential (GWP); freshwater use; and cumulative energy demand.

as these indicators were considered the most relevant for evaluating the environmental implications of manufacturing-intensive industrial systems.

The environmental data were used to:

- Define Life Cycle Inventory (LCI) flows; quantify environmental burdens associated with rejected products; and compare the AS-IS and TO-BE circular recovery scenarios.

3.3.4 Data Validation and Triangulation

To improve reliability and methodological consistency, data obtained from interviews, production records, and environmental sources were cross validated through triangulation.

This process included:

- Comparing stakeholder observations with operational data; validating scrap rates using multiple production records; and verifying environmental assumptions through standardized LCA datasets and EPD information.

The triangulation process strengthened the credibility of the AS-IS baseline and supported the development of realistic and operationally feasible TO-BE circular recovery scenarios.

3.4 Data Analysis

The analysis combined qualitative and quantitative approaches to evaluate manufacturing scrap generation, circular recovery opportunities, and environmental performance within the studied production system.

3.4.1 Qualitative Analysis

Qualitative information obtained from interviews was analysed using thematic interpretation methods. Interview data were reviewed and grouped into thematic categories representing:

- Rejection causes; recovery opportunities; operational constraints; and decision-making procedures.

Attention was given to identifying opportunities where rejected products could potentially be redirected toward repair, dismantling, or component recovery instead of direct recycling.

3.4.2 Quantitative Analysis

Quantitative production data were analysed to estimate:

- Scrap quantities; rejection rates; distribution of defect causes; and annual volumes of rejected AF contactors.

These calculations enabled the establishment of the AS-IS baseline describing the current scrap management workflow and associated material losses within the manufacturing system.

3.4.3 Scenario Development

Based on interview insights and production data analysis, alternative TO-BE circular recovery scenarios were developed.

The proposed scenarios included:

- Dismantling of rejected contactors; repair of recoverable units; component recovery strategies;

and design-for-repair considerations.

The feasibility of each scenario was evaluated according to:

- Technical constraints; workflow compatibility; operational practicality.

3.4.4 Integration with Environmental Modelling

The results from qualitative and quantitative analyses were integrated into the Life Cycle Assessment modelling process developed in SimaPro.

Quantitative production data were used to define inventory inputs and material flows, while qualitative insights supported the definition of operational assumptions related to dismantling, repair, transportation, and recovery operations.

This integrated analytical approach ensured consistency between industrial practice and environmental modelling while supporting the development of realistic circular manufacturing recommendations within the ABB Smart Power production environment.

3.5 Life Cycle Assessment (LCA)

Life Cycle Assessment (LCA) was applied as the primary environmental evaluation method to assess the impacts associated with internal manufacturing scrap and to quantify the potential environmental benefits of implementing circular recovery strategies within the ABB Smart Power production environment.

The LCA methodology follows the principles defined in ISO 14040 and ISO 14044 standards and is aligned with the EN 15804 + A2 framework commonly used in Environmental Product Declarations (EPDs). Environmental modelling was conducted using SimaPro software supported by datasets from the ecoinvent database.

The assessment compares the current AS-IS scrap management system with proposed TO-BE circular recovery scenarios involving dismantling and repair operations. The design-for-repair approach was considered as a long-term strategic eco-design perspective and was therefore evaluated qualitatively rather than fully quantified within the current LCA model.

3.5.1 Goal and Scope Definition

The goal of the Life Cycle Assessment is to quantify the environmental impacts associated with rejected AF contactors (sizes 4–6) and to evaluate the environmental improvement potential of circular recovery strategies implemented within internal manufacturing processes.

The assessment specifically compares:

- The current AS-IS scrap management system, where rejected contactors are directly sent to external recycling partners; and the proposed TO-BE circular recovery scenarios involving dismantling and repair operations.

The analysis focuses on the environmental burdens already embedded within rejected products during manufacturing stages and evaluates how circular recovery strategies can reduce material losses and environmental impacts.

3.5.2 Functional Unit

The functional unit for this study was defined as the management of one defective newly manufactured contactor generated during production and quality control operations, providing an equivalent recovery service through scrapping, dismantling, or repair pathways.

The functional unit was applied to three representative contactor models (AF116-146, AF190-205, and AF265-370) to enable a consistent comparison between the AS-IS and TO-BE scenarios. This approach allows the environmental performance of alternative recovery strategies to be assessed under an equivalent manufacturing function and scaled to annual production quantities.

3.5.3 System Boundaries

A cradle-to-gate system boundary was considered most appropriate because the study focuses on newly manufactured defective contactors rejected before market entry. Since the products do not reach the customer or use phase, downstream stages including product use, maintenance, and end-of-life treatment were excluded from the assessment.

The system includes:

- Raw material extraction; material processing; component manufacturing; transportation;

and assembly operations embedded within rejected contactors.

The use phase and end-of-life stages are excluded because the study focuses exclusively on internal manufacturing scrap generated within factory boundaries.

The AS-IS scenario represents the current industrial workflow in which rejected contactors are directly sent to external recycling without dismantling or repair operations.

The TO-BE scenarios evaluate circular recovery alternatives aimed at preserving component value and reducing environmental burdens associated with manufacturing-intensive materials and processes.

To clarify the scope of comparison: the cradle-to-gate boundary (EN 15804+A2 modules A1–A3) quantifies the environmental burden already embedded in each rejected contactor at the point of rejection. This embedded burden is taken from the product-stage modules of the AF116–146, AF190–205 and AF265–370 EPDs. The recovery operations evaluated in the TO-BE scenarios are not part of the EPD scope; they are modelled as additional foreground processes in SimaPro using ecoinvent unit processes and are layered onto the cradle-to-gate product burden rather than treated as a formal extension to end-of-life (C) modules. The AS-IS scenario treats the full A1–A3 embedded burden of the rejected unit as lost and includes external transport to the recycling partner and a generic metals/plastics recycling process (ecoinvent). The dismantling scenario retains the embedded burden of recovered reusable components while including manual dismantling, internal handling, and recycling of the non-recoverable fraction. The repair scenario retains the embedded burden of the whole unit except for the replaced housing, adding the A1–A3 production burden of one new housing and the energy of the standard functional re-test.

3.5.4 Data Sources

Environmental data used in this study were obtained from Environmental Product Declarations (EPDs), ecoinvent datasets, and Life Cycle Assessment databases integrated within SimaPro.

The EPDs provided standardized cradle-to-gate environmental information for representative AF contactor models developed in accordance with EN 15804 + A2 standards, while ecoinvent datasets were used to model dismantling, repair, recycling, transportation, and other recovery-related processes within the SimaPro environment.

Production data related to rejected contactors was obtained from ABB internal monitoring systems, including:

- Power BI dashboards;
- manufacturing records;
- and scrap reports provided by the quality department.

These datasets were used to define production quantities, rejected unit volumes, scrap generation rates, and scaling factors for comparative environmental assessment. The integration of standardized environmental datasets and real industrial production data contributed to improving the reliability and industrial relevance of the assessment.

3.5.5 Life Cycle Inventory (LCI)

The Life Cycle Inventory (LCI) was developed using environmental and production data associated with AF116-146, AF190-205, and AF265-370 contactors.

Inventory flows included:

- metallic materials;
- engineering polymers;
- manufacturing and transportation processes;
- dismantling activities;
- repair operations;
- and recycling processes.

Environmental datasets from the ecoinvent database were integrated into SimaPro to model material flows and operational recovery scenarios.

For the repair scenario, the inventory focused on the selective replacement of defective components while preserving the majority of the contactor structure and functional elements. This modelling approach reflects a component-level value-retention strategy, where embedded material and manufacturing value are preserved through targeted recovery operations.

3.5.6 Impact Assessment Method (LCIA)

Environmental impacts were evaluated using indicators consistent with the EN 15804 + A2 framework, commonly applied in industrial Environmental Product Declarations (EPDs). Impact assessment was performed using:

•The EN 15804 + A2 method; EN 15804 + A2 adapted EF 3.1 normalization and weighting; and Cumulative Energy Demand (CED) indicators.

Although several environmental impact categories were generated during the assessment, the study primarily focused on:

- Global Warming Potential (GWP); freshwater use; and cumulative energy demand.

These indicators were selected because they represent key environmental dimensions associated with manufacturing-intensive industrial systems, particularly regarding greenhouse gas emissions, energy consumption, material production, and water-related environmental burdens. Additional environmental indicators generated within SimaPro were used as complementary information for comparative interpretation.

3.5.7 Scenario Definition

To evaluate the environmental implications of circular manufacturing strategies, one AS-IS baseline scenario and three TO-BE scenarios were developed. The dismantling and repair scenarios were quantitatively assessed through LCA, while design for repair was considered qualitatively as a long-term eco-design strategy.

3.5.7.1. AS-IS Scenario

The AS-IS scenario represents the current scrap management system at ABB Smart Power, where rejected AF contactors generated during manufacturing and quality control are directly sent to external recycling without dismantling, repair, or component recovery.

3.5.7.2. TO-BE Scenario 1 – Dismantling

Rejected contactors undergo controlled dismantling to recover reusable components and separate materials before recycling non-recoverable parts.

3.5.7.3. TO-BE Scenario 2 – Repair

Rejected contactors with recoverable defects are repaired through selective replacement of defective components while preserving the majority of the product structure and functionality.

3.5.7.4. TO-BE Scenario 3 – Design for Repair

The design-for-repair scenario represents a long-term eco-design strategy aimed at improving modularity, accessibility, dismantling efficiency, and repairability in future product configurations. Since this strategy is future-oriented, it was analysed qualitatively rather than modelled within the current LCA framework.

3.5.8 Methodological Limitations

Several limitations should be acknowledged in this study. First, the analysis is based on a single industrial case study, providing context-specific insights related to the ABB Smart Power manufacturing environment. Due to confidentiality requirements, certain operational and environmental information could only be presented in aggregated form.

Additional limitations are associated with environmental modelling assumptions, as some material compositions, transportation distances, and operational parameters were estimated using representative datasets. Dismantling and repair scenarios were modelled using simplified assumptions based on expert knowledge and industrial observations rather than direct industrial-scale measurements.

Despite these limitations, the integration of standardized environmental datasets, industrial production data, and stakeholder validation provides a reliable basis for comparative environmental assessment.

4 RESULTS

4.1 Industrial Process Description

4.1.1 Manufacturing Process Overview

The production system analyzed in this study focuses on the manufacturing of AF4–6 contactors (AF116-146, AF190-205, and AF265-370) at ABB Smart Power. These electromechanical products are manufactured through sequential production stages including component manufacturing, assembly, functional testing, and quality control.

The production process combines automated and semi-automated systems, where metallic and polymer components are assembled and tested to ensure compliance with ABB quality and safety requirements.

During inspection and testing stages, products may be rejected due to:

• Manufacturing deviations; failed functional tests; assembly defects or quality non-conformities.

Rejected units are classified as internal manufacturing scrap within the production system.



Figur 3 Whole contactor Scrapped

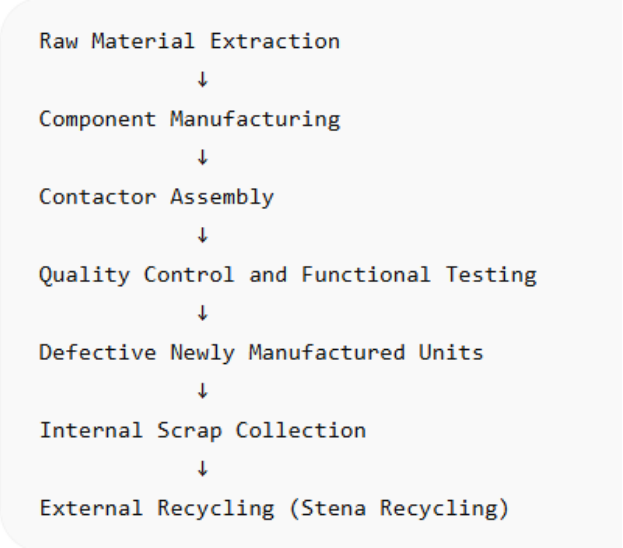
According to operational data collected for 2025, approximately 236,500 contactors were manufactured across the AF4–6 production lines, of which 485 units were rejected, corresponding to an estimated scrap rate of approximately 0.21%, see Table1.

Although relatively low from an operational perspective, rejected contactors represent substantial embedded material, manufacturing, and environmental value losses due to the material intensity and production complexity of the products.

Under the current AS-IS workflow as presented in Figure 3, rejected units are collected and sent directly to external recycling partners without dismantling, repair, or component

recovery operations as shown in Figure 2. While recycling enables partial material recovery, it does not preserve the functional value of reusable components or the embedded environmental impacts associated with upstream manufacturing.

This baseline assessment highlights an opportunity for implementing circular value-retention strategies such as repair, dismantling, and component recovery.



Figur 4 AS-IS Manufacturing and Scrap Management Flow

Tabell 1 Production and Scrap Overview

Parameter	Value
Total production (2025)	236,500 units
Rejected units	485 units
Scrap rate	0.21 %
Scrap destination	External recycling
Recycling partner	Stena Recycling

Table 1 summarises the key production and scrap indicators for the AF4–6 production line at ABB Smart Power in 2025. The data confirm that the 485 rejected units, representing a scrap rate of 0.21%, were entirely directed to external recycling via Stena Recycling, with no dismantling, repair, or component recovery operations carried out internally. This highlights that the current system operates as a linear scrap management model in which defective newly manufactured contactors exit the production system without recovery of their functional or material value. Despite the relatively low scrap rate, the absence of higher-value recovery pathways represents a measurable loss of embedded manufacturing value and environmental resources, which motivates the development of the TO-BE circular framework presented in this study."

4.2 AS-IS Baseline: Internal Production Scrap Assessment

4.2.1 System boundary and scope

The AS-IS baseline evaluates the current internal scrap management system implemented within the AF4–6 contactor production line at ABB Smart Power, focusing on rejected units generated during assembly, testing, and quality control operations.

The assessment follows a cradle-to-gate system boundary, covering raw material extraction, component manufacturing, assembly, and internal scrap generation within factory boundaries. Downstream stages, including product use and end-of-life treatment, were excluded since rejected units never entered the market.

Under the current operational workflow, rejected contactors are collected and directly sent to external recycling partners without dismantling, repair, or component recovery.

The baseline environmental assessment was developed using EPDs, ecoinvent datasets, and SimaPro modelling, providing the reference scenario for comparison with the proposed TO-BE circular recovery strategies.

4.2.2 Production performance

In 2025, the AF4–6 production line at ABB Smart Power manufactured approximately 236,500 contactors, of which 485 units were rejected during internal quality control processes and classified as manufacturing scrap. This corresponds to an estimated internal scrap rate of approximately 0.21%, see Table 2.

The rejected products included different contactor variants, primarily:

- AF116-146(Size 4);
- AF190-205 (Size 5);
- AF265-370-30 (Size 6).

Due to limited access to detailed scrap allocation data by product variant, rejected units were treated as a representative mixed product set reflecting the overall production system. This assumption enabled the establishment of a consistent AS-IS baseline for comparative environmental assessment.

Tabell 2 Production Performance Indicators (2025)

Indicator	Value	Unit	Description
Total production volume	236,500	units/year	Total number of AF4–6 contactors manufactured
Rejected units (internal scrap)	485	units/year	Units rejected during quality control

Indicator	Value	Unit	Description
Product range	AF116-146 / AF190-205 / AF265-370	–	Representative product models (Size 4–6)
Production system type	Mixed-model production	–	Multiple product variants on same line

4.2.3 Scrap rate analysis

The internal scrap rate is a key performance indicator used to evaluate production efficiency. It is defined as the ratio of rejected units to the total production volume.

The scrap rate is calculated using Equation (1):

$$\text{Scrap Rate (\%)} = \frac{\text{Number of Rejected Units}}{\text{Total Production Volume}} \times 100$$

Based on the production data presented in Section 4.2.2, the scrap rate for the AF4–6 production line in 2025 is calculated as follows:

$$\text{Scrap Rate} = \frac{485}{236{,}500} \times 100 = 0.205\%$$

This corresponds to approximately 2.05 rejected units per 1,000 units produced, as shown in Table 3.

From an operational perspective, this value indicates a high level of production efficiency and process stability. However, the scrap rate alone does not capture the magnitude of the associated material, economic, and environmental losses, which are analysed in the following sections.
However, even a low scrap rate may result in significant environmental losses when the rejected products are complex and resource intensive. Therefore, the magnitude of environmental impacts associated with rejected units must be evaluated in addition to production efficiency indicators.

Tabell 3 Scrap Rate Calculation

Parameter	Value	Unit
Total production	236,500	units
Rejected units	485	units
Scrap rate	0.205	%
Scrap intensity	2.05	units / 1,000 units

4.2.4 Physical material loss

To quantify the physical material loss associated with internal production scrap, the mass of representative contactor models was analyzed. The AF4–6 product family includes different sizes as presented in Table 4, with significant variations in weight due to differences in material composition and design.

Tabell 4 Product Weight of Representative AF Contactor Models

Model	Size	Total Weight (g)	Total Weight (kg)
AF116-146-30	Size 4	1952.18	1.952
AF190-205	Size 5	3458.22	3.458
AF265-370-30	Size 6	5635.30	5.635

Due to the absence of detailed production distribution data for each model, a simple arithmetic mean was used to estimate the average product mass, as expressed in Equation (2):

$$\text{Average Mass} = \frac{m_{AF146} + m_{AF205} + m_{AF265}}{3}$$

$$\text{Average Mass} = \frac{1.952 + 3.458 + 5.635}{3} = 3.68\ \text{kg}$$

Based on this estimated average mass, the total material loss associated with internal scrap was calculated by multiplying the average product mass by the number of rejected units, as shown in Equation (3):

$$\text{Total Material Loss} = \text{Average Mass} \times \text{Number of Rejected Units}$$

$$= 3.68 \times 485 = 1{,}784.8\ \text{kg} \approx 1.78\ \text{tonnes}$$

This corresponds to a total material loss of approximately 1.78 tonnes within the factory boundary in 2025.

To relate this loss to the overall production volume, a material loss intensity indicator was calculated, representing the average material loss per unit produced, as defined in Equation (4):

$$\text{Material Loss Intensity} = \frac{\text{Total Material Loss}}{\text{Total Production Volume}}$$

$$= \frac{1{,}784.8}{236{,}500} \approx 0.0076\ \text{kg/unit} = 7.6\ \text{g/unit}$$

This indicates that, on average, 7.6 grams of material are lost per unit produced due to internal scrap. As presented in Table 5.
Although this value appears relatively small at the unit level, its cumulative impact at annual production scale highlights the importance of improving resource efficiency through circular recovery strategies.

Tabell 5 Summary of Material Loss Indicators

Indicator	**Value**	**Unit**
Average product mass	3.68	kg/unit
Rejected units	485	units
Total material loss	1,784.8	kg
Total material loss	1.78	tonnes
Material loss intensity	7.6	g/unit

4.2.5 Economic impact

The economic impact associated with internal manufacturing scrap was estimated using internal ABB production cost information related to rejected contactors.

According to operational estimates, the annual economic loss associated with 485 rejected units was approximately 100,000 SEK**.** This value includes not only material losses but also manufacturing resources already invested in rejected products, including:

• Electricity consumption; water use; labor; machine operation; assembly activities; and associated production resources.

Since rejected contactors never reach functional use, these embedded manufacturing costs represent direct economic losses for the production system.

These findings demonstrate that internal manufacturing scrap generates both environmental and economic burdens**,** reinforcing the importance of circular recovery strategies aimed at preserving product value and reducing manufacturing losses.

4.2.6 Environmental Impact Assessment of the AS-IS Baseline

4.2.6.1. System boundary and calculation approach

The environmental impacts associated with internal production scrap were quantified using verified Environmental Product Declarations (EPDs) available for three representative AF contactor models: AF116-146-30**,** AF190-205, and AF265-370-30. These models represent different size categories within the AF4–6 product family and were selected to reflect variations in material content and manufacturing complexity.

Since rejected units do not proceed to downstream life-cycle stages and do not deliver any functional service, only the manufacturing-stage impacts reported in the EPDs were considered in the AS-IS environmental assessment. Impacts related to distribution, installation, use, and end-of-life phases were excluded.

The total number of rejected units recorded during the year 2025 was:

$$N_{rejected} = 485 \text{ units}$$

Due to the absence of precise information regarding the distribution of rejected units among the three product sizes, a balanced product-mix assumption was adopted as the primary reference case. In addition, a best-case and worst-case scenario were developed to evaluate the influence of product-size variability on the environmental results.

4.2.6.2. Balanced-case calculation

The manufacturing-phase Global Warming Potential (GWP) per unit obtained from EPD data is:

Tabell 6 Manufacturing-phase Global Warming Potential (GWP) per Contactor Model

Model	Manufacturing GWP
AF116-146-30	12.3 kg CO_2 eq
AF190-205	26.9 kg CO_2 eq
AF265-370-30	44.0 kg CO_2 eq

The average impact per rejected unit was calculated using Equation (1):

$$GWP_{avg} = \frac{GWP_{146} + GWP_{205} + GWP_{265}}{3}$$

Equation (1)

$$GWP_{avg} = \frac{12.3 + 26.9 + 44.0}{3}$$
$$GWP_{avg} = 27.73 \text{ kg } CO_2 \text{ eq/unit}$$

The annual embodied climate impact of internal scrap was calculated using Equation (2):

$$GWP_{annual} = GWP_{avg} \times N_{rejected}$$

Equation (2)

$$GWP_{annual} = 27.73 \times 485$$

$$GWP_{annual} = 13{,}450.67\ \text{kg}\ CO_2\ \text{eq/year}$$
$$GWP_{annual} \approx 13.5\ \text{t}\ CO_2\ \text{eq/year}$$

Primary energy demand calculation

Manufacturing-phase primary energy demand per unit:

Tabell 7 Manufacturing-phase energy demand per Contactor Model

Model	Energy demand
AF116-146-30	175 MJ
AF190-205	385 MJ
AF265-370-30	625 MJ

Average energy demand:

$$Energy_{avg} = \frac{175 + 385 + 625}{3}$$
$$Energy_{avg} = 395\ \text{MJ/unit}$$

Annual energy demand:

$$Energy_{annual} = 395 \times 485$$
$$Energy_{annual} = 191{,}575\ \text{MJ/year}$$

Conversion to MWh:

$$1\ \text{MWh} = 3600\ \text{MJ}$$
$$Energy_{annual} = \frac{191{,}575}{3600}$$
$$Energy_{annual} = 53.2\ \text{MWh/year}$$

Freshwater use calculation

Tabell 8 Manufacturing-phase Freshwater Consumption per Contactor Model

Model	Freshwater use
AF116-146-30	0.208 m³
AF190-205	0.460 m³
AF265-370-30	0.669 m³

Average freshwater use:

$$FW_{avg} = \frac{0.208 + 0.460 + 0.669}{3}$$
$$FW_{avg} = 0.446\ \mathrm{m}^3$$

Annual freshwater use:

$$FW_{annual} = 0.446 \times 485$$
$$FW_{annual} = 216\ \mathrm{m}^3/year$$

4.2.6.3.Sensitivity analysis: Best and Worst cases

To account for uncertainty in product-size distribution, two additional scenarios were evaluated.

Best-case scenario

Assumption:

All rejected units correspond to AF116-146-30, representing the lowest environmental burden.

$$GWP_{best} = 12.3 \times 485$$
$$GWP_{best} = 5965.5\ \mathrm{kg\ CO_2}$$
$$GWP_{best} \approx 6.0\ \mathrm{t\ CO_2}$$

Energy:

$$Energy_{best} = 175 \times 485 = 84{,}875\ \mathrm{MJ}$$
$$Energy_{best} = 23.6\ \mathrm{MWh}$$

Freshwater:

$$FW_{best} = 0.208 \times 485 = 101\ \mathrm{m}^3$$

Worst-case scenario

Assumption:

All rejected units correspond to AF265-370-30, representing the highest environmental burden.

$$GWP_{worst} = 44.0 \times 485 = 21{,}340\ \mathrm{kg\ CO_2}$$
$$GWP_{worst} \approx 21.3\ \mathrm{t\ CO_2}$$

Energy:

$$Energy_{worst} = 625 \times 485 = 303{,}125 \text{ MJ}$$
$$Energy_{worst} = 84.2 \text{ MWh}$$

Freshwater:

$$FW_{worst} = 0.669 \times 485 = 324 \text{ m}^3$$

Tabell 9Scenario-based environmental impact of rejected units (AS-IS baseline)

Scenario	GWP (t CO_2 eq/year)	Energy (MWh/year)	Freshwater (m^3/year)
Best-case	6.0	23.6	101
Balanced-case	13.5	53.2	216
Worst-case	21.3	84.2	324

The results indicate that internal manufacturing scrap represents a significant embodied environmental burden even under conservative assumptions. Depending on the distribution of rejected product sizes, the annual climate impact associated with rejected units ranges from approximately 6.0 to 21.3 t CO_2 eq/year, with the balanced-case estimate reaching approximately 13.5 t CO_2 eq/year as presented in Table6.

Similarly, the associated primary energy demand ranges from 23.6 to 84.2 MWh/year, while freshwater use varies between 101 and 324 m^3/year. These results confirm that rejected products represent not only a loss of material resources but also a loss of previously incurred environmental burdens associated with manufacturing processes.

From a circular manufacturing perspective, these findings provide strong quantitative justification for the development of recovery strategies such as dismantling, repair, and component reuse within the proposed TO-BE framework.

4.2.7 Summary of the As Is Baseline

The AS-IS baseline assessment shows that the AF4–6 production line operates with a relatively low scrap rate of approximately 0.205%, indicating a high level of operational efficiency. However, despite this low rejection rate, internal production scrap results in measurable material, environmental, and economic losses.

The analysis estimates an annual material loss of approximately 1.78 tonnes, associated with an estimated economic loss of approximately 100,000 SEK/year. From an environmental perspective, rejected units represent an embodied impact of approximately 13.5 t CO_2 eq/year, 53 MWh/year of primary energy demand, and 216 m^3/year of freshwater use under the balanced-case scenario.

The scenario-based sensitivity analysis further indicates that climate impacts may vary between approximately 6.0 and 21.3 t CO_2 eq/year, depending on the product mix of rejected units.

From a circular economy perspective, the current manufacturing system remains predominantly linear, as rejected contactors are directly sent to external recycling without repair, dismantling, or component recovery. While recycling enables partial material recovery, the functional value and embedded manufacturing effort of rejected products are lost.

These findings highlight the need for value-retention strategies capable of reducing environmental and economic losses through repair, dismantling, and component recovery, which are explored in the following TO-BE assessment.

4.3 Objective of the TO-BE Scenario

The objective of the TO-BE scenario is to investigate how the current internal manufacturing scrap management system at ABB Smart Power can be improved through the implementation of circular value-retention strategies aimed at reducing material loss, environmental impacts, and economic inefficiencies.

As demonstrated in the AS-IS baseline assessment, rejected contactors are currently directed to external recycling without dismantling, repair, or component recovery. Although recycling enables partial material recovery, it does not preserve the functional value of reusable components or the manufacturing effort already embedded in the products.

The proposed TO-BE framework was therefore developed to explore alternative circular recovery pathways capable of preserving product and component value before recycling is considered. The strategies investigated include:

• Controlled dismantling; component recovery; selective repair operations; and long-term design-for-repair considerations.

The framework was developed based on findings from the AS-IS baseline, stakeholder interviews, production data analysis, and Life Cycle Assessment (LCA) modelling performed in SimaPro. Rather than representing fully industrialized recovery systems, the proposed scenarios reflect operationally feasible circular improvement pathways evaluated through environmental and workflow analysis, As shown in Figure 4.

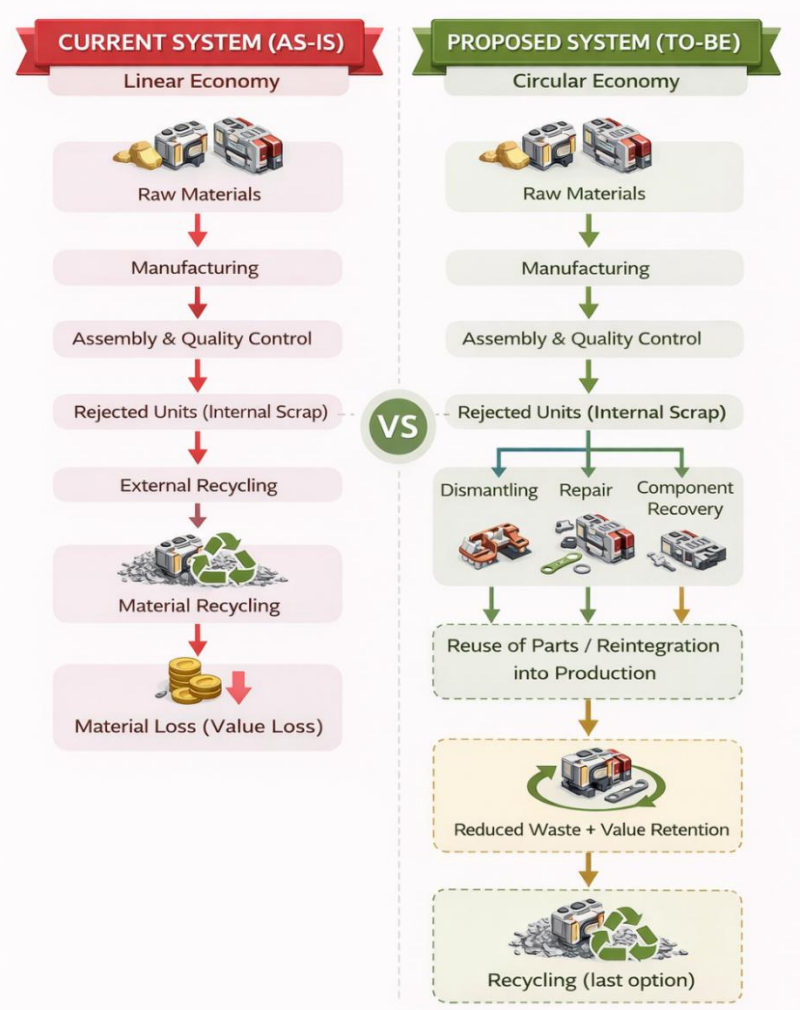
From a sustainability perspective, the TO-BE framework supports the transition from a linear to a more circular manufacturing system, aligned with resource efficiency and value-retention principles.

From a sustainability perspective, the TO-BE framework supports the transition from a linear manufacturing model toward a more circular and resource-efficient production system aligned with:

- SDG 9 (Industry, Innovation and Infrastructure);
- SDG 12 (Responsible Consumption and Production);
- and SDG 13 (Climate Action),

through the reduction of material losses, improved value retention, and mitigation of embodied environmental impacts associated with internal manufacturing scrap.

.



Figur 5 Transition from linear to circular management of internal production scrap

4.3.1 Circular Strategy Framework

The TO-BE framework is based on circular economy principles aimed at maximizing value retention by preserving products, components, and materials at their highest functional value within the manufacturing system.

In contrast to the current AS-IS configuration, where rejected contactors are directly sent to external recycling, the proposed framework prioritizes recovery strategies according to their capacity to preserve embedded product value before recycling is considered.

The framework follows a circular value-retention hierarchy:

• Repair and reuse – highest-value retention strategies;
• Dismantling and component recovery – intermediate value-retention strategies;
• Recycling – final recovery option when repair or recovery is not feasible.

The AS-IS baseline demonstrated that rejected AF contactors contain substantial embedded materials, energy, water, and manufacturing effort, indicating significant recovery potential within the production system.

Based on these findings, the proposed TO-BE framework introduces the following circular strategies:

1. Controlled dismantling: Rejected contactors undergo systematic dismantling to recover reusable components prior to recycling non-recoverable elements.
2. Selective repair : Rejected units presenting localized and recoverable defects are repaired through targeted replacement of defective components while preserving the remaining functional structure.
3. Component recovery and reuse :Validated functional components are reintegrated into the manufacturing system following inspection and quality verification.
4. Design for repair and circularity : Long-term eco-design improvements are proposed to improve modularity, accessibility, dismantling efficiency, and component replaceability in future product generations.
5. Recycling as a final recovery pathway :External recycling remains the last option for products or components that cannot be technically repaired or recovered.

The proposed framework benefits from the controlled conditions of industrial manufacturing, where product history, defect origin, and material composition are traceable, improving the feasibility of circular recovery operations.

4.3.2 Dismantling and Component Recovery

4.3.2.1. Dismantling Process Based on Industrial Procedure (ABB)

The dismantling and component recovery process is proposed as a practical circular strategy for improving the management of internal production scrap within the manufacturing system. The approach was inspired by dismantling and recovery practices observed in another ABB manufacturing facility, where rejected products are selectively disassembled in order to recover functional components before recycling.

However, the proposed TO-BE dismantling scenario does not directly replicate the dismantling process implemented in the other facility. Instead, the workflow was analysed and adapted according to the specific production conditions, operational organization, product configurations, and manufacturing constraints of the current factory environment. The proposed dismantling strategy therefore represents a locally adapted solution developed for the present manufacturing context while maintaining the same circular objective of improving component recovery and value retention.

The dismantling process is designed as a controlled and sequential workflow integrated within existing production operations. Rejected units are first identified during assembly or functional testing stages and are separated from conforming products to avoid unintended mixing within the production flow.

The proposed dismantling workflow consists of the following operational stages, see Table 7:

1. Rejected Unit Identification: Rejected products are detected during assembly or functional testing and isolated from conforming units.
2. Defect Screening: The rejected unit undergoes a basic evaluation to identify the type and severity of the detected defect.
3. Recovery Assessment: Based on predefined recovery rules, the product is classified according to its recovery potential.
4. Dismantling Operation: Products suitable for recovery undergo manual or semi-manual dismantling following standardized dismantling procedures designed to preserve component integrity.
5. Component Inspection: Recovered components are visually inspected and, when necessary, subjected to basic functional verification.
6. Component Classification: Recovered parts are classified as reusable or non-recoverable according to predefined quality and safety requirements.
7. Reintegration or Recycling: Reusable components are reintroduced into the manufacturing system, while non-recoverable parts are directed to recycling streams.

To support dismantling operations under current factory conditions, a simplified defect identification system is proposed. Instead of implementing a complex traceability infrastructure, the proposed approach relies on small color-coded mini labels attached directly to rejected products after quality-control inspection, see Figure 5.

The labelling process is integrated within the existing quality-control workflow. Once a defect is identified, the operator assigns the corresponding recovery category and applies a mini label directly into the rejected contactor.

Three recovery categories are proposed:

- Green label: high recovery potential;
- Yellow label: medium recovery potential;
- Red label: no recovery potential.

The mini-label system enables dismantling operators to rapidly identify the appropriate recovery pathway without repeating detailed defect diagnosis during dismantling operations. Compared with advanced traceability systems, this simplified solution offers several operational advantages, including reduced handling time, lower implementation cost,

simplified operator decision-making, and easier integration into existing manufacturing workflows.

The labelling and sorting operation is expected to require approximately 1.25 minutes per rejected unit, while the dismantling operation itself is estimated at approximately 2 minutes per unit under current factory conditions. The total operational handling time is therefore estimated at approximately 3.25 minutes per rejected contactor.



Figur 6Labelling process with colours distinction in the contactors.

For the 485 rejected units identified in 2025, the total dismantling and labeling workload is estimated at approximately 26.3 working hours per year, corresponding to an operational cost of around 10,000 SEK annually. These findings indicate that the proposed dismantling workflow is operationally feasible and requires relatively limited labour resources. From an environmental perspective, dismantling supports component-level value retention by recovering reusable parts before recycling, thereby reducing material losses and avoiding part of the environmental impacts associated with manufacturing new components.

Tabell 10 Defect code regarding the dismantling process

Color Code	Defect Category	Typical Defect	Recovery Level	Dismantling Decision	Final Destination
🟢	D01 – Marking defect	Incorrect or missing labeling, minor visual non-conformity	High recovery potential	Full dismantling	Component reuse
🟢	D02 – Spring defect	Missing or defective spring	High recovery potential	Targeted dismantling	Component reuse
🟡	D03 – Contact defect	Contact misalignment or positioning issue	Medium recovery potential	Partial dismantling	Reuse + partial recycling

Color Code	Defect Category	Typical Defect	Recovery Level	Dismantling Decision	Final Destination
🟡	D04 – Plastic damage	Cracks or deformation in housing	Medium recovery potential	Selective dismantling	Reuse + partial recycling
🟡	D05 – Functional failure	Product fails electrical or mechanical test	Medium recovery potential	Inspection and selective dismantling	Reuse if compliant / recycling if not
🔴	D06 – Structural defect	Severe deformation or irreversible damage	No recovery potential	No dismantling	Direct recycling

4.3.2.2. Operational Dismantling Sequence

Based on internal observations conducted under current factory conditions, the manual dismantling of one rejected contactor (size 4/6) is estimated to require approximately 2 minutes per unit under standard working conditions as presented in Table 8. The dismantling operation is performed manually by trained operators using standard workshop tools and follows a simplified dismantling sequence intended to recover reusable components while preserving their functional integrity.

The dismantling process mainly involves separating the main subassemblies and sorting reusable and non-recoverable parts before reintegration or recycling. Due to the relatively limited number of rejected units generated annually, the dismantling activity remains small-scale and operationally manageable within the existing production environment.

Tabell 11 Labour cost estimation for dismantling operations

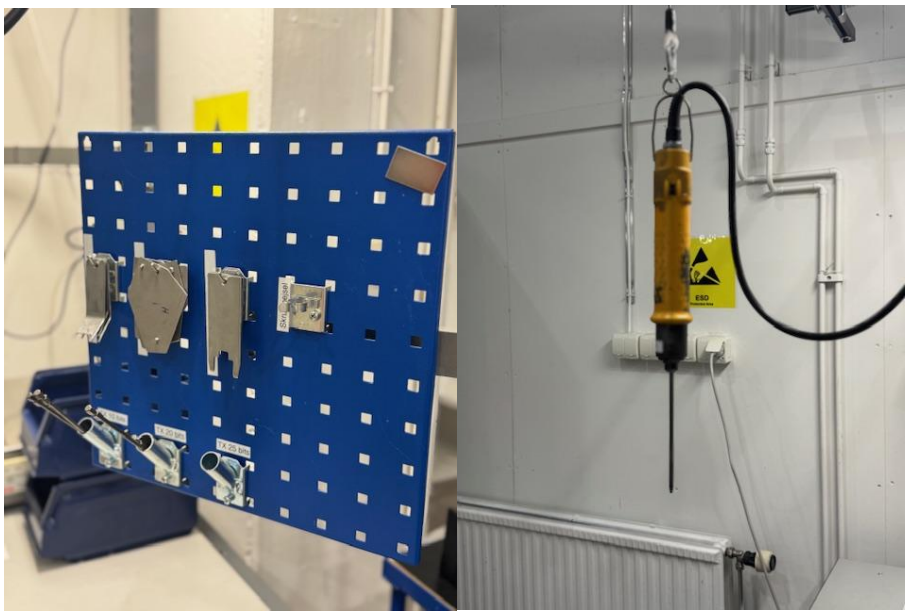
Parameter	Value
Labour rate	380 SEK/hour
Cost per minute	6.33 SEK/min
Dismantling time per contactor	2 min
Estimated labour cost per unit	12.7 SEK

4.3.2.3. Operational Capacity and Annual Workload

In 2025, approximately 485 rejected contactors were identified within the production system. Based on an estimated dismantling time of 2 minutes per unit, the total annual dismantling workload corresponds to approximately 970 minutes (16.2 working hours/year), as presented in Table 9.

Assuming dismantling activities are performed twice per week for one hour per session, the estimated operational capacity reaches approximately 3120 contactors per year, significantly exceeding the observed annual scrap volume.

These findings indicate that the proposed dismantling process is operationally feasible and could be integrated into existing factory operations without causing significant disruption or production bottlenecks. Its manual dismantling with tools shown in the Figure 6.



Figur 7 Labour cost estimation for dismantling operations

Tabell 12 Annual dismantling workload estimation

Parameter	Value
Internal scrap in 2025	485 units
Dismantling time per contactor	2 min/unit
Total dismantling time	970 min
Total annual workload	16.2 hours
Capacity per session	30 units
Weekly dismantling capacity	60 units/week

Parameter	Value
Annual theoretical capacity	3120 units/year
Estimated processing duration for 2025 scrap	≈ 8 weeks

From an operational perspective, the dismantling workload remains relatively limited compared with the existing production capacity. This suggests that dismantling activities could potentially be integrated using existing personnel and infrastructure without requiring major organizational modifications or dedicated dismantling resources.

4.3.2.4. Component Recovery and Controlled Reuse

Following dismantling, components presenting acceptable physical and functional conditions may be reintegrated into the manufacturing system through a controlled component recovery and reuse strategy. Instead of directly recycling all dismantled elements, selected reusable components are given a controlled second life cycle within the production environment.

Recovered components are visually inspected according to predefined quality criteria. Components showing no visible damage or functional risk may be approved for reuse. Typical recoverable components may include:

• Contact systems; coils; magnetic cores; terminals; and selected internal mechanical elements.

To ensure traceability and quality control, validated components receive a dedicated visual identification mark before reintegration into internal storage. A marking system applied to selected housing or plastic elements may facilitate distinction between reused and newly manufactured parts during assembly and testing, as shown in Figure 7.

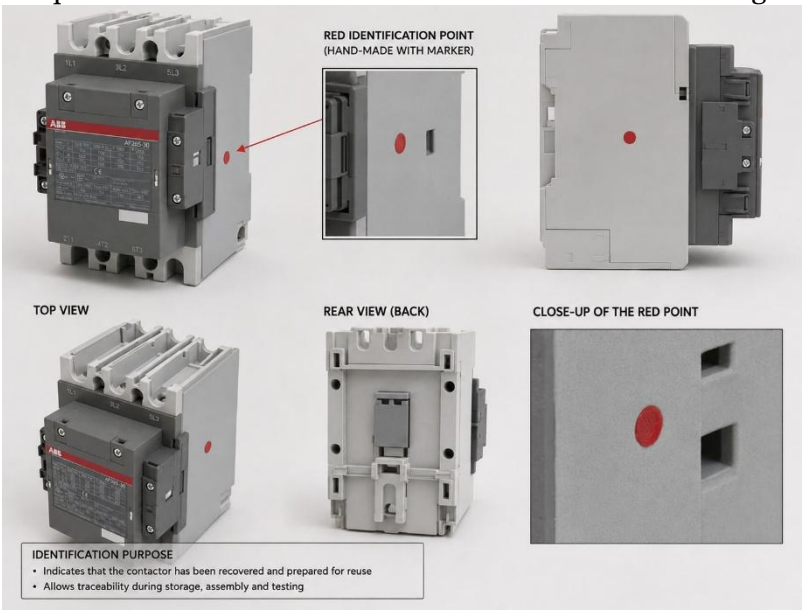
The proposed strategy follows a single-cycle reuse principle, meaning recovered components are authorized for only one secondary usage cycle. Products containing reused components must pass standard ABB quality and functional testing. Components identified as defective after reuse are removed from circulation and sent to recycling.

From a circular economy perspective, this strategy improves value retention by preserving the functional and manufacturing value embedded in reusable components. Environmentally, component reuse may reduce the need for new material extraction, manufacturing operations, energy consumption, freshwater use, and greenhouse gas emissions.

Operationally, the strategy remains relatively simple to implement, relying primarily on manual inspection, visual identification, existing quality-control procedures, and controlled storage reintegration.

Consequently, the strategy can potentially be integrated into the current production environment without requiring major technological modifications or complex digital traceability infrastructure.

Overall, the proposed component recovery and controlled reuse strategy represents a practical and industrially feasible circular solution that strengthens the overall TO-BE framework by extending component lifetime while maintaining strict operational and quality-control requirements within the ABB Smart Power manufacturing environment.



Figur 8 Traceability for the recovery components.

4.3.3 Repair and Product Life Extension

The repair strategy represents the highest-value circular recovery approach proposed within the TO-BE framework because it preserves the rejected contactor as an almost complete functional product while requiring only limited corrective intervention.

In the proposed repair scenario, repair operations are primarily based on localized component replacement rather than complete product reconstruction. The objective is to preserve the majority of the original product structure and embedded manufacturing value while replacing only the defective component responsible for the rejection.

For example, certain rejected contactors may present defects affecting only the external housing while the remaining internal components remain fully functional. In such cases, the defective housing component can be selectively replaced while preserving:

The coil; the contact system; terminals; magnetic components; and the internal electromechanical structure of the contactor.

After replacement of the defective component, the repaired contactor undergoes standard functional testing and quality verification before reintegration into the production flow.

This approach significantly differs from conventional recycling because it avoids the disposal of the entire product despite the fact that most internal components remain fully operational and compliant with industrial performance requirements.

From a circular economy perspective, this repair logic enables product-level value retention by preserving:

- material resources; manufacturing effort; energy consumption; and embedded environmental impacts already invested during production.

Within the ABB Smart Power manufacturing environment, repair is considered a selective and controlled strategy applicable only to rejected products presenting:

- Minor; localized; and clearly recoverable defects.

Due to strict industrial quality and safety requirements, repair operations cannot be applied universally to all rejected contactors and must remain limited to products that can be restored without compromising:

- Electrical safety; product reliability; functional performance; or operational efficiency.

The proposed repair strategy was developed according to the operational conditions and quality requirements of the current production environment. The objective is not to establish a separate remanufacturing system, but rather to integrate targeted corrective interventions into existing manufacturing workflows when technically feasible, as shown in Figure 8.

The repair workflow begins following defect detection during assembly or functional testing operations. Once a rejected contactor is identified, the defect is evaluated to determine whether the unit is technically suitable for repair or whether it should instead be redirected toward dismantling or recycling.

The proposed repair workflow consists of the following operational stages:

1. **Defect Identification:** Rejected products are inspected to determine the type, localization, and severity of the defect.
2. **Repair Eligibility Assessment:** Products presenting recoverable and localized defects are classified as suitable candidates for repair.
3. **Selective Component Replacement or Adjustment**: Corrective interventions are performed to restore the intended product configuration and functionality while preserving the remaining functional structure.
4. **Functional Validation**: Repaired contactors undergo electrical and mechanical testing together with quality-control verification.
5. **Reintegration into Production**: Successfully validated products are reintegrated into the manufacturing or delivery flow.

Typical repair operations may include:

- Replacement of defective housing elements.
- Repositioning incorrectly assembled components.

- Correcting assembly deviations.
- Replacing small defective parts or correcting minor visual non-conformities.

Repair interventions are performed manually by trained operators using standardized industrial procedures to ensure consistency, traceability, and product reliability.

Following repair, each contactor undergoes the same functional validation procedures applied to newly manufactured products. This step is essential to ensure that repaired units satisfy ABB quality, safety, and operational requirements before reintegration into the production flow.

Products successfully passing validation are reintegrated into production or released as finished products depending on their manufacturing stage. Products failing validation are redirected toward dismantling or external recycling to avoid the risk of transferring non-conforming products to customers.

The simplified color-coded defect labelling system introduced in the dismantling framework also supports repair decision-making:

🟢 Green-label defects represent the highest repair potential;

🟡 Yellow-label defects may require selective dismantling depending on severity;

🔴 Red-label defects are excluded from repair due to severe structural or functional damage.

From an environmental engineering perspective, repair offers major environmental advantages because it avoids complete product disposal and significantly reduces the need to manufacture entirely new contactors.

By preserving most of the original product structure, repair minimizes:

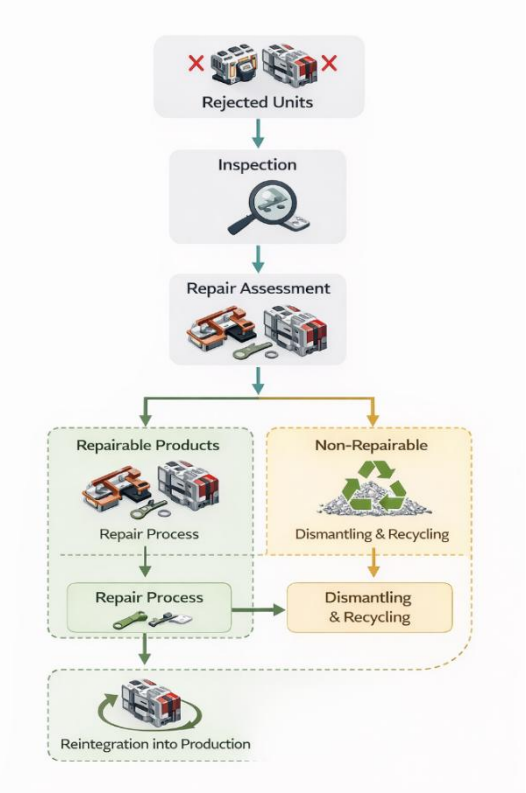
- Virgin material consumption; manufacturing energy demand; freshwater use; greenhouse gas emissions; and manufacturing-related environmental burdens.

From a life-cycle perspective, repaired products directly substitute newly manufactured contactors within the production system. Consequently, repair generates avoided environmental impacts associated with:

- Raw material extraction; component manufacturing; assembly operations; transportation; and industrial processing activities.

From an operational perspective, repair activities remain relatively limited because only a fraction of rejected contactors is expected to present defects compatible with rapid repair intervention under current factory conditions.

Nevertheless, even limited implementation of repair operations can contribute significantly by reducing material losses; improving production efficiency; preserving high-value products; and reducing environmental impacts associated with internal manufacturing scrap.



Figur 9 Diagram of the repair scenario

4.3.4 Design for Repair and Circularity

Design for repair is introduced as a long-term circular strategy aimed at improving product recoverability, reducing material losses, and supporting more sustainable manufacturing practices. While dismantling and repair mainly address internal production scrap at the operational level, eco-design provides a more preventive approach by integrating circularity principles directly into product development.

The current contactor architecture is primarily optimized for technical performance, safety, and reliability. However, repairability and dismantling efficiency are not systematically integrated, meaning products with minor defects may still become scrap due to limited accessibility, complex assemblies, or difficult component replacement.

Observations from the dismantling and repair workflows revealed opportunities to improve future product circularity, particularly for components such as housings, coils, contact systems, and mechanical assemblies, which may be recoverable if product accessibility and modularity are improved.

The proposed TO-BE framework therefore introduces several design-for-repair principles as shown in the Figure 9, including:

- improved accessibility to critical components;
- reduced dismantling complexity;
- simplified component replacement;

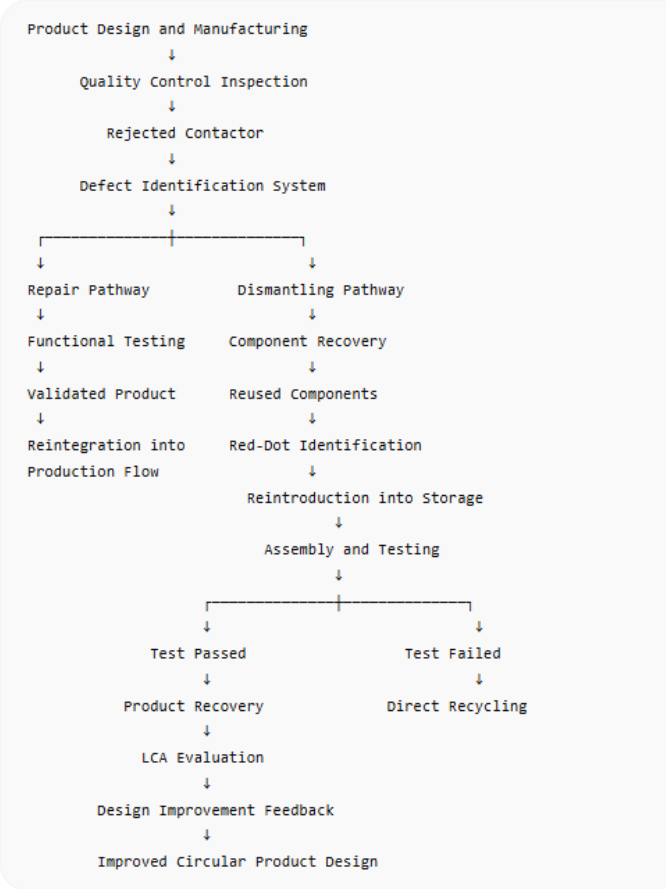
• increased modularity of subassemblies;
• standardized fastening systems;
• and improved traceability of recovered components.

To support traceability, reusable components may receive a visual identification mark (e.g., a small red dot on selected housing or plastic elements) before reintegration into storage. This enables operators to distinguish reused components from new ones during assembly and testing while maintaining controlled quality verification.

The proposed reuse logic follows a single-cycle recovery principle**:** reused components are validated through standard functional testing and removed from circulation if performance issues are detected. This approach improves reliability while avoiding uncontrolled reuse loops.

From a circular manufacturing perspective, design for repair complements operational recovery strategies by improving the feasibility of dismantling, repair, and component reuse**,** while reducing dependence on new materials and manufacturing effort. In the long term, this strategy may contribute to lower material losses, reduced energy demand, and lower embodied environmental impacts.

Overall, design for repair represents a proactive circular strategy supporting greater resource efficiency, product recoverability, and industrial sustainability while remaining compatible with quality and manufacturing requirements.



Figur 10 Ecodesign and Circular recovery integration Framework

4.3.5 Environmental Results — TO-BE Scenarios

This chapter presents the Life Cycle Assessment (LCA) results obtained from the comparison of three end-of-life and recovery strategies for defective newly manufactured ABB contactors.

The study evaluates the environmental performance of the following scenarios:

- **AS IS**: scrapping of a defective newly manufactured contactor.
- **Dismantling**: dismantling and material recovery of the defective contactor.
- **Repair**: repair of the defective contactor through replacement of defective components while preserving functional parts.

Three contactor models were analysed: AF116-146, AF190-205, AF265-370

The results are presented through: characterization assessment; single score assessment; comparative environmental analysis.

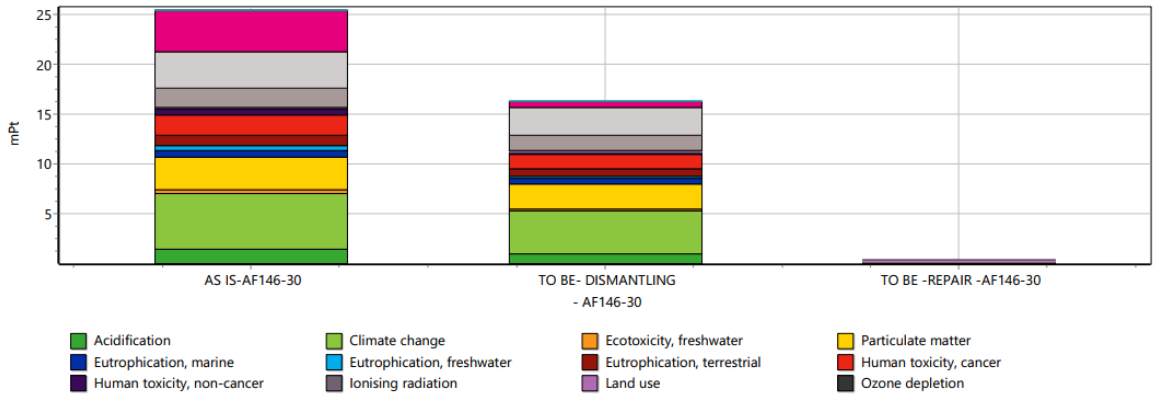
4.3.5.1. Results AF116-146

The characterization results for the AF116–146 contactor showed clear environmental differences between the three evaluated scenarios as shown in the Figure 10.

The AS-IS scenario generated the highest environmental impacts, as the defective contactor was entirely discarded, resulting in the loss of embedded materials, manufacturing energy, and industrial resources. The main impacts were associated with raw material extraction, manufacturing, energy use, transportation, and waste treatment.

The Dismantling scenario reduced environmental impacts through component and material recovery. However, transportation, dismantling activities, recycling processes, and electricity consumption still contributed to environmental burdens.

The Repair scenario showed the lowest environmental impacts, since most components were preserved and only defective parts were replaced. In this case, the repair scenario considered only the replacement of the defective plastic housing, while the remaining components were retained and reused, significantly reducing the need for virgin material production.



Figur 11 The chart bar of single score results on the AF116-146.

- **Single Score Results**

The total single score results for AF116-146 are presented below.

Tabell 13 Single Score Results for AF116-146

Scenario	Single Score (mPt)
AS-IS	25.5
TO-BE – Dismantling	16.3
TO-BE – Repair	0.375

The repair scenario reduced the environmental impact by approximately:

$$\frac{25.5 - 0.375}{25.5} \times 100 \approx 98.5\%$$

Similarly, the dismantling scenario reduced environmental impacts by approximately:

$$\frac{25.5 - 16.3}{25.5} \times 100 \approx 36.1\%$$

These findings demonstrate that both circular strategies improved environmental performance compared with the current AS-IS scenario. However, repair emerged as the most environmentally favourable option, as it preserved the greatest proportion of embedded product value and avoided most of the environmental burdens associated with virgin material extraction and remanufacturing.

4.3.5.2. Results for AF190-205

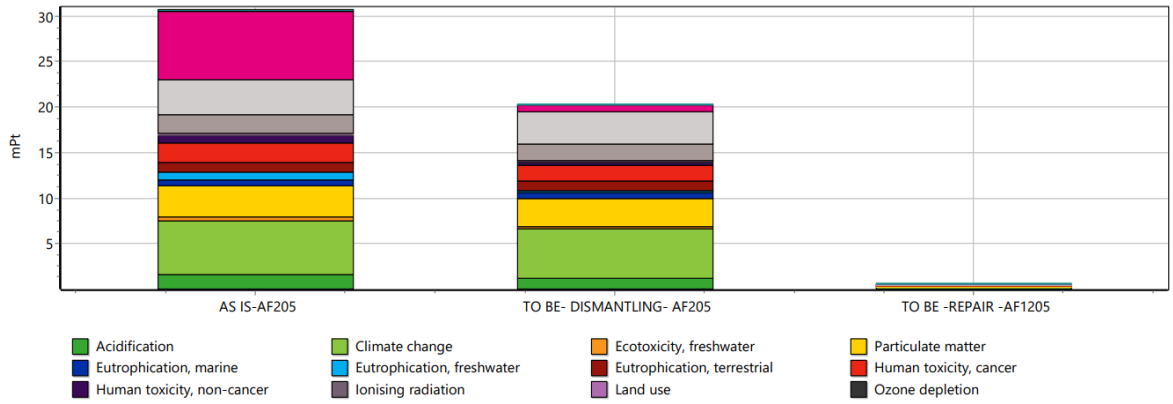
- **Characterization Results**

Similar environmental trends were observed for the AF190-205 contactor as shown in the Figure 11.

The AS-IS scenario generated the highest environmental impacts, as the defective contactor was entirely scrapped, resulting in the loss of embedded materials, manufacturing energy, and industrial resources. The main environmental burdens were associated with raw material extraction, copper and steel production, industrial processing, and energy consumption.

The Dismantling scenario reduced environmental impacts through material and component recovery. However, transportation, recycling treatments, dismantling operations, and electricity consumption still contributed to environmental burdens.

The Repair scenario again demonstrated the best environmental performance. In this case, only the defective plastic housing was considered for replacement, while the remaining functional components were preserved and reused, significantly reducing the need for additional manufacturing and virgin material consumption.



Figur 12 The chart bar of single score results on the AF190-205

Single Score Results

Tabell 14 Single Score Results for AF190-205

Scenario	Single Score (mPt)
AS-IS	30.7
TO-BE – Dismantling	20.4
TO-BE – Repair	0.638

Table11. Single Score Results for AF190-205

The repair strategy reduced the environmental impact by approximately:

$$\frac{30.7 - 0.638}{30.7} \times 100 \approx 97.9\%$$

Similarly, the dismantling scenario reduced environmental impacts by approximately:

$$\frac{30.7 - 20.4}{30.7} \times 100 \approx 33.6\%$$

Compared with the AS-IS scenario. These results demonstrate that repair is the most

environmentally favorable strategy for reducing environmental impacts and improving resource efficiency within manufacturing systems.

4.3.5.3. Results for AF265-370

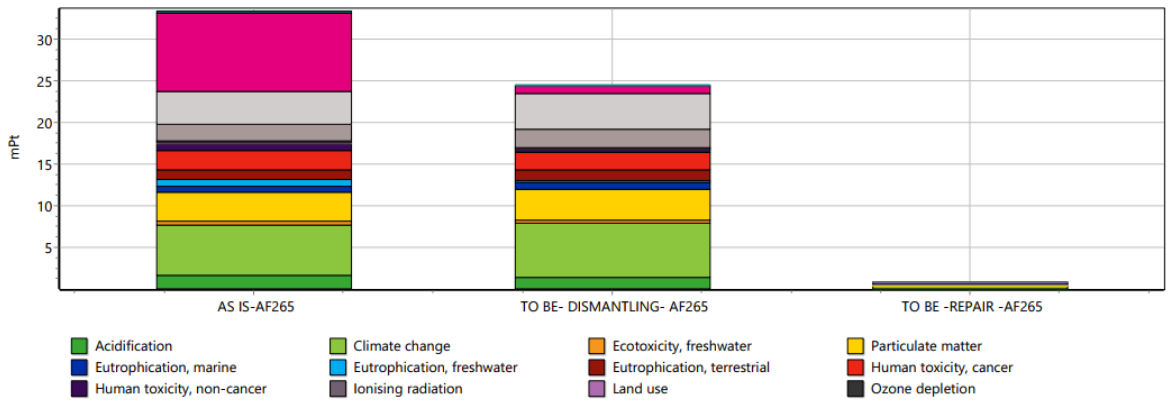
Characterization Results

The AF265-370 contactor generated the highest environmental burdens among the three analyzed products due to its larger dimensions, higher material content, and increased manufacturing complexity as presented in the Figure 12.

In the AS-IS scenario, the defective newly manufactured contactor was fully scrapped, resulting in the loss of embedded materials, manufacturing energy, and industrial resources, which led to the highest environmental impacts.

The Dismantling scenario reduced environmental impacts through material and component recovery. However, transportation, dismantling activities, recycling processes, and electricity consumption still contributed to environmental burdens.

The Repair scenario achieved the lowest environmental impacts. In this case, only the defective plastic housing was considered for replacement, while the remaining functional components were preserved and reused, significantly reducing the environmental burdens associated with new material production.



Figur 13 The chart bar of single score results on the AF256

Single Score Results

Tabell 15 Single Score Results for AF256

Scenario	Single Score (mPt)
AS-IS	33.3
TO-BE – Dismantling	24.4
TO-BE – Repair	0.845

The repair strategy reduced the environmental impact by approximately:

$$\frac{33.3 - 0.845}{33.3} \times 100 \approx 97.5\%$$

Similarly, the dismantling scenario reduced environmental impacts by approximately:

$$\frac{33.3 - 24.4}{33.3} \times 100 \approx 26.7\%$$

These results further confirm that repair is the most environmentally favorable strategy, as it preserves the highest level of embedded product value while minimizing additional manufacturing requirements.

4.3.5.4. Comparative Analysis

The comparative analysis of the three contactor models revealed consistent environmental trends across all evaluated scenarios.

For all products, the AS-IS scenario generated the highest environmental impacts, as defective newly manufactured contactors were entirely scrapped, resulting in the loss of embedded materials, manufacturing energy, and industrial resources.

The Dismantling scenario reduced environmental impacts through component and material recovery. However, dismantling operations still required transportation, electricity, industrial sorting, and recycling activities, which contributed to remaining environmental burdens.

The Repair scenario consistently represented the most environmentally favorable strategy. In this study, the repair scenario considered only the replacement of the defective plastic housing, while preserving the remaining functional components. This significantly reduced the need for virgin material extraction and additional manufacturing activities.

The results also showed that environmental benefits became more significant for larger contactors due to their increased material content and manufacturing complexity. Across all products, the dominant impact categories in the AS-IS scenario included climate change, resource use, human toxicity, and ecotoxicity.

Overall, the findings confirm that repairing defective newly manufactured contactors is environmentally preferable to scrapping and replacing them, supporting the implementation of repair-oriented circular economy strategies in industrial electrical equipment manufacturing.

4.3.5.5. Comparison of Single Score Results

The total single score results clearly illustrate the environmental differences between the evaluated scenarios.see tabell 13.

Tabell 16 Environmental differences between the evaluated scenarios.

Contactor	AS-IS (mPt)	TO-BE – Dismantling (mPt)	TO-BE – Repair (mPt)
AF116-146-30	25.5	16.3	0.375
AF190-205	30.7	20.4	0.638
AF265-370	33.3	24.4	0.845

4.3.5.6. Environmental Interpretation and Comparative Conclusion

The comparative assessment demonstrates that the environmental performance of defective newly manufactured contactors is strongly influenced by the selected recovery strategy.

The AS-IS scenario generated the highest environmental impacts, as defective contactors were entirely discarded, resulting in the loss of embedded materials, manufacturing energy, and industrial resources. In contrast, the Dismantling scenario reduced environmental impacts through material and component recovery, although transportation, dismantling operations, recycling activities, and electricity consumption still contributed to remaining environmental burdens.

Among all evaluated scenarios, repair demonstrated the most environmentally favorable performance. In this study, the repair scenario considered only the replacement of the defective plastic housing, while the remaining functional components were preserved and reused. This significantly reduced virgin material demand, avoided unnecessary manufacturing activities, and minimized waste generation.

The environmental benefits of repair became particularly significant for larger and more material-intensive contactors, such as AF265-370, where embodied manufacturing impacts are higher. These findings suggest that the environmental advantages of repair-oriented strategies increase with product size and material intensity.

Overall, the findings support a transition from conventional scrapping toward higher value-retention strategies, particularly selective repair for minor and recoverable defects, while dismantling and controlled component recovery remain relevant alternatives for non-repairable units in accordance with ABB quality, reliability, and safety requirements. However, the results should be interpreted within the scope of this study, where repair was intentionally limited to localized and recoverable defects under a cradle-to-gate assessment perspective.

4.3.6 Integrated Circular Workflow

The TO-BE scenario integrates repair, dismantling, component recovery, and recycling into a unified decision-based workflow for managing defective newly manufactured contactors as

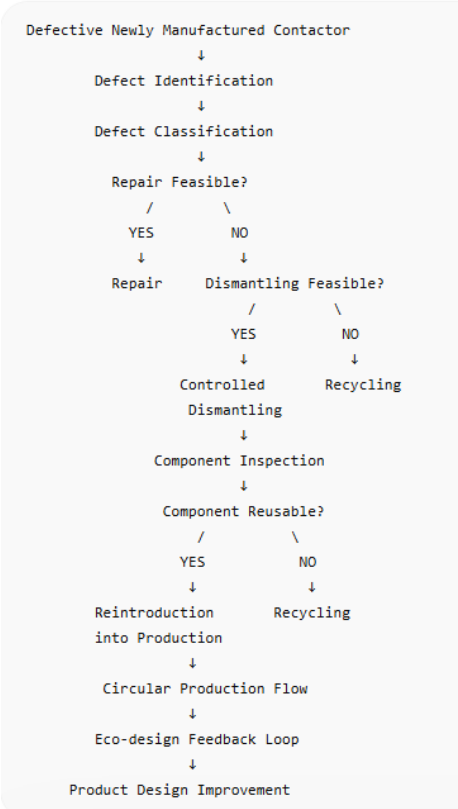
shown in the Figure 13. Rejected units are first classified according to defect type, severity, and location, enabling faster and more standardized decision-making.

The first option evaluated is repairability. Contactors with minor and localized defects are directed toward selective repair, which in this study involved replacing only the defective plastic housing while preserving the remaining functional components.

If repair is not feasible, the workflow evaluates dismantling suitability, allowing reusable components to be recovered and reintegrated into production. Non-recoverable elements are redirected to recycling.

Recycling remains the final recovery pathway, applied only when repair or component recovery is not technically feasible. In parallel, the workflow integrates eco-design feedback to improve future product repairability, modularity, and recoverability.

Overall, the workflow supports higher value retention, reduced material losses, and improved resource efficiency, while remaining compatible with ABB quality and production requirements.



Figur 14 Integrated Circular Workflow for defective newly manufactured contactors.

4.3.7 Expected Economic and Operational Benefits

In addition to environmental improvements identified through the Life Cycle Assessment, the proposed TO-BE circular framework demonstrates important economic and operational benefits compared with the current AS-IS manufacturing system.

Under the existing linear scrap management approach, defective newly manufactured contactors are fully scrapped, resulting in the loss of embedded material, labour, energy, machine operation, and manufacturing value already invested during production.

Based on internal ABB estimates, current internal production scrap represents an annual manufacturing-related loss of approximately 100,000 SEK, including materials, electricity, water use, labour, machine operation, and assembly-related activities. Reducing the number of fully scrapped units through circular recovery pathways may therefore contribute to measurable cost savings and improved manufacturing efficiency.

In addition, rejected contactors represent a substantial embedded commercial product value. Based on an estimated average market value of approximately 7,000 SEK per contactor, the 485 rejected units generated annually correspond to an estimated affected product value of approximately 3.4 million SEK/year**.** Although this value does not represent a direct financial loss equivalent to unrealized sales revenue, it highlights the strategic importance of implementing higher-value recovery strategies before scrapping.

Furthermore, dismantling and component recovery may reduce procurement requirements for selected reusable components and material-intensive elements such as copper, steel, and engineering polymers. Additional benefits may also emerge through reduced waste-handling activities and lower dependence on external recycling processes.

Overall, the proposed circular framework may improve manufacturing resilience, material efficiency, and resource utilization, while supporting more economically sustainable manufacturing operations.

4.3.8 Summary of the TO-BE Scenario

The TO-BE scenario represents a transition from a linear scrap-management system toward a more circular and resource-efficient manufacturing framework**.** Unlike the AS-IS system, where rejected contactors are directly sent to recycling, the proposed approach prioritizes repair, dismantling, component recovery, and controlled recycling to preserve embedded product value.

At the operational level, selective repair and dismantling reduce material losses through the recovery of functional products and reusable components. The integration of defect classification and traceability systems further supports standardized recovery decisions and improved process efficiency.

From an environmental and economic perspective, the TO-BE framework demonstrates strong potential to reduce manufacturing-related impacts, improve material utilization, reduce production losses, and preserve product value.

Overall, the proposed framework combines short-term recovery strategies with long-term eco-design improvements, providing a practical and scalable approach for improving circularity and sustainability within industrial manufacturing systems.

4.4 Environmental Results and Comparative Assessment

4.4.1 Comparative Environmental Assessment of Contactors

The environmental performance of three management scenarios was evaluated for the ABB contactors AF116-146, AF190-205, and AF265-370 using the EN15804 + A2 adapted / EF 3.1 method in SimaPro.

The assessed scenarios included:

- **AS-IS**: defective manufactured contactors are scrapped and replaced by newly manufactured products;
- **Dismantling**: component and material recovery through controlled dismantling;
- **Repair**: selective replacement of defective components while preserving functional parts.

The objective of the assessment was to evaluate how circular recovery strategies can reduce the environmental burdens associated with scrapping defective contactors and replacing them with newly manufactured units.

4.4.2 Comparative Interpretation and Single-Score Analysis

The comparative assessment demonstrated a consistent environmental ranking across all analyzed contactors. See Table 14.

Tabell 17 Comparative interpretation between scenarios.

Scenario	Environmental Performance
TO-BE – Repair	Lowest impacts
TO-BE – Dismantling	Intermediate impacts
AS-IS	Highest impacts

For all contactors, the AS-IS scenario generated the highest environmental burdens, as defective products were fully discarded and replaced by newly manufactured units, resulting in additional raw material extraction, manufacturing activities, energy consumption, transportation, and waste generation.

The Dismantling scenario reduced impacts through component and material recovery prior to recycling. However, dismantling remained more environmentally intensive than repair because additional operations such as inspection, separation, transportation, and recycling treatment were still required.

Among all evaluated strategies, repair consistently demonstrated the best environmental performance. In this study, repair considered only the replacement of the defective plastic housing, while preserving the remaining functional components, thereby minimizing additional manufacturing activities and environmental burdens.

The single-score assessment further confirmed these findings, see tabell 15:

Tabell 18 Comparative single-score results across evaluated scenarios.

Contactor	AS-IS (mPt)	TO-BE – Dismantling (mPt)	TO-BE – Repair (mPt)
AF116-146-30	25.5	16.3	0.375
AF190-205	30.7	20.4	0.638
AF265-370	33.3	24.4	0.845

The environmental benefits of repair became increasingly significant for larger and more material-intensive contactors, demonstrating the strong environmental potential of component-level recovery strategies within industrial manufacturing systems.

It is important to note that the near-complete impact reduction observed for the repair scenario (approximately 97–99 % compared with scrapping) is a direct consequence of the modelling assumption: only the defective plastic housing is replaced while all other components are preserved. This represents defect class D04 (plastic damage) in the classification framework of Table 7 and constitutes an upper-bound, best-case illustration of product-level value retention. The remaining defect classes are routed through dismantling (D01–D03, D05) or direct recycling (D06). Because the per-class breakdown of the 485 rejected units was not separately available, the share of units eligible for housing-only repair is treated as a modelling assumption rather than a measured proportion. The magnitude of the repair benefit should therefore be interpreted as indicative of the environmental potential of targeted, component-level recovery rather than a guaranteed outcome for all rejected products.

4.5 Circularity and Industrial Implications

The results demonstrate that circular manufacturing strategies can substantially reduce the environmental burdens associated with defective industrial contactors.

Unlike the conventional AS-IS approach, where rejected units are scrapped, the proposed TO-BE framework prioritizes repair, dismantling, and component recovery to preserve embedded product value before recycling is considered.

From a circular economy perspective, repair represents the highest value-retention strategy, while dismantling provides an intermediate recovery pathway for products unsuitable for direct repair. In this study, repair was limited to the replacement of the defective plastic housing, while preserving the remaining functional components. Recycling remains the final recovery option when neither repair nor component recovery is technically feasible.

However, the practical implementation of repair strategies should remain limited to contactors presenting minor and recoverable defects and must comply with ABB quality, reliability, and safety requirements. For non-repairable units, dismantling and controlled component recovery remain relevant circular alternatives.

Overall, the findings support a transition from a linear scrap-oriented manufacturing system toward a more circular, resource-efficient, and value-retention-based production model.

It must be emphasised that any repaired unit or recovered component is a candidate for reintroduction only after it has passed the same electrical-safety, dielectric, mechanical-endurance and functional tests applied to newly manufactured products, together with traceability marking and a single-cycle reuse limit. This quality- and safety-gating is treated as a mandatory precondition of the proposed framework, not as an outcome of it: a unit that fails any verification step is removed from circulation and redirected to recycling.

5 DISCUSSION

5.1 Interpretation of the Results

The findings reveal significant untapped circular potential within the current manufacturing system. Although the AS-IS baseline showed a relatively low scrap rate (0.205%), rejected units still represent measurable losses of materials, energy, and embedded manufacturing value at an annual production scale.

The results indicate that the current system remains predominantly linear, where rejected contactors are mainly directed to recycling. While recycling enables partial material recovery, it does not preserve the functional value of products and components, resulting in the loss of previously invested manufacturing effort and environmental resources.

The proposed TO-BE scenario demonstrates that higher-value circular recovery strategies are both technically feasible and operationally compatible with existing industrial conditions. In particular, repair and dismantling enable a transition from product-level disposal toward greater value retention, where functional products and reusable components are preserved before recycling is considered.

These findings should be interpreted within the context of scenario-based modelling. While the results indicate strong practical potential, confirmed operational compatibility would require pilot implementation with measured process data.

The Life Cycle Assessment confirmed the environmental advantages of circular recovery pathways. Across all analyzed contactors **(**AF116-146, AF190-205, and AF265-370**),** repair consistently generated the lowest environmental impacts, mainly due to the preservation of functional components and the avoidance of additional manufacturing. In this study, repair was limited to the replacement of the defective plastic housing, while preserving the remaining functional parts.

Although dismantling significantly reduced environmental burdens compared with the AS-IS scenario**,** additional operations such as inspection, transportation, and recycling activities resulted in higher impacts than repair.

Overall, the findings suggest that higher value-retention strategies, particularly selective repair and controlled component recovery, provide greater environmental benefits than recycling alone**,** supporting the transition toward more circular and resource-efficient manufacturing systems.

5.2 Implications for Circular Manufacturing

The findings provide important insights into how circular economy principles can be integrated into industrial manufacturing systems. Although the current AS-IS configuration demonstrates high operational efficiency, it remains predominantly linear, as rejected products are mainly directed to recycling, resulting in the loss of embedded product value.

The results suggest that circular manufacturing should move beyond recycling-oriented recovery and prioritize higher value-retention strategies, particularly repair, dismantling, and component recovery. These approaches preserve functional products and components, reduce material losses, and improve resource efficiency.

The proposed recovery workflows demonstrate that circular strategies can be integrated within existing industrial systems without major structural modifications, primarily through improved decision logic, defect classification, and standardized recovery pathways. In this study, repair was limited to the replacement of the defective plastic housing, illustrating how selective interventions can significantly reduce environmental burdens while preserving functional product value.

The study also highlights the importance of traceability and decision-support systems in enabling scalable circular manufacturing. The proposed color-coded defect identification system illustrates how relatively simple interventions may improve operational consistency, reduce diagnostic effort, and support standardized recovery decisions.

From a long-term perspective, integrating eco-design principles may further strengthen circularity by improving product repairability, dismantling efficiency, and component accessibility. This suggests that circularity should be considered not only during scrap treatment but also during the product development stage.

Overall, the findings demonstrate that significant improvements in circularity can be achieved by shifting from waste management toward value retention, improving environmental performance, operational efficiency, and long-term manufacturing resilience.

5.3 Environmental Implications (LCA Perspective)

The Life Cycle Assessment (LCA) results strongly support the environmental relevance of the proposed TO-BE circular framework. The findings confirm that material production and manufacturing stages represent the main environmental hotspots, particularly regarding greenhouse gas emissions, energy demand, and resource consumption.

The AS-IS baseline demonstrates that internal production scrap represents not only a loss of physical materials but also a loss of embedded environmental value accumulated during raw material extraction, processing, and assembly. Although the internal scrap rate remained relatively low (0.205%), the estimated annual material loss of approximately 1.78 tonnes highlights a measurable environmental inefficiency.

The results further indicate that higher value-retention strategies provide greater environmental benefits than recycling alone. While recycling enables partial material recovery, it still requires additional industrial processing and energy consumption. In contrast, repair and dismantling preserve functional products and reusable components, thereby avoiding part of the environmental burdens associated with virgin material extraction and replacement manufacturing. In this study, repair was limited to the replacement of the defective plastic housing, while preserving the remaining functional components.

The assessment also showed that environmental benefits increased with product size and material intensity, suggesting that circular recovery strategies may be particularly effective for larger and more resource-intensive industrial products.

From a long-term perspective, the integration of eco-design principles may further improve environmental performance by enhancing repairability, dismantling efficiency, and material circularity.

Although the estimated benefits are based on modeled recovery scenarios and operational assumptions, the findings consistently indicate that even partial implementation of circular strategies can generate measurable environmental improvements.

Overall, the results confirm that transitioning from a linear scrap-oriented system toward circular value-retention strategies offers strong potential to reduce environmental impacts within industrial manufacturing systems.

5.4 Methodological Reflection and Limitations

This study adopted a qualitative industrial case study approach supported by quantitative production data and Life Cycle Assessment (LCA) to evaluate circular recovery opportunities within a real manufacturing environment. A key methodological strength lies in the integration of industrial practice with environmental assessment, ensuring that the proposed

TO-BE framework remained grounded in operational reality while enabling the identification of environmental hotspots and the comparative evaluation of circular recovery strategies.

However, several limitations should be acknowledged. First, the study relied on modeled recovery scenarios and simplified operational assumptions, introducing uncertainty regarding the exact magnitude of environmental and economic benefits. In particular, the repair scenario was intentionally simplified, representing a realistic industrial configuration in which only the defective plastic housing was replaced while the remaining functional components were preserved.

Second, the study focused on a single industrial case study involving AF4–6 contactors at ABB, which may limit the broader generalizability of the findings to other manufacturing contexts, products, or industrial sectors.

Third, the proposed circular strategies were evaluated through workflow modelling rather than full industrial implementation, meaning that parameters such as recovery efficiency, labour requirements, cycle times, and production integration remain partially estimated.

Furthermore, although the environmental assessment enabled the identification of major comparative trends, the simplified LCA modelling and selected sensitivity assumptions may not fully capture all operational variations, material variability, and recovery uncertainties.

Despite these limitations, the methodology remains appropriate for the exploratory and development-oriented nature of the study, providing a structured and realistic basis for assessing the integration of circular economy strategies within industrial manufacturing systems. The findings should therefore be interpreted as indicative rather than definitive, while offering a strong basis for future industrial validation and methodological refinement.

5.5 Relation to Literature and Generalization

The findings of this study are consistent with existing research on circular economy and sustainable manufacturing, particularly regarding the importance of prioritizing higher value-retention strategies over conventional recycling. Previous studies have shown that repair, reuse, and remanufacturing generally provide greater environmental benefits than recycling, as they preserve product functionality and reduce demand for new manufacturing. The present findings support this perspective, demonstrating that repair and component recovery substantially reduce environmental burdens compared with direct scrapping and recycling alone.

The study also aligns with literature emphasizing that the implementation of circular manufacturing is often constrained more by organizational and operational barriers than technical limitations. The results indicate that structured workflows, defect classification, and standardized recovery pathways are important enablers for integrating circular practices into industrial production systems.

In addition, the proposed color-coded defect identification system supports previous research highlighting the importance of traceability and decision-support mechanisms for scalable circular manufacturing. Similarly, the integration of eco-design principles reinforces literature emphasizing the role of modularity, accessibility, and material selection in improving repairability and recoverability.

From an environmental perspective, the LCA findings confirm previous studies identifying material production and manufacturing stages as dominant environmental hotspots, reinforcing the value of Life Cycle Assessment (LCA) as a decision-support tool for circular manufacturing.

Although this research is based on a single industrial case study involving AF4–6 contactors at ABB, the proposed framework may be transferable to other industries involving complex assembled products, particularly in electrical equipment, machinery, and electromechanical systems. However, implementation remains context-dependent, influenced by product architecture, manufacturing complexity, and organizational readiness.

Overall, this study contributes to bridging the gap between theoretical circular economy principles and practical industrial implementation, providing a realistic framework for integrating circular strategies within manufacturing systems.

6 CONCLUSION

This study investigated how circular economy principles can be integrated into industrial manufacturing systems to improve the management of internal production scrap, using the AF4–6 contactor production system at ABB Smart Power as an industrial case study.

The findings revealed that, despite a relatively low scrap rate, the current AS-IS system remains predominantly linear, as rejected products are mainly directed to external recycling. While recycling enables material recovery, it results in the loss of functional components and embedded manufacturing value accumulated during earlier production stages.

To address this limitation, a structured TO-BE circular framework was developed, integrating repair, dismantling, component recovery, and eco-design principles. The findings indicate that repair, dismantling, and controlled component recovery represent the most relevant circular strategies for improving the management of defective newly manufactured contactors. Among these, repair demonstrated the highest value-retention potential, while dismantling provided an intermediate recovery pathway for non-repairable units.

From an operational perspective, the study demonstrated that circular strategies can be integrated into existing industrial systems through structured recovery workflows, defect classification, and traceability mechanisms, without requiring major disruption to production operations. This confirms the technical feasibility of implementing circular recovery pathways under realistic industrial conditions.

This assessment is based on scenario modelling using representative production data and expert knowledge. Full confirmation of technical feasibility would require pilot-scale validation, including measured recovery rates, quality-control outcomes, and operational cost data.

The environmental assessment confirmed that material production and manufacturing stages represent the dominant environmental hotspots within the product life cycle. The results further demonstrated that repair-oriented strategies provide the greatest environmental

benefits, as they preserve product functionality while minimizing the need for additional manufacturing activities. In this study, repair was limited to the replacement of the defective plastic housing, while preserving the remaining functional components. Dismantling also generated important environmental improvements through controlled component and material recovery prior to recycling.

Overall, this study demonstrates that the transition from a linear manufacturing model toward a circular and value-retention-oriented system is both technically feasible and environmentally beneficial. By combining short-term operational improvements with long-term design-oriented strategies, the proposed framework provides a realistic pathway for improving sustainability, resource efficiency, and manufacturing resilience in industrial production systems.

It should be noted, however, that confirmed industrial feasibility still requires pilot-scale validation and measured operational data. The strategies evaluated in this study were assessed through scenario modelling based on representative data and expert knowledge rather than full pilot implementation.

Finally, this research contributes to bridge the gap between circular economy theory and industrial implementation, offering a practical framework that may support future applications of circular manufacturing strategies in other industrial contexts.

7 FUTURE RESEARCH DIRECTIONS

Building on the findings of this study, the following four directions are identified as priority areas for future research and improvement:

- Pilot validation: A small-scale dismantling and component-recovery cell targeting non-functional defect classes (D01–D03) should be trialled at ABB Smart Power to collect quantitative operational data — including recovery time, reuse rates, and component yield — and confirm the technical assumptions of the proposed circular workflow under real manufacturing conditions.
- Digital data infrastructure: An integrated circular data management system should be developed, incorporating RFID traceability, IoT-based monitoring, and digital twin simulation, to enable intelligent and adaptive recovery decision-making and reduce reliance on manual operator judgment.
- Eco-design integration: Future contactor generations should incorporate design-for-disassembly principles — including modular product configurations, standardized replaceable modules, and reversible fastening systems — to improve long-term repairability, dismantling efficiency, and component recoverability across the product life cycle.

- Extended environmental assessment: The LCA framework should be expanded from cradle-to-gate to full cradle-to-grave system boundaries, integrating dynamic and process-specific impact models, to provide a more comprehensive evaluation of the long-term sustainability performance of circular manufacturing strategies.

Based on the findings of this study, a progressive circular manufacturing roadmap can be proposed for future industrial implementation.

Tabell 19 Circular Manufacturing Roadmap for Future Industrial Implementation

Phase	**Main Objective**	**Circular Strategies**	**Digital & Data Solutions**	**Expected Outcomes**
Phase 1 – Operational Circularity	Improve internal recovery processes	Defect classification, repair workflow, dismantling, component reuse	Color-coded labels, barcode systems, recovery database	Reduced scrap, improved traceability, faster decision-making
Phase 2 – Integrated Circular Systems	Strengthen circular process integration	Design for disassembly, modularity, reusable components, circular inventory management	RFID tracking, centralized circular data platform, digital traceability	Higher recovery efficiency, improved resource utilization
Phase 3 – Smart Circular Manufacturing	Develop intelligent circular operations	Predictive repair, remanufacturing, closed-loop recovery systems	AI defect detection, machine learning, digital twin simulation, IoT monitoring	Optimized recovery pathways, predictive decision-making
Phase 4 – Advanced Circular Ecosystem	Achieve scalable data-driven circular manufacturing	Autonomous recovery systems, industrial symbiosis, circular supply chains	Real-time LCA, digital product passports, blockchain traceability, smart factory systems	Near-zero waste manufacturing, maximum value retention, sustainable industrial ecosystems

The proposed roadmap demonstrates that the transition toward advanced circular manufacturing systems will not rely solely on physical recovery operations, but increasingly on the ability to generate, structure, analyze, and exploit high-quality manufacturing and recovery-related data.

Overall, future research should focus on strengthening the integration between environmental engineering, circular economy strategies, digital technologies, product design optimization, and intelligent data-driven decision-making systems. Advancing these areas will support the development of scalable and sustainable circular manufacturing systems capable of reducing environmental impacts while improving industrial resource efficiency and long-term operational resilience.

8 DECLARATION OF AI USE

ChatGPT (OpenAI) was used to support structuring of the report, brainstorming of ideas and for minor grammar improvements. It was not used to generate or support any claims, data, analysis or interpretations in this work. All interpretations, calculations and findings presented in this report are the author's own work. The author takes responsibility for all content presented.

9 REFERENCES

ABB. (n.d.). Environmental Product Declaration: AF116–146 contactors [Product Environmental Profile]. ABB.

ABB. (n.d.). Environmental Product Declaration: AF190–205 contactors [Product Environmental Profile]. ABB.

ABB. (n.d.). Environmental Product Declaration: AF265–370 contactors [Product Environmental Profile]. ABB.

Acerbi, F., Sassanelli, C., Terzi, S., & Taisch, M. (2021). A systematic literature review on data and information required for circular manufacturing strategies adoption. Sustainability, 13(4), 2047. https://doi.org/10.3390/su13042047

Boldoczki, S., Thorenz, A., & Tuma, A. (2020). The environmental impacts of preparation for reuse: A case study of WEEE reuse in Germany. Journal of Cleaner Production, 252, 119736. https://doi.org/10.1016/j.jclepro.2019.119736

Bovea, M. D., Ibáñez-Forés, V., Juan, P., Pérez-Belis, V., & Braulio-Gonzalo, M. (2018). Variables that affect the environmental performance of small electrical and electronic equipment: Methodology and case study. Journal of Cleaner Production, 203, 1067–1084. https://doi.org/10.1016/j.jclepro.2018.08.240

Bovea, M. D., Ibáñez-Forés, V., & Pérez-Belis, V. (2020). Repair vs. replacement: Selection of the best end-of-life scenario for small household electric and electronic equipment based on life cycle assessment. Journal of Environmental Management, 254, 109679. https://doi.org/10.1016/j.jenvman.2019.109679

Bressanelli, G., Saccani, N., Pigosso, D. C. A., & Perona, M. (2020). Circular economy in the WEEE industry: A systematic literature review and a research agenda. Sustainable Production and Consumption, 23, 174–188. https://doi.org/10.1016/j.spc.2020.05.007

Chu, T., Ma, J., Zhong, Y., Sun, H., & Jia, W. (2024). Shared recycling model for waste electrical and electronic equipment based on the targeted responsibility system in the

context of China. Humanities and Social Sciences Communications, 11(1), 1–26. https://doi.org/10.1057/s41599-024-02901-0

Corsini, F., Gusmerotti, N. M., & Frey, M. (2020). Consumers' circular behaviors in relation to the purchase, extension of life, and end-of-life management of electrical and electronic products: A review. Sustainability, 12(24), 10443. https://doi.org/10.3390/su122410443

González, X. M., Rodríguez, M., & Pena-Boquete, Y. (2017). The social benefits of WEEE re-use schemes: A cost–benefit analysis for PCs in Spain. Waste Management, 64, 202–213. https://doi.org/10.1016/j.wasman.2017.03.009

Iraldo, F., Facheris, C., & Nucci, B. (2017). Is product durability better for environment and for economic efficiency? A comparative assessment applying LCA and LCC to two energy-intensive products. Journal of Cleaner Production, 140, 1353–1364. https://doi.org/10.1016/j.jclepro.2016.10.017

Kissling, R., Coughlan, D., Fitzpatrick, C., Boeni, H., Luepschen, C., Andrew, S., & Dickenson, J. (2013). Success factors and barriers in re-use of electrical and electronic equipment. Resources, Conservation and Recycling, 80, 21–31. https://doi.org/10.1016/j.resconrec.2013.07.009

Maniati, E., Chroni, C., Boikou, K., Lasaridi, K., & Abeliotis, K. (2025). Repair and preparing for reuse as a driver for circular economy. Societal Impacts, 100132. https://doi.org/10.1016/j.socimp.2025.100132

McMahon, K., Johnson, M., & Fitzpatrick, C. (2019). Enabling preparation for re-use of waste electrical and electronic equipment in Ireland: Lessons from other EU member states. Journal of Cleaner Production, 232, 1005–1017. https://doi.org/10.1016/j.jclepro.2019.05.339

Messmann, L., Boldoczki, S., Thorenz, A., & Tuma, A. (2019). Potentials of preparation for reuse: A case study at collection points in the German state of Bavaria. Journal of Cleaner Production, 211, 1534–1546. https://doi.org/10.1016/j.jclepro.2018.11.264

Pan, X., Wong, C. W. Y., & Li, C. (2022). Circular economy practices in the waste electrical and electronic equipment (WEEE) industry: A systematic review and future research agendas. Journal of Cleaner Production, 365, 132671. https://doi.org/10.1016/j.jclepro.2022.132671

Parajuly, K., Habib, K., & Liu, G. (2017). Waste electrical and electronic equipment (WEEE) in Denmark: Flows, quantities and management. Resources, Conservation and Recycling, 123, 85–92. https://doi.org/10.1016/j.resconrec.2016.08.004

Pehlken, A., Dawel, L., & Meyer, O. (2024). Digital twins: Enhancing circular economy through digital tools. Procedia CIRP, 122, 563–568. https://doi.org/10.1016/j.procir.2024.01.082

Pérez-Belis, V., Bovea, M. D., & Ibáñez-Forés, V. (2015). An in-depth literature review of the waste electrical and electronic equipment context: Trends and evolution. Waste Management & Research, 33(1), 3–29. https://doi.org/10.1177/0734242X14557382

Pini, M., Lolli, F., Balugani, E., Gamberini, R., Neri, P., Rimini, B., & Ferrari, A. M. (2019). Preparation for reuse activity of waste electrical and electronic equipment: Environmental performance, cost externality and job creation. Journal of Cleaner Production, 222, 77–89. https://doi.org/10.1016/j.jclepro.2019.03.004

Prieto-Sandoval, V., Jaca, C., & Ormazabal, M. (2018). Towards a consensus on the circular economy. Journal of Cleaner Production, 179, 605–615. https://doi.org/10.1016/j.jclepro.2017.12.224

Psarommatis, F., May, G., & Azamfirei, V. (2025). Product reuse and repurpose in circular manufacturing: A critical review of key challenges, shortcomings and future directions. Journal of Remanufacturing, 15(2), 273–310. https://doi.org/10.1007/s13243-025-00153-y

Reike, D., Vermeulen, W. J. V., & Witjes, S. (2018). The circular economy: New or refurbished as CE 3.0? Exploring controversies in the conceptualization of the circular economy through a focus on history and resource value retention options. Resources, Conservation and Recycling, 135, 246–264. https://doi.org/10.1016/j.resconrec.2017.08.027

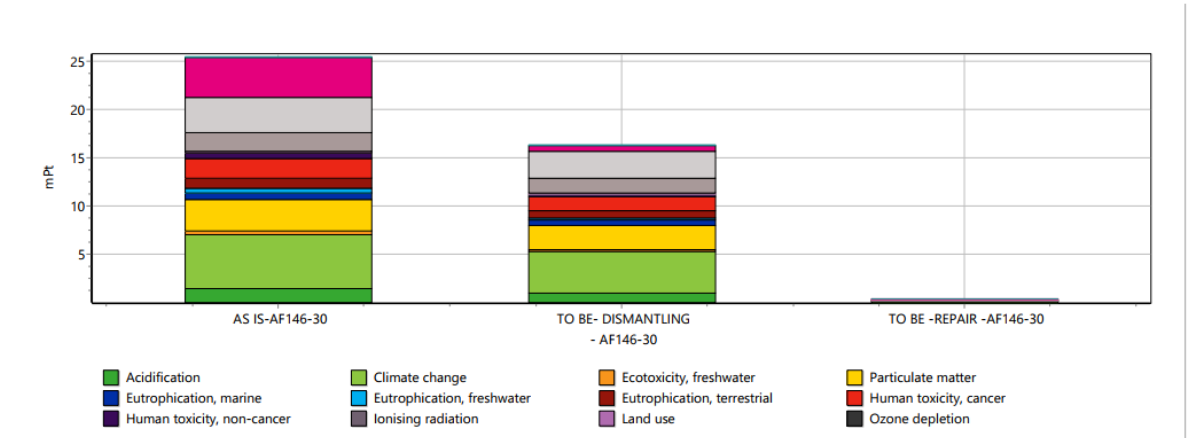
Wang, F., Huisman, J., Stevels, A., & Baldé, C. P. (2013). Enhancing e-waste estimates: Improving data quality by multivariate input–output analysis. Waste Management, 33(11), 2397–2407. https://doi.org/10.1016/j.wasman.2013.07.0

Appendix A

Calculation:	Compare
Results:	Impact assessment
Product 1:	1 p AS IS-AF146-30 (of project AS IS BASELINE)
Product 2:	1 p TO BE- DISMANTLING- AF146-30 (of project AS IS BASELINE)
Product 3:	1 p TO BE -REPAIR -AF146-30 (of project AS IS BASELINE)
Method:	EN 15804 + A2 (adapted) V1.01 / EF 3.1 normalization and weighting set
Indicator:	Single score
Skip categories:	Never
Default units:	No
Exclude infrastructure processes:	No
Exclude long-term emissions:	No
Per impact category:	No
Sorted on item:	Damage category
Sort order:	Ascending

Dam	Unit	AS IS-AF146-30	TO BE- DISMANT	TO BE -REPAIR -/
Tota	mPt	25,5	16,3	0,375
Acid	mPt	1,4	0,953	0,0215
Clim	mPt	5,66	4,36	0,125
Ecot	mPt	0,351	0,2	0,00531
Part	mPt	3,23	2,48	0,0388
Eutr	mPt	0,698	0,539	0,00913
Eutr	mPt	0,501	0,185	0,00741
Eutr	mPt	1,05	0,815	0,00708
Hum	mPt	1,99	1,38	0,00575
Hum	mPt	0,565	0,251	0,00146
Ioni	mPt	0,0549	0,041	0,00545
Lanc	mPt	0,164	0,126	0,000804
Ozoi	mPt	0,00475	0,00375	3,29E-5
Phot	mPt	1,94	1,5	0,0142
Resi	mPt	3,67	2,83	0,0932
Resi	mPt	4,07	0,595	0,00839
Wat	mPt	0,137	0,0678	0,0315

Calculation:	Compare
Results:	Impact assessment
Product 1:	1 p AS IS-AF146-30 (of project AS IS BASELINE)
Product 2:	1 p TO BE- DISMANTLING- AF146-30 (of project AS IS BASELINE)
Product 3:	1 p TO BE -REPAIR -AF146-30 (of project AS IS BASELINE)
Method:	EN 15804 + A2 (adapted) V1.01 / EF 3.1 normalization and weighting set
Indicator:	Single score
Unit:	mPt
Skip categories:	Never
Default units:	No
Exclude infrastructure processes:	No
Exclude long-term emissions:	No
Per impact category:	No
Sorted on item:	Damage category
Sort order:	Ascending

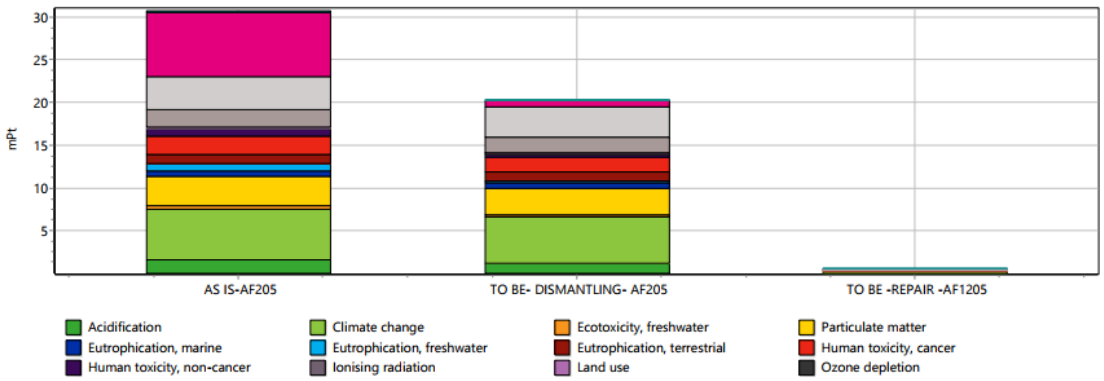


od: EN 15804 + A2 (adapted) V1.01 / EF 3.1 normalization and weighting set / Single score
paring 1 p 'AS IS-AF146-30', 1 p 'TO BE- DISMANTLING- AF146-30' and 1 p 'TO BE -REPAIR -AF146-30';

Calculation:	Compare
Results:	Impact assessment
Product 1:	1 p AS IS-AF205 (of project COMPARAISON AF 190-205)
Product 2:	1 p TO BE- DISMANTLING- AF205 (of project COMPARAISON AF 190-205)
Product 3:	1 p TO BE -REPAIR -AF1205 (of project COMPARAISON AF 190-205)
Method:	EN 15804 + A2 (adapted) V1.01 / EF 3.1 normalization and weighting set
Indicator:	Single score
Skip categories:	Never
Default units:	No
Exclude infrastructure processes:	No
Exclude long-term emissions:	No
Per impact category:	No
Sorted on item:	Damage category
Sort order:	Ascending

Dam	Unit	AS IS-AF205	TO BE- DISMANT	TO BE -REPAIR -/
Tota	mPt	30,7	20,4	0,638
Acid	mPt	1,65	1,19	0,037
Clim	mPt	5,88	5,42	0,217
Ecot	mPt	0,446	0,249	0,00685
Part	mPt	3,34	3,1	0,0676
Eutr	mPt	0,723	0,673	0,0159
Eutr	mPt	0,797	0,229	0,0117
Eutr	mPt	1,09	1,02	0,0121
Hun	mPt	2,13	1,72	0,00826
Hun	mPt	0,856	0,31	0,00217
Ioni	mPt	0,0645	0,0526	0,0083
Lan	mPt	0,172	0,157	0,00122
Ozo	mPt	0,00484	0,00468	4,85E-5
Phot	mPt	2	1,88	0,0241
Res	mPt	3,82	3,53	0,159
Res	mPt	7,55	0,739	0,0116
Wat	mPt	0,189	0,0844	0,0554

Calculation:	Compare
Results:	Impact assessment
Product 1:	1 p AS IS-AF205 (of project COMPARAISON AF 190-205)
Product 2:	1 p TO BE- DISMANTLING- AF205 (of project COMPARAISON AF 190-205)
Product 3:	1 p TO BE -REPAIR -AF1205 (of project COMPARAISON AF 190-205)
Method:	EN 15804 + A2 (adapted) V1.01 / EF 3.1 normalization and weighting set
Indicator:	Single score
Unit:	mPt
Skip categories:	Never
Default units:	No
Exclude infrastructure processes:	No
Exclude long-term emissions:	No
Per impact category:	No
Sorted on item:	Damage category
Sort order:	Ascending

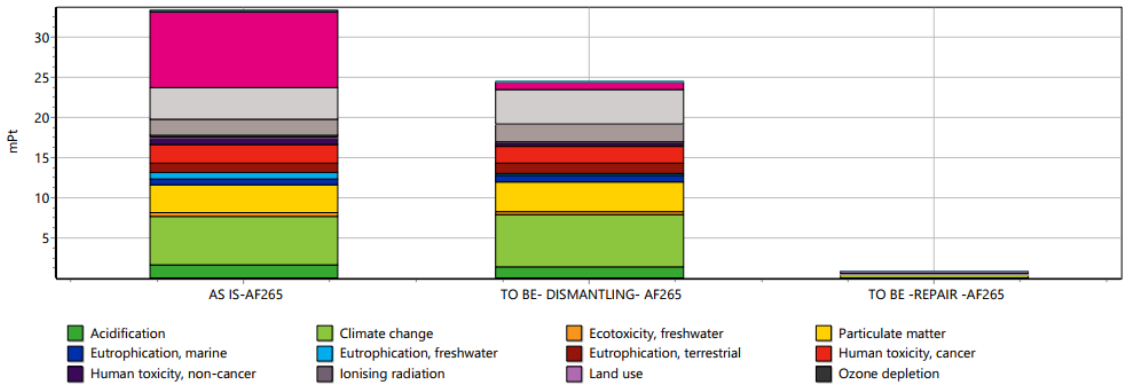


od: EN 15804 + A2 (adapted) V1.01 / EF 3.1 normalization and weighting set / Single score
aring 1 p 'AS IS-AF205', 1 p 'TO BE- DISMANTLING- AF205' and 1 p 'TO BE -REPAIR -AF1205';

Calculation:	Compare
Results:	Impact assessment
Product 1:	1 p AS IS-AF265 (of project COMPARAISON AF 265)
Product 2:	1 p TO BE- DISMANTLING- AF265 (of project COMPARAISON AF 265)
Product 3:	1 p TO BE -REPAIR -AF265 (of project COMPARAISON AF 265)
Method:	EN 15804 + A2 (adapted) V1.01 / EF 3.1 normalization and weighting set
Indicator:	Single score
Skip categories:	Never
Default units:	No
Exclude infrastructure processes:	No
Exclude long-term emissions:	No
Per impact category:	No
Sorted on item:	Damage category
Sort order:	Ascending

Dam	Unit	AS IS-AF265	TO BE- DISMANT	TO BE -REPAIR -/
Tota	mPt	33,3	24,4	0,845
Acid	mPt	1,65	1,42	0,0488
Clim	mPt	6,03	6,51	0,286
Ecot	mPt	0,493	0,299	0,0121
Part	mPt	3,4	3,72	0,0883
Eutr	mPt	0,738	0,807	0,0208
Eutr	mPt	0,817	0,275	0,0165
Eutr	mPt	1,11	1,22	0,016
Hum	mPt	2,43	2,07	0,0125
Hum	mPt	0,813	0,372	0,00318
Ioni	mPt	0,0673	0,0682	0,00955
Lanc	mPt	0,175	0,189	0,00166
Ozoi	mPt	0,00493	0,00562	7,21E-5
Phot	mPt	2,03	2,25	0,0321
Resc	mPt	3,93	4,24	0,208
Resc	mPt	9,43	0,888	0,0181
Wat	mPt	0,211	0,102	0,0717

Calculation:	Compare
Results:	Impact assessment
Product 1:	1 p AS IS-AF265 (of project COMPARAISON AF 265)
Product 2:	1 p TO BE- DISMANTLING- AF265 (of project COMPARAISON AF 265)
Product 3:	1 p TO BE -REPAIR -AF265 (of project COMPARAISON AF 265)
Method:	EN 15804 + A2 (adapted) V1.01 / EF 3.1 normalization and weighting set
Indicator:	Single score
Unit:	mPt
Skip categories:	Never
Default units:	No
Exclude infrastructure processes:	No
Exclude long-term emissions:	No
Per impact category:	No
Sorted on item:	Damage category
Sort order:	Ascending



od: EN 15804 + A2 (adapted) V1.01 / EF 3.1 normalization and weighting set / Single score
aring 1 p 'AS IS-AF265', 1 p 'TO BE- DISMANTLING- AF265' and 1 p 'TO BE -REPAIR -AF265';

Appendix B

PRODUCT ENVIRONMENTAL PROFILE

Environmental Product Declaration

AF116-146 Contactors



REGISTRATION NUMBER ABBG-00028-V01.01	IN COMPLIANCE WITH PCR-ED4-EN-2021 09 06 SUPPLEMENTED BY PSR-0005-ED2-EN-2016 03 29
VERIFIER ACCREDITATION NUMBER VH42	INFORMATION AND REFERENCE DOCUMENTS WWW.PEP-ECOPASSPORT.ORG
DATE OF ISSUE 09-2022	VALIDITY PERIOD 5 YEARS
INDEPENDENT VERIFICATION OF THE DECLARATION AND DATA, IN COMPLIANCE WITH ISO 14025: 2006	
INTERNAL ☐ / EXTERNAL ☒	
THE PCR REVIEW WAS CONDUCTED BY A PANEL OF EXPERTS CHAIRED BY JULIE ORGELET (DDEMAIN)	
PEP ARE COMPLIANT WITH XP C08-100-1 :2016 OR EN 50693:2019 THE ELEMENTS OF THE PRESENT PEP CANNOT BE COMPARED WITH ELEMENTS FROM ANOTHER PROGRAM.	
DOCUMENT IN COMPLIANCE WITH ISO 14025: 2006 « ENVIRONMENTAL LABELS AND DECLARATIONS. TYPE III ENVIRONMENTAL DECLARATIONS »	

© Copyright 2022 ABB. All rights reserved.

EPD Owner	ABB Electrification Sweden AB, Motorgraend 20 Bnr 415, Vaesteras, Sweden www.abb.com
Manufacturer name and address	ABB Electrification Sweden AB, Motorgraend 20 Bnr 415, Vaesteras, Sweden
Company contact	Johan Lundstrom- johan.lundstrom@se.abb.com R&D Engineer
Reference product	AF146-30 Contactor
Description of the product	AF146-30 is 1000V IEC or 600V UL contactor with auxiliary contacts and double clamps. 75kW / 400VAC, AC-3 or 100 hp / 480V UL, wide control rage 20-500V AC/DC.
Functional unit	The Functional Unit is to switch on and off during 20 years electrical power supply of a downstream installation with an electrical and/or mechanical control. The functional unit is characterized by type 3NO, a control circuit voltage U_c, a power circuit voltage U_p and a maximum allowed intensity by the power circuit I_p. Power Circuit Voltage U_p [V]: 1000V Maximum allowed intensity by the power circuit I_p [A]: 146A Rated Control Circuit Voltage U_c [V]: 20-500V Number of poles: 3
Other products covered	The PEP covers offerings for 3-pole variants including PLC variants (AF116/140/146-30-*, AF116B/140B-30-*, AF140N4-30-*, AF140N4R-30-*, AFS116/140/146-30-*) of AF116-140 and AF146 Contactors and 4-pole variants (AF116/140-40-*, AF116B/140B-40-*) of AF116-140 Contactors.
Reference lifetime	20 years
Product category	Electrical, Electronic and HVAC-R Products
Use Scenario	The use phase has been modeled based on the sales mix data (2021), and the corresponding low voltage electricity countries mix
Geographical representativeness	Raw materials & Manufacturing: [Europe / Global] Assembly: [Sweden] Distribution / Use: [Global] specific sales mix EoL: [Global]
Technological representativeness	Materials and processes data are specific for the production of AF146-30 Contactors
LCA Study	This study is based on the LCA study described in the LCA report 1SFC100091D0201
EPD type	Representative Product
EPD scope	“Cradle to grave”
Year of reported primary data	2021
LCA software	SimaPro 9.3.0.3 (2022)
LCI database	ecoinvent v3.8 (2021)
LCIA methodology	EN 50693:2019

© Copyright 2022 ABB. All rights reserved.

Contents

ABB Purpose & Embedding Sustainability **4**

General Information **4**
- **AF Contactors product cluster** 5

Constituent Materials **5**
- **Functional unit and Reference Flow** 7
- **System boundaries and life cycle stages** 7
- **Temporal and geographical boundaries** 7
- **Boundaries in the life cycle** 8
- **Data quality** 8
- **Environmental impact indicators** 8
- **Allocation rules** 8
- **Limitations and simplifications** 8
- **Energy Models** 9

Inventory analysis **9**

Manufacturing stage **10**

Distribution **10**

Installation **11**

Use **11**

End of life **12**

Environmental impacts **13**

Additional environmental information **17**

References **18**

© Copyright 2022 ABB. All rights reserved.

ABB Purpose & Embedding Sustainability

ABB is a leading global technology company that energizes the transformation of society and industry to achieve a more productive, sustainable future. By connecting software to its electrification, robotics, automation and motion portfolio, ABB pushes the boundaries of technology to drive performance to new levels. With a history of excellence stretching back more than 130 years, ABB's success is driven by about 110 thousand talented employees in over 100 countries.
ABB's Electrification business offers a wide-ranging portfolio of products, digital solutions and services, from substation to socket, enabling safe, smart and sustainable electrification. Offerings encompass digital and connected innovations for low voltage and medium voltage, including EV infrastructure, solar inverters, modular substations, distribution automation, power protection, wiring accessories, switchgear, enclosures, cabling, sensing and control.
ABB is committed to continually promoting and embedding sustainability across its operations and value chain, aspiring to become a role model for others to follow. With its ABB Purpose, ABB is focusing on reducing harmful emissions, preserving natural resources and championing ethical and humane behavior.

General Information

The ABB facility located in Vasteras, is for the Smart Power Division, the competence center for Contactors >100A, Softstarters and one of leading manufacturing site.
All the products are produced in an automated system with precision and accuracy. These are supplied to utilities, industrial, and tertiary sector customers.
ABB adopts and implements for its own activities an integrated Quality/Environmental/Health Management System in compliance with the following
standards:
SS-EN ISO 9001: 2015 - Quality Management Systems- Requirements
SS-EN ISO 14001: 2015 - Environmental management systems - Requirements with guidance for use
ISO 45001:2018 - Occupational Health and Safety Assessment Series
SS-EN ISO 50001: 2018 - Energy management systems

AF contactors are most commonly used for controlling electric motors and switch various none-motor loads. Contactors can break current over a wide range of currents, from a few amperes to thousands of amperes, and voltages upto 1000V AC.

© Copyright 2022 ABB. All rights reserved.

AF Contactors product cluster

Product cluster declared in this LCA includes the following AF116-146 Contactors:

Product Range	Product	Number of poles	Power Circuit Voltage [U_p]	Maximum allowed intensity by the power circuit [I_p]	Rated Control Circuit Voltage [U_c]
AF116-146	**AF116-AF140**	3	690V	116-140A	20-500V
	AF116-AF140 (PLC)	3	690V	116-140A	20-500V
	AF116-AF140	4	690V	116-140A	20-500V
	AF146	3	1000V	146A	20-500V
	AF146 (PLC)	3	1000V	146A	20-500V

Table 1: Technical characteristics of AF116-146 Contactors.

The accessories associated with these products are also included in the study.

Reference Product:

The reference product for the LCA of the complete range of AF116-146 is AF146-30.

Constituent Materials

The AF146-30 weights about 1.9kg including its installed accessories, packaging, and paper documentation.

AF146-30				
Materials	**Name**	**IEC 62474 MC**	**[g]**	**%**
Metals	Cu and Cu alloys	M-121	294.30	15.07%
	Steel	M-119	863.64	44.24%
	Stainless Steel	M-100	10.38	0.54%
	Precious Metals	M-159	11.16	0.58%
Plastics	Unsaturated Polyester	M-301	35.22	1.80%
	Polyamide	M-258	465.21	23.83%
	PBT	M-261	0.18	0.01%
	Rubber	M-320	1.16	0.06%
	Polyurethane	M-300	34.00	1.74%
	Polyethylene	M-251	101.35	5.20%
Other	Paper / Cardboard	M-341	105.58	5.40%
	Other	N/A	30.00	1.53%
		Total	1952.18	100%

Table 2: Weight of materials AF146

© Copyright 2022 ABB. All rights reserved.

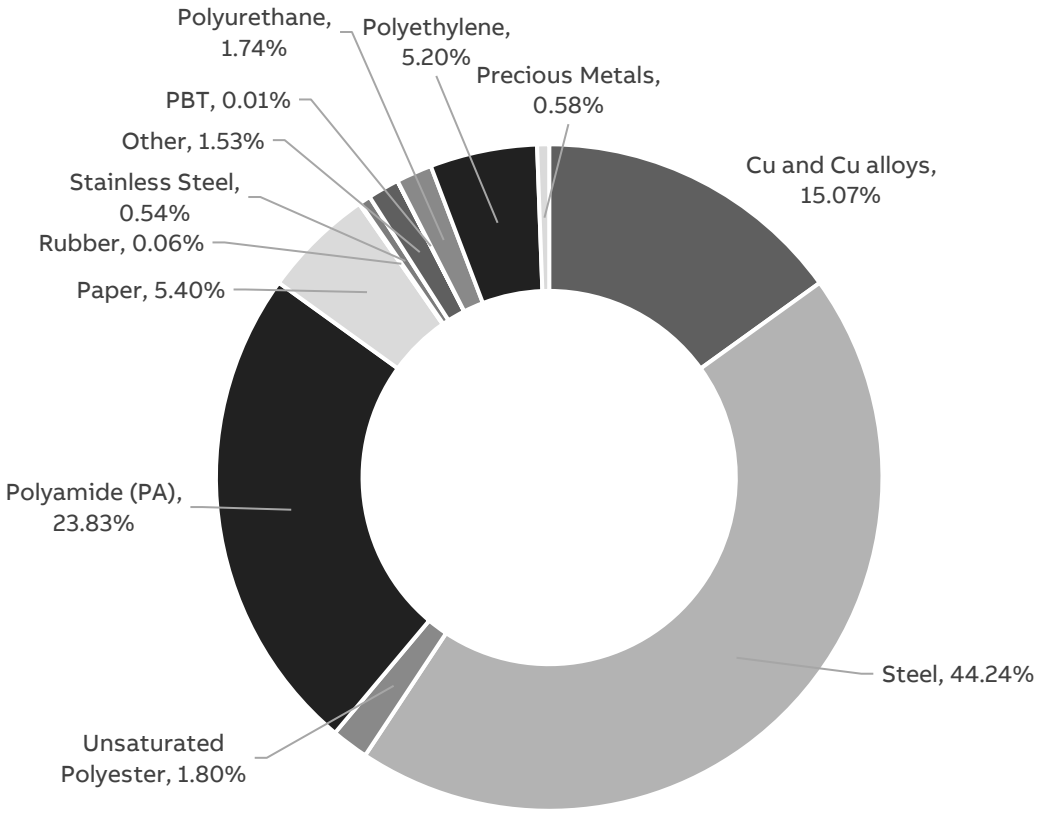


Figure 1: Composition of AF146

Along the whole AF116-146 product cluster, a set of different build configurations have been covered by this analysis.

Packaging weighs 80g, with the following substance composition:

Material	Unit	Total	%
Corrugated Cardboard	g	80	4.0%

Table 3: Weight of Packaging for AF146

Official declarations 2CMT2021-006277 [10] and 2CMT2021-006202 [11] states compliance of ABB AF Contactors respectively to RoHS and REACH regulations; 2CMT2021-006277 [10] provides exemptions considered for RoHS while 2CMT2021-006202 [11] lists REACH substances present in a concentration above 0.1% adding reference to products where involved parts are mounted.

© Copyright 2022 ABB. All rights reserved.

Functional unit and Reference Flow

The Functional Unit is to switch on and off during 20 years electrical power supply of a downstream installation with an electrical and/or mechanical control. The functional unit is characterized by type 3NO, a control circuit voltage U_c, a power circuit voltage U_p and a maximum allowed intensity by the power circuit I_p. (table 1)

The Reference Flow of the study is a single Contactor (including its packaging and accessories) with mass described, table 2.

System boundaries and life cycle stages

The life cycle of the Contactor, an EEPS (Electronic and Electrical Products and Systems), is a “from cradle to grave” analysis and covers the following main life cycle stages: manufacturing, including the relevant acquisition of raw material, preparation of semi-finished goods, etc. and processing steps; distribution; installation, including the relevant steps for the preparation of the product for use; use including the required maintenance steps within the RSL (reference service life of the product) associated to the reference product; end-of-life stage, including the necessary steps until final disposal or recovery of the product system.

The following table shows the stages of the product life cycle and the information stages according to EN 50693:2019 [3] for the evaluation of electronic and electrical products and systems.

Manufacturing	Distribution	Installation	Use	End-of-Life (EoL)
Acquisition of raw materials Transport to manufacturing site Components/parts manufacturing Assembly Packaging EoL treatment of generated waste	Transport to distributor/ logistic center Transport to place of use	Installation EoL treatment of generated waste (packaging)	Usage Maintenance	Deinstallation Collection and transport EoL treatment

Table 4: Phases for the evaluation of construction products according to EN50693:2019 [3].

Temporal and geographical boundaries

The ABB component suppliers are sourced all over the world. All primary data collected are from 2021, which is a representative production year. Secondary data are also representative for this year, as provided by ecoinvent [6].

© Copyright 2022 ABB. All rights reserved.

The selected ecoinvent [6] processes in the LCA model have a global representativeness, due to the unclear origin of each component. In this way, a conservative approach has been adopted.

Boundaries in the life cycle

As indicated in the PCR capital goods such as buildings, machinery, tools and infrastructure, the packaging for internal transport which cannot be allocated directly to the production of the reference product, may be excluded from the system boundary.

Infrastructures, when present, such as processes deriving from the ecoinvent [6] database have not been excluded.

Data quality

In this LCA, both primary and secondary data are used. Site specific foreground data have been provided by ABB. Main data sources are the bill of materials & drawings which are available on the ERP (SAP) & Windchill. For all processes for which primary are not available, generic data originating from the ecoinvent database [6], allocation cut-off by classification, are used. The ecoinvent database available in the SimaPro software [7] is used for the calculations.

The data quality characterized by quantitative and qualitative aspects, is presented in Ap-pendix 1. Each data quality parameter has been rated according to DQR tables from Chapter 7.19.2.2 of the Product Environmental Footprint Guide v.6.3 to give an indication of geography, technology and temporal representativeness.

Environmental impact indicators

The information obtained from the inventory analysis is aggregated according to the effects related to the various environmental issues. According to “PCR-ed4-EN-2021 09 06” and EN 50693 [3] the environmental impact indicators must be determined using the characterization factors and impact assessment methods specified in EN 15804:2012+A2:2019 [8].

PCR-ed4-EN-2021 09 06 and the EN 50693:2019 [3] standard establish four indicators for GWP: GWP (total) which includes all greenhouse gases; GWP (fossil fuels); GWP (biogenic) which includes the emissions and absorption of biogenic carbon dioxide and biogenic carbon stored in the product; GWP (land use) - land use and land use transformation. Other indicators as per the PCR[1].

Allocation rules

Allocation coefficients are based on the AF146 line’s occupancy area for electricity consumption since, apart from assembly processes, the whole production line is temperature-regulated throughout the year. The allocation of the total amount of waste generated by the production line and water consumption, has been based on this criterion.

Limitations and simplifications

Raw materials life cycle stage includes the extraction of raw materials as well as the transport distances to the manufacturing suppliers. These distances are assumed to be 1000 km as per PCR. This distance has been added to the one already included in the market processes used for the model, as a result of a conservative choice made by the LCA operators.

© Copyright 2022 ABB. All rights reserved.

Surface treatments like galvanizing, tin and silver plating as well as their related transport processes (back and forth from the finishing suppliers) have been considered in the LCA model.

Scraps for metal working and plastic processes are included when already defined in ecoinvent[6].

Energy Models

LCA Stage	EN 15804:2012 +A2:2019 module	Energy model	Notes
Raw material extraction and processing	A1-A2	Electricity, {RER}| market group for | Cut-off Electricity, {GLO}| market group for | Cut-off	Based on materials and supplier locations
Manufacturing	A3	Electricity, {SE}| market for | Cut-off	Specific Energy model for ABB Vasteras manufacturing plant, 100% renewable
Installation (Packaging EoL)	A5	Electricity, {GLO}| market group for | Cut-off	
Use Stage	B1	Electricity, [country]x | market for | Cut-off, S **	Low voltage, based on 2021 country sales mix
EoL	C1-C4	Electricity, {GLO}| market group for | Cut-off	

Table 5: Energy models used in each LCA stage

** Please refer the use phase page 11 for further description

Inventory analysis

In this LCA, both primary and secondary data are used. Site specific foreground data have been provided by ABB. For data collection, Bills of Material (BOM) extracted from ABB's internal SAP software were used. They are a list of all the components and assemblies that constitute the finished product, organized by level. Each item is matched with its code, quantity, weight and supplier. The BOMs were then processed, adding material, surface area and other weight data, taken from technical drawings. Finally, the manufacturing process and surface treatment were assigned, according to information provided by R&D personnel. Road distances between the suppliers and ABB were calculated using Google Maps, and marine distances using Distances & Time (Searates).

All primary data collected from ABB are from 2021, which was a representative production year. The ecoinvent v3.8 cut-off by classification system processes [6] are used to model the background system of the processes.

Due to a lot of components in the Contactors, raw material inputs have been modelled with data from ecoinvent representing either a European [RER] or Global [RoW] market coverage based on the supplier's location. These datasets are assumed to be representative.

© Copyright 2022 ABB. All rights reserved.

Manufacturing stage

The Contactors are composed of a multitude of components, all of which are made from of numerous materials. Most of the inputs to the products' manufacturing stage are already produced component parts.

The single use packaging as well as paper documentation are also included in the analysis in the manufacturing stage. ABB receives packaging components from outside suppliers and packages the Contactors before shipping them.

Most of the inputs to the products' manufacturing stage are already produced component parts from the supply chain. In the ABB manufacturing plant, the different components and subassemblies are assembled into the Contactor. All the semi-finished and ancillary products are produced by ABB's suppliers.

The entire AF Contactors suppliers' network has been modelled with the calculation of each transportation stage: from the first manufacturing supplier to the next. All the distances from the last subassembly suppliers' factories to the ABB manufacturing facility have been calculated.
All the distances from the last subassembly suppliers' factories to the ABB manufacturing facility have been calculated.
In the ABB factory, the different components and subassemblies are assembled into the Contactor. All the semi-finished and ancillary products are produced by ABB's suppliers.

The energy mix used for the production phase is representative for ABB Vasteras production site and includes renewable energy only (Hydroelectric).

The complete energy mix has been modeled considering the certificate on Guarantee of origins provided to ABB for the year 2021.

Distribution

The transport distances from ABB manufacturing plant to the distribution centers (regional distribution centers / local sales organizations) have been calculated considering the specific 2021 sales mix data for AF116-146 Contactors product cluster (SAP ERP sales data as a source).
Since no specific data is available for the transport distances from the Distribution Centre to place of actual use (Customer site), distances of 1000 km are assumed (local/domestic transport by lorry, according to PCR [1]).

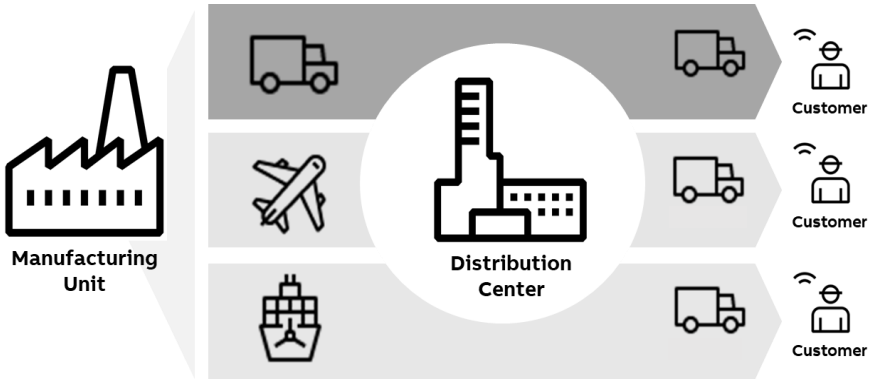


Figure 2: Distribution Methodology

© Copyright 2022 ABB. All rights reserved.

Installation

The installation phase only implies manual activities, and no energy is consumed. This phase also includes the disposal of the packaging and paper technical documentation of the AF Contactor.

For the disposal of the packaging and documentation after installation of the AF Contactor at the end of its life, a transport distance of 1000 km (according to PCR[1]) was assumed. The chosen transportation datasets from Ecoinvent [6].

The actual disposal site is unknown and is managed by the customer.

Use

Use and maintenance are modelled according to the PCR [1].

During the use phase, AF Contactor, dissipates some electricity due to power losses. They are calculated according to the data provided in the catalogue of the AF Contactor and following the PCR [1] & PSR [2] rules:

Parameters		
Iu	[A]	146
Iu	[%]	50
h/year	[h]	8760
RSL	[years]	20
Time operating coefficient α	[%]	50

Table 6: Use phase parameters

The formula for the calculation of the electricity consumed is shown below and it is described as follows, where P_{use} is the power consumed by the Contactor at a given value of current:

$$E_{use}\ [kWh] = \frac{P_{use} * 8760 * RSL * \alpha}{1000}$$

The above calculations have been performed according to the number of poles (3) on which relevant current flows during use phase.

The Energy model used for this phase has been modeled based on the 2021 actual sales mix data (SAP ERP sales data as a source). From the Ecoinvent [6] database, the low voltage electricity country mix for each country(x) has been selected with its respective percentage on the total sales mix (Electricity, low voltage [country]x | market for | Cut-off, S).

Since no maintenance happens during the use phase, the environmental impacts linked to this procedure have been considered as null in the analysis.

© Copyright 2022 ABB. All rights reserved.

End of life

The end-of-life stage is modelled according to PCR [1] and IEC/TR 62635 [9]. The percentages for end-of-life treatments of materials are taken from IEC/TR 62635 [9].

Since no specific data is available, the transport distances from the place of use to the place of disposal are assumed to be 1000 km (local/domestic transport by lorry, according to PCR [1]).

Disassembly manuals can be provided to the customer to support product disposal.

© Copyright 2022 ABB. All rights reserved.

Environmental impacts

The following tables show the environmental impact indicators of the life cycle of a single Contactor, as indicated by PEP Ecopassport PCR and EN 50693:2019 [3]. The indicators are divided into the contribution of the processes to the different modules (upstream, core and downstream) and stages (manufacturing, distribution, installation, use and end-of-life).

AF146-30

Impact category	Unit	Total	Manufacturing	Distribution	Installation	Use	End of Life
GWP-total	kg CO2 eq	4.12E+02	1.23E+01	4.22E-01	4.63E-02	3.99E+02	4.48E-01
GWP-fossil	kg CO2 eq	4.03E+02	1.23E+01	4.22E-01	1.48E-02	3.89E+02	4.35E-01
GWP-biogenic	kg CO2 eq	8.49E+00	3.05E-02	3.53E-04	3.14E-02	8.41E+00	1.32E-02
GWP-luluc	kg CO2 eq	7.77E-01	1.46E-02	1.79E-04	5.55E-06	7.62E-01	3.04E-04
ODP	kg CFC11 eq	2.19E-05	7.24E-07	9.84E-08	3.51E-09	2.10E-05	4.78E-08
AP	mol H+ eq	2.10E+00	2.26E-01	4.09E-03	7.75E-05	1.86E+00	2.40E-03
EP-freshwater	kg P eq	2.51E-01	1.96E-02	2.43E-05	9.83E-07	2.31E-01	9.74E-05
EP-marine	kg N eq	3.62E-01	2.64E-02	1.18E-03	4.10E-05	3.34E-01	7.55E-04
EP-terrestrial	mol N eq	3.56E+00	2.85E-01	1.30E-02	2.89E-04	3.25E+00	5.82E-03
POCP	kg NMVOC eq	9.79E-01	8.07E-02	3.63E-03	9.01E-05	8.93E-01	1.69E-03
ADP-m&m	kg Sb eq	1.03E-02	7.82E-03	9.08E-07	3.46E-08	2.48E-03	5.08E-07
ADP-fossil	MJ	6.06E+03	1.75E+02	6.41E+00	2.31E-01	5.87E+03	5.17E+00
WDP	m3	1.03E+02	7.63E+00	2.07E-02	1.18E-03	9.58E+01	3.50E-02
PENRE	MJ	6.03E+03	1.55E+02	6.41E+00	2.31E-01	5.87E+03	5.17E+00
PENRM	MJ	1.95E+01	1.95E+01	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PENRT	MJ	6.05E+03	1.75E+02	6.41E+00	2.31E-01	5.87E+03	5.17E+00
PERE	MJ	1.03E+03	1.55E+01	7.62E-02	3.11E-03	1.01E+03	3.52E-01
PERM	MJ	1.67E+00	1.67E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PERT	MJ	1.03E+03	1.72E+01	7.62E-02	3.11E-03	1.01E+03	3.52E-01
SM	kg	6.09E-01	6.09E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00
RSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
NRSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
FW	m3	4.31E+00	2.08E-01	7.08E-04	3.95E-05	4.10E+00	1.46E-03
HWD	kg	6.33E-03	1.95E-03	1.42E-05	5.55E-07	4.36E-03	7.33E-06
N-HWD	kg	2.77E+01	2.10E+00	5.18E-01	2.94E-02	2.46E+01	4.16E-01
RWD	kg	2.42E-02	4.60E-04	4.36E-05	1.55E-06	2.36E-02	2.46E-05
MfR	kg	2.13E+00	2.90E-01	0.00E+00	1.44E-01	0.00E+00	1.70E+00
MfER	kg	2.70E-02	7.25E-03	0.00E+00	8.00E-03	0.00E+00	1.18E-02
Efp	disease inc.	1.16E-05	9.21E-07	4.44E-08	1.77E-09	1.06E-05	4.24E-08
IrHH	kBq U-235 eq	9.59E+01	1.16E+00	3.20E-02	1.17E-03	9.47E+01	3.24E-02
ETX FW	CTUe	8.40E+03	2.14E+03	4.86E+00	2.30E-01	6.24E+03	8.24E+00
HTX CE	CTUh	1.67E-07	3.92E-08	1.62E-10	5.77E-12	1.27E-07	4.61E-10
HTX N-CE	CTUh	6.15E-06	1.98E-06	5.07E-09	2.52E-10	4.14E-06	2.78E-08
IrLS	Pt	1.27E+03	1.03E+02	6.43E+00	2.64E-01	1.16E+03	4.15E+00

Table 7: Impact indicators for AF146-30

© Copyright 2022 ABB. All rights reserved.

Impact category	Unit	Total
Biogenic Carbon content of the product	kg	8.39E-02
Biogenic Carbon content of the associated packaging	kg	1.6E-02

Table 8: Inventory Flow indicators of AF146-30.

Environmental impact indicators

GWP-total	Global Warming Potential total (Climate change)
GWP-fossil	Global Warming Potential fossil
GWP-biogenic	Global Warming Potential biogenic
GWP-luluc	Global Warming Potential land use and land use change
ODP	Depletion potential of the stratospheric ozone layer
AP	Acidification potential
EP-freshwater	Eutrophication potential - freshwater compartment
EP-marine	Eutrophication potential - fraction of nutrients reaching marine end compartment
EP-terrestrial	Eutrophication potential -Accumulated Exceedance
POCP	Formation potential of tropospheric ozone
ADP-m&m	Abiotic Depletion for non-fossil resources potential
ADP-fossil	Abiotic Depletion for fossil resources potential, WDP
WDP	Water deprivation potential.

Resource use indicators

PENRE	Use of non-renewable primary energy excluding renewable primary energy resources used as raw material
PENRM	Use of non-renewable primary energy resources used as raw material
PENRT	Total use of non-renewable primary energy resources (primary energy and primary energy resources used as raw materials)
PERE	Use of renewable primary energy excluding non-renewable primary energy resources used as raw material
PERM	Use of renewable primary energy resources used as raw material
PERT	Total use of renewable primary energy resources (primary energy and primary energy resources used as raw materi-als)

Secondary materials, water and energy resources

SM	Use of secondary materials
RSF	Use of renewable secondary fuels

© Copyright 2022 ABB. All rights reserved.

NRSF	Use of non-renewable secondary fuels
FW	FW: Net use of fresh water

Waste category indicators

HWD	Hazardous waste disposed
N-HWD	Non-hazardous waste disposed
RWD	Radioactive waste disposed

Output flow indicators

MfR	Materials for recycling
MfER	Materials for energy recovery

Others indicators

Efp	Emissions of Fine particles
IrHH	Ionizing radiation, human health
ETX FW	Ecotoxicity, freshwater
HTX CE	Human toxicity, carcinogenic effects
HTX N-CE	Human toxicity, non-carcinogenic effects

© Copyright 2022 ABB. All rights reserved.

Extrapolation for Homogeneous environmental family

This LCA covers different build configurations than the representative product from the IEC and UL types. All the analyzed configurations have the same main functionality, product standards and manufacturing technology

The different life cycle stages can be extrapolated to other products of the same homogeneous environmental family by applying a rule of proportionality to the parameters in the following tables, divided by different life cycle stages

Contactors	LCA Stage	GWP-total	GWP-fossil	GWP-biogenic	GWP-luluc	ODP	AP	EP-freshwater	EP-marine	EP-terrestrial	POCP	ADP-minerals & metals	ADP-fossil	WDP
AF116-AF140 (3 Pole)	Manu.	1.07	1.07	1.18	1.08	1.04	1.06	1.14	1.11	1.12	1.10	1.21	1.06	1.05
	EoL	0.98	0.98	1.00	0.96	1.00	0.99	0.97	0.96	0.99	0.99	0.98	0.99	0.97
AF116-AF140 (3 Pole + PLC)	Manu.	1.15	1.15	1.43	1.20	1.12	1.22	1.22	1.20	1.22	1.22	1.28	1.11	1.11
	EoL	0.97	0.97	1.08	0.95	0.97	0.99	0.97	0.94	0.98	0.98	0.96	0.97	0.96
AF116-AF140 (4 Pole)	Manu.	1.62	1.63	-1.96	1.86	1.78	1.70	2.09	1.92	2.00	1.85	2.38	1.57	1.54
	EoL	1.52	1.48	2.62	1.55	1.31	1.53	1.62	1.40	1.43	1.42	1.33	1.45	1.52
AF146 (3 Pole)	Manu.	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	EoL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
AF146 (3 Pole + PLC)	Manu.	1.03	1.03	0.91	1.02	0.98	1.08	0.91	0.96	0.94	1.00	0.82	1.03	1.04
	EoL	0.49	0.48	0.84	0.21	0.88	0.37	0.11	0.69	0.58	0.59	0.79	0.53	0.35

Table 9: Extrapolation factors for AF146

Reference product: AF146-30 -Manufacturing / End of Life

Contactor	LCA Stage	Factor
AF116-AF140 (3 Pole)	Distribution	1.00
AF116-AF140 (3 Pole+ PLC)		1.02
AF116-AF140 (4 Pole)		1.44
AF146 (3 Pole)		1.00
AF146 (3 Pole+ PLC)		1.03

Table10: Extrapolation factors for AF146.

Reference product: AF146-30 -Distribution

© Copyright 2022 ABB. All rights reserved.

Contactor	LCA Stage	Factor
AF116 (3 Pole- with & without PLC)	Use Phase	0.63
AF140 (3 Pole- with & without PLC)		0.91
AF146 (3 Pole- with & without PLC)		1.00
AF116 (4 Pole)		1.18
AF140 (4 Pole)		1.74

Table 11: Extrapolation factors for AF146
Reference product: AF146-30-Use Phase

Additional environmental information

According to the waste treatment scenario calculation in Simapro, based on the recycling rate in the technical report IEC/TR 62635 Edition 1.0 [9] Table D.6, the following recyclability potentials were calculated. The recyclability potential is calculated based on the product weight (excluding packaging).

	AF146-30
Recyclability potential	87.83%

Table 12: Recyclability potential of AF146-30.

© Copyright 2022 ABB. All rights reserved.

References

[1] PEP Ecopassport PCR-ed4-EN-2021 09 06 “Product Category Rules for Electrical, Electronic and HVAC-R Products”
(published: 6th September 2021)

[2] PEP Ecopassport PSR-0005-ed2-EN-2016 03 29 “Product Specific Rules for Electrical Switchgear and Control gear Solutions” (published: May 2016)

[3] EN 50693:2019 - Product category rules for life cycle assessments of electronic and electrical products and systems

[4] ISO 14040:2006 - Environmental management -Life cycle assessment - Principles and framework

[5] ISO 14044:2006 - Environmental management - Life cycle assessment - Requirements and guidelines

[6] ecoinvent v3.8 (2021). ecoinvent version 3.8. Swiss, Centre for Life Cycle Inventories, Dübendorf, Switzerland

[7] PRé Consultants, 2021. Software SimaPro versione 9.3.1 (www.pre.nl).

[8] UNI EN 15804:2012+A2:2019: Sustainability of constructions - Environmental product declarations (September 2019).

[9] IEC/TR 62635 - Guidelines for end-of-life information provided by manufacturers and recyclers and for recyclability rate calculation of electrical and electronic equipment - Edition 1.0 2012-10

[10] 2CMT2021-006277- RoHS

[11] 2CMT2021-006202- REACh

[12] 1SFC100091D0201-LCA Report

© Copyright 2022 ABB. All rights reserved.

PRODUCT ENVIRONMENTAL PROFILE

Environmental Product Declaration

AF190-205 Contactors



SGS Italia S.p.A.
Via Caldera, 21
20153 MILANO
VERIFIED
DATE: 2022-09-27

REGISTRATION NUMBER ABBG-00029-V01.01	IN COMPLIANCE WITH PCR-ED4-EN-2021 09 06 SUPPLEMENTED BY PSR-0005-ED2-EN-2016 03 29
VERIFIER ACCREDITATION NUMBER VH42	INFORMATION AND REFERENCE DOCUMENTS www.pep-ecopassport.org
DATE OF ISSUE 09-2022	VALIDITY PERIOD 5 years
INDEPENDENT VERIFICATION OF THE DECLARATION AND DATA, IN COMPLIANCE WITH ISO 14025: 2006	
INTERNAL ☐ / EXTERNAL ☒	
THE PCR REVIEW WAS CONDUCTED BY A PANEL OF EXPERTS CHAIRED BY JULIE ORGELET (DDEMAIN)	
PEP ARE COMPLIANT WITH XP C08-100-1 :2016 OR EN 50693:2019 THE ELEMENTS OF THE PRESENT PEP CANNOT BE COMPARED WITH ELEMENTS FROM ANOTHER PROGRAM.	
DOCUMENT IN COMPLIANCE WITH ISO 14025: 2006 « ENVIRONMENTAL LABELS AND DECLARATIONS. TYPE III ENVIRONMENTAL DECLARATIONS »	

© Copyright 2022 ABB. All rights reserved.

EPD Owner	ABB Electrification Sweden AB, Motorgraend 20 Bnr 415, Vaesteras, Sweden www.abb.com
Manufacturer name and address	ABB Electrification Sweden AB, Motorgraend 20 Bnr 415, Vaesteras, Sweden
Company contact	Johan Lundstrom- johan.lundstrom@se.abb.com R&D Engineer
Reference product	AF205-30 Contactor
Description of the product	AF205-30 is a 3 pole - 1000V IEC or 600V UL contactor with pre-mounted auxiliary contacts and Main Circuit Bars, controlling motors up to 110kW / 400V AC (AC-3) or 150hp / 480V UL and switching power circuits up to 350A (AC-1) or 300A UL general use. The contactor has a wide control voltage range (20-500V 50/60Hz and DC), managing large control voltage variations, reducing panel energy consumptions and ensuring distinct operations in unstable networks.
Functional unit	The Functional Unit is to switch on and off during 20 years electrical power supply of a downstream installation with an electrical and/or mechanical control. The functional unit is characterized by type 3NO, a control circuit voltage U_c, a power circuit voltage U_p and a maximum allowed intensity by the power circuit I_p. Power Circuit Voltage U_p [V]: 1000V Maximum allowed intensity by the power circuit I_p [A]: 205A Rated Control Circuit Voltage U_c [V]: 20-500V Number of poles: 3
Other products covered	The PEP covers offerings for the 3-pole variants including PLC variants (AF190/205-30-*, AF190B/205B-30-*, AF190R/205R-30-*, AFS190/AFS205-30-*) and for the 4-pole variants (AF190/205-40-*) of AF190 and AF205 Contactors.
Reference lifetime	20 years
Product category	Electrical, Electronic and HVAC-R Products
Use Scenario	The use phase has been modeled based on the sales mix data (2021), and the corresponding low voltage electricity countries mix
Geographical representativeness	Raw materials & Manufacturing: [Europe / Global] Assembly: [Sweden] Distribution / Use: [Global] specific sales mix EoL: [Global]
Technological representativeness	Materials and processes data are specific for the production of AF205-30 Contactors
LCA Study	This study is based on the LCA study described in the LCA report 1SFC100091D0201
EPD type	Representative Product
EPD scope	“Cradle to grave”
Year of reported primary data	2021
LCA software	SimaPro 9.3.0.3 (2022)
LCI database	ecoinvent v3.8 (2021)
LCIA methodology	EN 50693:2019

© Copyright 2022 ABB. All rights reserved.

Contents

ABB Purpose & Embedding Sustainability **4**

General Information **4**
- **AF Contactors product cluster** 5

Constituent Materials **5**
- **Functional unit and Reference Flow** 7
- **System boundaries and life cycle stages** 7
- **Temporal and geographical boundaries** 7
- **Boundaries in the life cycle** 8
- **Data quality** 8
- **Environmental impact indicators** 8
- **Allocation rules** 8
- **Limitations and simplifications** 8
- **Energy Models** 9

Inventory analysis **9**

Manufacturing stage **10**

Distribution **10**

Installation **11**

Use **11**

End of life **12**

Environmental impacts **13**

Additional environmental information **17**

References **18**

© Copyright 2022 ABB. All rights reserved.

ABB Purpose & Embedding Sustainability

ABB is a leading global technology company that energizes the transformation of society and industry to achieve a more productive, sustainable future. By connecting software to its electrification, robotics, automation and motion portfolio, ABB pushes the boundaries of technology to drive performance to new levels. With a history of excellence stretching back more than 130 years, ABB's success is driven by about 110 thousand talented employees in over 100 countries.
ABB's Electrification business offers a wide-ranging portfolio of products, digital solutions and services, from substation to socket, enabling safe, smart and sustainable electrification. Offerings encompass digital and connected innovations for low voltage and medium voltage, including EV infrastructure, solar inverters, modular substations, distribution automation, power protection, wiring accessories, switchgear, enclosures, cabling, sensing and control.
ABB is committed to continually promoting and embedding sustainability across its operations and value chain, aspiring to become a role model for others to follow. With its ABB Purpose, ABB is focusing on reducing harmful emissions, preserving natural resources and championing ethical and humane behavior.

General Information

The ABB facility located in Vasteras, is for the Smart Power Division, the competence center for Contactors >100A, Softstarters and one of leading manufacturing site.
All the products are produced in an automated system with precision and accuracy. These are supplied to utilities, industrial, and tertiary sector customers.
ABB adopts and implements for its own activities an integrated Quality/Environmental/Health Management System in compliance with the following
standards:
SS-EN ISO 9001: 2015 - Quality Management Systems- Requirements
SS-EN ISO 14001: 2015 - Environmental management systems - Requirements with guidance for use
ISO 45001:2018 - Occupational Health and Safety Assessment Series
SS-EN ISO 50001: 2018 - Energy management systems

AF contactors are most commonly used for controlling electric motors and switch various none-motor loads. Contactors can break current over a wide range of currents, from a few amperes to thousands of amperes, and voltages upto 1000V AC.

© Copyright 2022 ABB. All rights reserved.

AF Contactors product cluster

Product cluster declared in this LCA includes the following AF190-205 Contactors.

Product Range	Product	Number of poles	Power Circuit Voltage [U_p]	Maximum allowed intensity by the power circuit [I_p]	Rated Control Circuit Voltage [U_c]
AF190-205	AF190-AF205	3	1000V	190-205A	20-500V
	AF190-AF205 (PLC)	3	1000V	190-205A	20-500V
	AF190-AF205	4	1000V	190-205A	20-500V

Table 1: Technical characteristics of AF AF190-205 Contactors.

The accessories associated with these products are also included in the study.

Reference Product:

The reference product for the LCA of the complete range of AF 190-205 is AF205-30.

Constituent Materials

The AF205-30 weights about 3.4kg including its installed accessories, packaging, and paper documentation.

AF205-30				
Materials	**Name**	**IEC 62474 MC**	**[g]**	**%**
Metals	Cu and Cu alloys	M-121	633.66	18.32%
	Steel	M-119	1129.42	32.66%
	Stainless Steel	M-100	116.56	3.37%
	Precious Metals	M-159	18.76	0.54%
Plastics	Unsaturated Polyester	M-301	396.04	11.45%
	Polyamide (PA)	M-258	831.02	24.03%
	Polyethylene	M-251	108.09	3.13%
	PBT	M-261	0.18	0.01%
	Polyurethane	M-300	34.00	0.98%
	Rubber	M-320	2.52	0.07%
Other	Paper / Cardboard	M-341	157.97	4.57%
	Other	N/A	30.00	0.87%
Total			**3458.22**	**100%**

Table 2: Weight of materials AF205-30

© Copyright 2022 ABB. All rights reserved.

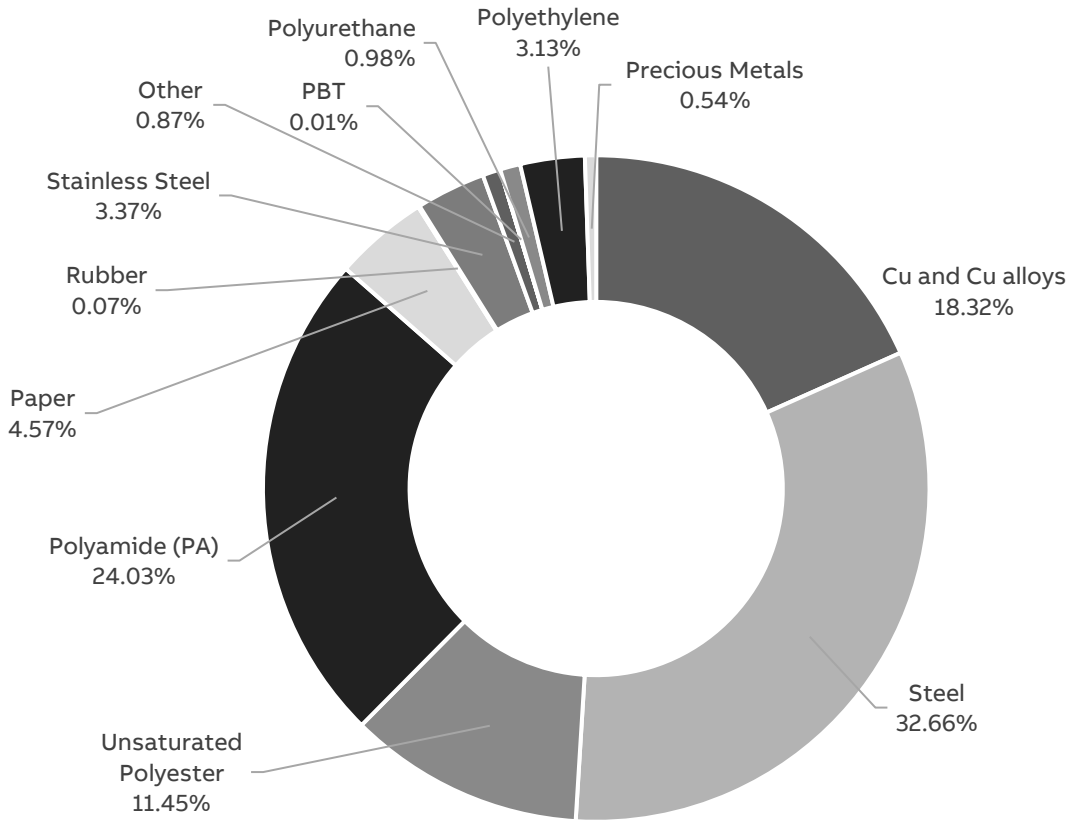


Figure 1: Composition of AF205-30

Along the whole AF190-205 product cluster, a set of different build configurations have been covered by this analysis.

Packaging weighs 120g, with the following substance composition:

Material	Unit	Total	%
Corrugated Cardboard	g	120	3.3%

Table 3: Weight of Packaging for AF205-30

Official declarations 2CMT2021-006277 [10] and 2CMT2021-006202 [11] states compliance of ABB AF Contactors respectively to RoHS and REACH regulations; 2CMT2021-006277 [10] provides exemptions considered for RoHS while 2CMT2021-006202 [11] lists REACH substances present in a concentration above 0.1% adding reference to products where involved parts are mounted.

© Copyright 2022 ABB. All rights reserved.

Functional unit and Reference Flow

The Functional Unit is to switch on and off during 20 years electrical power supply of a downstream installation with an electrical and/or mechanical control. The functional unit is characterized by a type 3NO, a control circuit voltage U_c, a power circuit voltage U_p and a maximum allowed intensity by the power circuit I_p. (table 1)

The Reference Flow of the study is a single Contactor (including its packaging and accessories) with mass described, table 2.

System boundaries and life cycle stages

The life cycle of the Contactor, an EEPS (Electronic and Electrical Products and Systems), is a “from cradle to grave” analysis and covers the following main life cycle stages: manufacturing, including the relevant acquisition of raw material, preparation of semi-finished goods, etc. and processing steps; distribution; installation, including the relevant steps for the preparation of the product for use; use including the required maintenance steps within the RSL (reference service life of the product) associated to the reference product; end-of-life stage, including the necessary steps until final disposal or recovery of the product system.

The following table shows the stages of the product life cycle and the information stages according to EN 50693:2019 [3] for the evaluation of electronic and electrical products and systems.

Manufacturing	Distribution	Installation	Use	End-of-Life (EoL)
Acquisition of raw materials Transport to manufacturing site Components/parts manufacturing Assembly Packaging EoL treatment of generated waste	Transport to distributor/ logistic center Transport to place of use	Installation EoL treatment of generated waste (packaging)	Usage Maintenance	Deinstallation Collection and transport EoL treatment

Table 4: Phases for the evaluation of construction products according to EN50693:2019 [3].

Temporal and geographical boundaries

The ABB component suppliers are sourced all over the world. All primary data collected are from 2021, which is a representative production year. Secondary data are also representative for this year, as provided by ecoinvent [6].

© Copyright 2022 ABB. All rights reserved.

The selected ecoinvent [6] processes in the LCA model have a global representativeness, due to the unclear origin of each component. In this way, a conservative approach has been adopted.

Boundaries in the life cycle

As indicated in the PCR capital goods such as buildings, machinery, tools and infrastructure, the packaging for internal transport which cannot be allocated directly to the production of the reference product, may be excluded from the system boundary.

Infrastructures, when present, such as processes deriving from the ecoinvent [6] database have not been excluded.

Data quality

In this LCA, both primary and secondary data are used. Site specific foreground data have been provided by ABB. Main data sources are the bill of materials & drawings which are available on the ERP (SAP) & Windchill. For all processes for which primary are not available, generic data originating from the ecoinvent database [6], allocation cut-off by classification, are used. The ecoinvent database available in the SimaPro software [7] is used for the calculations.

The data quality characterized by quantitative and qualitative aspects, is presented in Ap-pendix 1. Each data quality parameter has been rated according to DQR tables from Chapter 7.19.2.2 of the Product Environmental Footprint Guide v.6.3 to give an indication of geography, technology and temporal representativeness.

Environmental impact indicators

The information obtained from the inventory analysis is aggregated according to the effects related to the various environmental issues. According to “PCR-ed4-EN-2021 09 06” and EN 50693 [3] the environmental impact indicators must be determined using the characterization factors and impact assessment methods specified in EN 15804:2012+A2:2019 [8].

PCR-ed4-EN-2021 09 06 and the EN 50693:2019 [3] standard establish four indicators for GWP: GWP (total) which includes all greenhouse gases; GWP (fossil fuels); GWP (biogenic) which includes the emissions and absorption of biogenic carbon dioxide and biogenic carbon stored in the product; GWP (land use) - land use and land use transformation. Other indicators as per the PCR[1].

Allocation rules

Allocation coefficients are based on the AF205 line’s occupancy area for electricity consumption since, apart from assembly processes, the whole production line is temperature-regulated throughout the year. The allocation of the total amount of waste generated by the production line and water consumption, has been based on this criterion.

Limitations and simplifications

Raw materials life cycle stage includes the extraction of raw materials as well as the transport distances to the manufacturing suppliers. These distances are assumed to be 1000 km as per PCR. This distance has been added to the one already included in the market processes used for the model, as a result of a conservative choice made by the LCA operators.

© Copyright 2022 ABB. All rights reserved.

Surface treatments like galvanizing, tin and silver plating as well as their related transport processes (back and forth from the finishing suppliers) have been considered in the LCA model.

Scraps for metal working and plastic processes are included when already defined in ecoinvent[6].

Energy Models

LCA Stage	EN 15804:2012 +A2:2019 module	Energy model	Notes
Raw material extraction and processing	A1-A2	Electricity, {RER}| market group for | Cut-off Electricity, {GLO}| market group for | Cut-off	Based on materials and supplier locations
Manufacturing	A3	Electricity, {SE}| market for | Cut-off	Specific Energy model for ABB Vasteras manufacturing plant, 100% renewable
Installation (Packaging EoL)	A5	Electricity, {GLO}| market group for | Cut-off	
Use Stage	B1	Electricity, [country]x | market for | Cut-off, S **	Low voltage, based on 2021 country sales mix
EoL	C1-C4	Electricity, {GLO}| market group for | Cut-off	

Table 5: Energy models used in each LCA stage

** Please refer the use phase page 11 for further description

Inventory analysis

In this LCA, both primary and secondary data are used. Site specific foreground data have been provided by ABB. For data collection, Bills of Material (BOM) extracted from ABB's internal SAP software were used. They are a list of all the components and assemblies that constitute the finished product, organized by level. Each item is matched with its code, quantity, weight and supplier. The BOMs were then processed, adding material, surface area and other weight data, taken from technical drawings. Finally, the manufacturing process and surface treatment were assigned, according to information provided by R&D personnel. Road distances between the suppliers and ABB were calculated using Google Maps, and marine distances using Distances & Time (Searates).

All primary data collected from ABB are from 2021, which was a representative production year. The ecoinvent v3.8 cut-off by classification system processes [6] are used to model the background system of the processes.

Due to a lot of components in the Contactors, raw material inputs have been modelled with data from ecoinvent representing either a European [RER] or Global [RoW] market coverage based on the supplier's location. These datasets are assumed to be representative.

© Copyright 2022 ABB. All rights reserved.

Manufacturing stage

The Contactors are composed of a multitude of components, all of which are made from of numerous materials. Most of the inputs to the products' manufacturing stage are already produced component parts.

The single use packaging as well as paper documentation are also included in the analysis in the manufacturing stage. ABB receives packaging components from outside suppliers and packages the Contactors before shipping them.

Most of the inputs to the products' manufacturing stage are already produced component parts from the supply chain. In the ABB manufacturing plant, the different components and subassemblies are assembled into the Contactor. All the semi-finished and ancillary products are produced by ABB's suppliers.

The entire AF Contactors suppliers' network has been modelled with the calculation of each transportation stage: from the first manufacturing supplier to the next. All the distances from the last subassembly suppliers' factories to the ABB manufacturing facility have been calculated.
All the distances from the last subassembly suppliers' factories to the ABB manufacturing facility have been calculated.
In the ABB factory, the different components and subassemblies are assembled into the Contactor. All the semi-finished and ancillary products are produced by ABB's suppliers.

The energy mix used for the production phase is representative for ABB Vasteras production site and includes renewable energy only (Hydroelectric).The complete energy mix has been modeled considering the certificate on Guarantee of origins provided to ABB for the year 2021.

Distribution

The transport distances from ABB manufacturing plant to the distribution centers (regional distribution centers / local sales organizations) have been calculated considering the specific 2021 sales mix data for AF190-205 Contactors product cluster (SAP ERP sales data as a source).
Since no specific data is available for the transport distances from the Distribution Centre to place of actual use (Customer site), distances of 1000 km are assumed (local/domestic transport by lorry, according to PCR [1]).

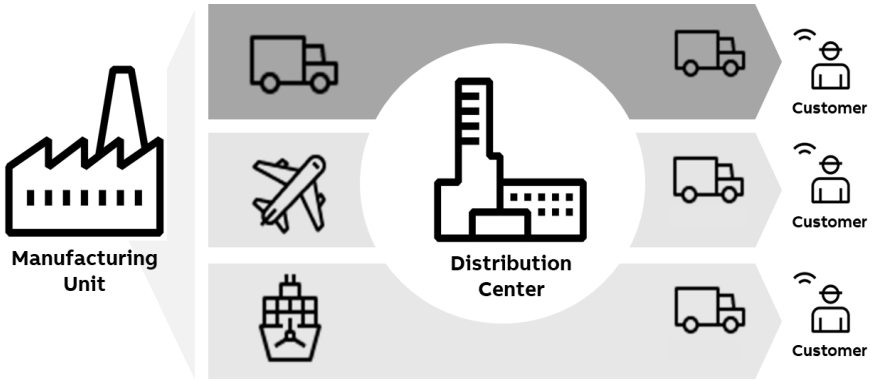


Figure 2: Distribution Methodology

© Copyright 2022 ABB. All rights reserved.

Installation

The installation phase only implies manual activities, and no energy is consumed. This phase also includes the disposal of the packaging and paper technical documentation of the AF Contactor.

For the disposal of the packaging and documentation after installation of the AF Contactor at the end of its life, a transport distance of 1000 km (according to PCR[1]) was assumed. The chosen transportation datasets from Ecoinvent [6].

The actual disposal site is unknown and is managed by the customer.

Use

Use and maintenance are modelled according to the PCR [1].

During the use phase, AF Contactor, dissipates some electricity due to power losses. They are calculated according to the data provided in the catalogue of the AF Contactor and following the PCR [1] & PSR [2] rules:

Parameters		
Iu	[A]	205
Iu	[%]	50
h/year	[h]	8760
RSL	[years]	20
Time operating coefficient	[%]	50

Table 6: Use phase parameters

The formula for the calculation of the electricity consumed is shown below and it is described as follows, where P_{use} is the power consumed by the Contactor at a given value of current:

$$E_{use}\ [kWh] = \frac{P_{use} * 8760 * RSL * \alpha}{1000}$$

The above calculations have been performed according to the number of poles (3) on which relevant current flows during use phase.

The Energy model used for this phase has been modeled based on the 2021 actual sales mix data (SAP ERP sales data as a source). From the Ecoinvent [6] database, the low voltage electricity country mix for each country(x) has been selected with its respective percentage on the total sales mix (Electricity, low voltage [country]x | market for | Cut-off, S).

Since no maintenance happens during the use phase, the environmental impacts linked to this procedure have been considered as null in the analysis.

© Copyright 2022 ABB. All rights reserved.

End of life

The end-of-life stage is modelled according to PCR [1] and IEC/TR 62635 [9]. The percentages for end-of-life treatments of materials are taken from IEC/TR 62635 [9].

Since no specific data is available, the transport distances from the place of use to the place of disposal are assumed to be 1000 km (local/domestic transport by lorry, according to PCR [1]).

Disassembly manuals can be provided to the customer to support product disposal.

© Copyright 2022 ABB. All rights reserved.

Environmental impacts

The following tables show the environmental impact indicators of the life cycle of a single Contactor, as indicated by PEP Ecopassport PCR and EN 50693:2019 [3]. The indicators are divided into the contribution of the processes to the different modules (upstream, core and downstream) and stages (manufacturing, distribution, installation, use and end-of-life).

AF205-30

Impact category	Unit	Total	Manufacturing	Distribution	Installation	Use	End of Life
GWP-total	kg CO2 eq	3.70E+02	2.69E+01	7.82E-01	8.19E-02	3.42E+02	9.24E-01
GWP-fossil	kg CO2 eq	3.62E+02	2.65E+01	7.81E-01	3.47E-02	3.34E+02	9.01E-01
GWP-biogenic	kg CO2 eq	7.46E+00	3.00E-01	5.72E-04	4.72E-02	7.10E+00	2.15E-02
GWP-luluc	kg CO2 eq	1.06E+00	3.25E-02	3.59E-04	1.28E-05	1.02E+00	6.13E-04
ODP	kg CFC11 eq	1.67E-05	1.77E-06	1.79E-07	8.26E-09	1.46E-05	8.68E-08
AP	mol H+ eq	2.11E+00	4.83E-01	9.62E-03	1.79E-04	1.61E+00	4.70E-03
EP-freshwater	kg P eq	2.86E-01	4.47E-02	4.29E-05	2.25E-06	2.41E-01	1.98E-04
EP-marine	kg N eq	3.54E-01	5.55E-02	2.65E-03	8.31E-05	2.94E-01	2.01E-03
EP-terrestrial	mol N eq	3.46E+00	6.13E-01	2.93E-02	6.70E-04	2.80E+00	1.10E-02
POCP	kg NMVOC eq	9.40E-01	1.70E-01	8.02E-03	2.06E-04	7.59E-01	3.20E-03
ADP-m&m	kg Sb eq	1.96E-02	1.76E-02	1.61E-06	8.06E-08	2.02E-03	9.45E-07
ADP-fossil	MJ	5.66E+03	3.85E+02	1.17E+01	5.41E-01	5.25E+03	9.91E+00
WDP	m3	8.47E+01	1.68E+01	3.60E-02	2.44E-03	6.78E+01	7.33E-02
PENRE	MJ	5.62E+03	3.45E+02	1.17E+01	5.41E-01	5.25E+03	9.91E+00
PENRM	MJ	3.93E+01	3.93E+01	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PENRT	MJ	5.65E+03	3.85E+02	1.17E+01	5.41E-01	5.25E+03	9.91E+00
PERE	MJ	8.75E+02	3.72E+01	1.33E-01	7.15E-03	8.37E+02	7.17E-01
PERM	MJ	2.50E+00	2.50E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PERT	MJ	8.77E+02	3.97E+01	1.33E-01	7.15E-03	8.37E+02	7.17E-01
SM	kg	9.35E-01	9.35E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00
RSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
NRSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
FW	m3	4.49E+00	4.60E-01	1.23E-03	8.25E-05	4.02E+00	3.03E-03
HWD	kg	6.66E-03	3.41E-03	2.43E-05	1.30E-06	3.22E-03	1.32E-05
N-HWD	kg	2.97E+01	5.40E+00	8.53E-01	6.23E-02	2.24E+01	1.02E+00
RWD	kg	2.36E-02	1.20E-03	7.95E-05	3.65E-06	2.22E-02	4.56E-05
MfR	kg	3.55E+00	4.94E-01	0.00E+00	3.44E-01	0.00E+00	2.72E+00
MfER	kg	7.07E-02	2.38E-02	0.00E+00	1.20E-02	0.00E+00	3.48E-02
Efp	disease inc.	8.13E-06	1.99E-06	7.57E-08	4.15E-09	5.98E-06	8.10E-08
IrHH	kBq U-235 eq	9.22E+01	3.26E+00	5.76E-02	2.74E-03	8.88E+01	6.33E-02
ETX FW	CTUe	1.05E+04	4.77E+03	8.68E+00	4.98E-01	5.66E+03	1.65E+01
HTX CE	CTUh	2.14E-07	1.02E-07	3.19E-10	1.29E-11	1.11E-07	9.33E-10
HTX N-CE	CTUh	8.79E-06	4.83E-06	8.77E-09	5.45E-10	3.90E-06	5.65E-08
IrLS	Pt	1.19E+03	2.38E+02	1.07E+01	6.20E-01	9.32E+02	7.79E+00

Table 7: Impact indicators for AF205-30

© Copyright 2022 ABB. All rights reserved.

Impact category	Unit	Total
Biogenic Carbon content of the product	kg	1.87E-01
Biogenic Carbon content of the associated packaging	kg	2.4E-02

Table 8: Inventory Flow indicators of AF205-30.

Environmental impact indicators

GWP-total	Global Warming Potential total (Climate change)
GWP-fossil	Global Warming Potential fossil
GWP-biogenic	Global Warming Potential biogenic
GWP-luluc	Global Warming Potential land use and land use change
ODP	Depletion potential of the stratospheric ozone layer
AP	Acidification potential
EP-freshwater	Eutrophication potential - freshwater compartment
EP-marine	Eutrophication potential - fraction of nutrients reaching marine end compartment
EP-terrestrial	Eutrophication potential -Accumulated Exceedance
POCP	Formation potential of tropospheric ozone
ADP-m&m	Abiotic Depletion for non-fossil resources potential
ADP-fossil	Abiotic Depletion for fossil resources potential, WDP
WDP	Water deprivation potential.

Resource use indicators

PENRE	Use of renewable primary energy excluding renewable primary energy resources used as raw material
PENRM	Use of re-newable primary energy resources used as raw material
PENRT	Total use of renewable primary energy resources (primary energy and primary energy resources used as raw materials)
PERE	Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw material
PERM	Use of non-renewable primary energy resources used as raw material
PERT	Total use of non-renewable primary energy resources (primary energy and primary energy resources used as raw materi-als)

Secondary materials, water and energy resources

SM	Use of secondary materials
RSF	Use of renewable secondary fuels

© Copyright 2022 ABB. All rights reserved.

NRSF	Use of non-renewable secondary fuels
FW	FW: Net use of fresh water

Waste category indicators

HWD	Hazardous waste disposed
N-HWD	Non-hazardous waste disposed
RWD	Radioactive waste disposed

Output flow indicators

MfR	Materials for recycling
MfER	Materials for energy recovery

Others indicators

Efp	Emissions of Fine particles
IrHH	Ionizing radiation, human health
ETX FW	Ecotoxicity, freshwater
HTX CE	Human toxicity, carcinogenic effects
HTX N-CE	Human toxicity, non-carcinogenic effects

© Copyright 2022 ABB. All rights reserved.

Extrapolation for Homogeneous environmental family

This LCA covers different build configurations than the representative product. All the analyzed configurations have the same main functionality, product standards and manufacturing technology

The different life cycle stages can be extrapolated to other products of the same homogeneous environmental family by applying a rule of proportionality to the parameters in the following tables, divided by different life cycle stages

Contactors	LCA Stage	GWP-total	GWP-fossil	GWP-biogenic	GWP-luluc	ODP	AP	EP-freshwater	EP-marine	EP-terrestrial	POCP	ADP-minerals & metals	ADP-fossil	WDP
AF190-AF205 (3 Pole)	Manu.	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	EoL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
AF190-AF205 (3 Pole +PLC)	Manu.	1.09	1.09	1.03	1.08	1.14	1.10	1.05	1.08	1.08	1.10	1.04	1.09	1.06
	EoL	1.02	1.02	1.05	1.02	1.03	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02
AF190-AF205 (4 Pole)	Manu.	1.24	1.24	1.11	1.21	1.23	1.21	1.24	1.26	1.25	1.24	1.27	1.22	1.24
	EoL	1.24	1.21	2.34	1.22	1.21	1.21	1.22	1.24	1.21	1.21	1.21	1.21	1.22

Table 9: Extrapolation factors for AF205

Reference product: AF205-30 -Manufacturing / End of Life

Contactor	LCA Stage	Factor
AF190-AF205 (3 Pole)	Distribution	1.00
AF190-AF205 (3 Pole+ PLC)		1.05
AF190-AF205 (4 Pole)		1.20

Table10: Extrapolation factors for AF205.

Reference product: AF205-30 -Distribution

© Copyright 2022 ABB. All rights reserved.

Contactor	LCA Stage	Factor
AF190 (3 Pole- with & without PLC)	Use Phase	0.89
AF205 (3 Pole- with & without PLC)		1.00
AF190 (4 Pole)		2.36
AF205 (4 Pole)		3.87

Table 11: Extrapolation factors for AF205
Reference product: AF205-30-Use Phase



Additional environmental information

According to the waste treatment scenario calculation in Simapro, based on the recycling rate in the technical report IEC/TR 62635 Edition 1.0 [9] Table D.6, the following recyclability potentials were calculated. The recyclability potential is calculated based on the product weight (excluding packaging).

	AF205-30
Recyclability potential	79.19%

Table 12: Recyclability potential of AF205-30.

© Copyright 2022 ABB. All rights reserved.

References

[1] PEP Ecopassport PCR-ed4-EN-2021 09 06 “Product Category Rules for Electrical, Electronic and HVAC-R Products”
(published: 6th September 2021)

[2] PEP Ecopassport PSR-0005-ed2-EN-2016 03 29 “Product Specific Rules for Electrical Switchgear and Control gear Solutions” (published: May 2016)

[3] EN 50693:2019 - Product category rules for life cycle assessments of electronic and electrical products and systems

[4] ISO 14040:2006 - Environmental management -Life cycle assessment - Principles and framework

[5] ISO 14044:2006 - Environmental management - Life cycle assessment - Requirements and guidelines

[6] ecoinvent v3.8 (2021). ecoinvent version 3.8. Swiss, Centre for Life Cycle Inventories, Dübendorf, Switzerland

[7] PRé Consultants, 2021. Software SimaPro versione 9.3.1 (www.pre.nl).

[8] UNI EN 15804:2012+A2:2019: Sustainability of constructions - Environmental product declarations (September 2019).

[9] IEC/TR 62635 - Guidelines for end-of-life information provided by manufacturers and recyclers and for recyclability rate calculation of electrical and electronic equipment - Edition 1.0 2012-10

[10] 2CMT2021-006277- RoHS

[11] 2CMT2021-006202- REACh

[12] 1SFC100091D0201-LCA Report

© Copyright 2022 ABB. All rights reserved.

PRODUCT ENVIRONMENTAL PROFILE

Environmental Product Declaration

AF265-370 Contactors



REGISTRATION NUMBER ABBG-00041-V01.01	IN COMPLIANCE WITH PCR-ED4-EN-2021 09 06 SUPPLEMENTED BY PSR-0005-ED2-EN-2016 03 29
VERIFIER ACCREDITATION NUMBER VH42	INFORMATION AND REFERENCE DOCUMENTS WWW.PEP-ECOPASSPORT.ORG
DATE OF ISSUE 01-2023	VALIDITY PERIOD 5 YEARS
INDEPENDENT VERIFICATION OF THE DECLARATION AND DATA, IN COMPLIANCE WITH ISO 14025: 2006	
INTERNAL ☐ EXTERNAL ☒	
THE PCR REVIEW WAS CONDUCTED BY A PANEL OF EXPERTS CHAIRED BY JULIE ORGELET (DDEMAIN)	
PEP ARE COMPLIANT WITH XP C08-100-1 :2016 OR EN 50693:2019 THE ELEMENTS OF THE PRESENT PEP CANNOT BE COMPARED WITH ELEMENTS FROM ANOTHER PROGRAM.	
DOCUMENT IN COMPLIANCE WITH ISO 14025: 2006 « ENVIRONMENTAL LABELS AND DECLARATIONS. TYPE III ENVIRONMENTAL DECLARATIONS »	

© Copyright 2022 ABB. All rights reserved.

EPD Owner	ABB Electrification Sweden AB, Motorgraend 20 Bnr 415, Vaesteras, Sweden www.abb.com
Manufacturer name and address	ABB Electrification Sweden AB, Motorgraend 20 Bnr 415, Vaesteras, Sweden
Company contact	EPD_ELSP@in.abb.com
Reference product	AF265-30 Contactor
Description of the product	AF265-30 is 1000V IEC or 600V UL contactor with auxiliary contacts and double clamps. 132kW / 400VAC, AC-3 or 200 hp / 480V UL and switching power circuits up to 400A (AC-1) or 350A UL general use. The contactor has a wide control voltage range (20-500V 50/60Hz and DC), managing large control voltage variations, reducing panel energy consumptions and ensuring distinct operations in unstable networks.
Functional unit	The Functional Unit is to switch on and off during 20 years electrical power supply of a downstream installation with an electrical and/or mechanical control. The functional unit is characterized by type 3NO, a control circuit voltage U_c, a power circuit voltage U_p and a maximum allowed intensity by the power circuit I_p. Power Circuit Voltage U_p [V]: 1000V Maximum allowed intensity by the power circuit I_p [A]: 265A Rated Control Circuit Voltage U_c [V]: 20-500V Number of poles: 3
Other products covered	The PEP covers offerings for 3-pole variants including PLC variants (AF265/305/370-30-*, AF265B/305B/370B-30-*, AF265N5-30-*, AF265N5R-30-*, AFS265/305/370-30-*, AF265R/305R/370R-30-*) of AF265-370 Contactors and 4-pole variants (AF265/AF305/370-40-*, AF265B/305B/370B-40-*) of AF265-AF370 Contactors.
Reference lifetime	20 years
Product category	Electrical, Electronic and HVAC-R Products
Use Scenario	The use phase has been modeled based on the sales mix data (2021), and the corresponding low voltage electricity countries mix
Geographical representativeness	Raw materials & Manufacturing: [Europe / Global] Assembly: [Sweden] Distribution / Use: [Global] specific sales mix EoL: [Global]
Technological representativeness	Materials and processes data are specific for the production of AF265-30 Contactors
LCA Study	This study is based on the LCA study described in the LCA report 1SFC100101D0201
EPD type	Product Family Declaration
EPD scope	“Cradle to grave”
Year of reported primary data	2021
LCA software	SimaPro 9.3.0.3 (2022)
LCI database	ecoinvent v3.8 (2021)
LCIA methodology	EN 50693:2019

© Copyright 2022 ABB. All rights reserved.

Contents

ABB Purpose & Embedding Sustainability **4**

General Information **4**
- **AF Contactors product cluster** 5

Constituent Materials **5**
- **Functional unit and Reference Flow** 7
- **System boundaries and life cycle stages** 7
- **Temporal and geographical boundaries** 8
- **Boundaries in the life cycle** 8
- **Data quality** 8
- **Environmental impact indicators** 8
- **Allocation rules** 8
- **Limitations and simplifications** 9
- **Energy Models** 9

Inventory analysis **9**

Manufacturing stage **10**

Distribution **10**

Installation **11**

Use **11**

End of life **12**

Environmental impacts **13**

Additional environmental information **17**

References **18**

© Copyright 2022 ABB. All rights reserved.

ABB Purpose & Embedding Sustainability

ABB is a leading global technology company that energizes the transformation of society and industry to achieve a more productive, sustainable future. By connecting software to its electrification, robotics, automation and motion portfolio, ABB pushes the boundaries of technology to drive performance to new levels. With a history of excellence stretching back more than 130 years, ABB's success is driven by about 110 thousand talented employees in over 100 countries.
ABB's Electrification business offers a wide-ranging portfolio of products, digital solutions and services, from substation to socket, enabling safe, smart and sustainable electrification. Offerings encompass digital and connected innovations for low voltage and medium voltage, including EV infrastructure, solar inverters, modular substations, distribution automation, power protection, wiring accessories, switchgear, enclosures, cabling, sensing and control.
ABB is committed to continually promoting and embedding sustainability across its operations and value chain, aspiring to become a role model for others to follow. With its ABB Purpose, ABB is focusing on reducing harmful emissions, preserving natural resources and championing ethical and humane behavior.

General Information

The ABB facility located in Vasteras, is for the Smart Power Division, the competence center for Contactors >100A, Softstarters and one of leading manufacturing site.
All the products are produced in an automated system with precision and accuracy. These are supplied to utilities, industrial, and tertiary sector customers.
ABB adopts and implements for its own activities an integrated Quality/Environmental/Health Management System in compliance with the following
standards:
SS-EN ISO 9001: 2015 - Quality Management Systems- Requirements
SS-EN ISO 14001: 2015 - Environmental management systems - Requirements with guidance for use
ISO 45001:2018 - Occupational Health and Safety Assessment Series
SS-EN ISO 50001: 2018 - Energy management systems

AF contactors are most commonly used for controlling electric motors and switch various none-motor loads. Contactors can break current over a wide range of currents, from a few amperes to thousands of amperes, and voltages upto 1000V AC.

© Copyright 2022 ABB. All rights reserved.

AF Contactors product cluster

Product cluster declared in this LCA includes the following AF265-370 Contactors:

Product Range	Product	Number of poles	Power Circuit Voltage [U_p]	Maximum allowed intensity by the power circuit [I_p]	Rated Control Circuit Voltage [U_c]
AF265-370	AF265-AF370	3	1000V	265-370A	20-500V
	AF265-AF370 (PLC)	3	1000V	265-370A	20-500V
	AF265-AF370	4	1000V	265-370A	20-500V

Table 1: Technical characteristics of AF265-370 Contactors.

The accessories associated with these products are also included in the study.

Reference Product:

The reference product for the LCA of the complete range of AF265-370 is AF265-30.

Constituent Materials

The AF265-30 weights about 5.6kg including its installed accessories, packaging, and paper documentation.

AF265-30				
Materials	**Name**	**IEC 62474 MC**	**[g]**	**%**
Metals	Steel	M-119	2277.2	40.4%
	Cu Alloys	M-121	567.9	10.1%
	Silver Alloys	M-124	40.9	0.7%
	Stainless Steel	M-100	22.5	0.4%
Plastics	Polyamide	M-258	1073.5	19.0%
	Unsaturated Polyester	M-301	902	16.0%
	Polyethylene	M-251	179.2	3.2%
	Polyurethane	M-300	81.2	1.4%
	Rubber	M-320	5.7	0.1%
	PBT	M-261	0.2	<0.1%
Other	Paper / Cardboard	M-341	413.1	7.3%
	Other	N/A	71.9	1.3%
		Total	**5635.3**	100%

Table 2: Weight of materials AF265-30

© Copyright 2022 ABB. All rights reserved.

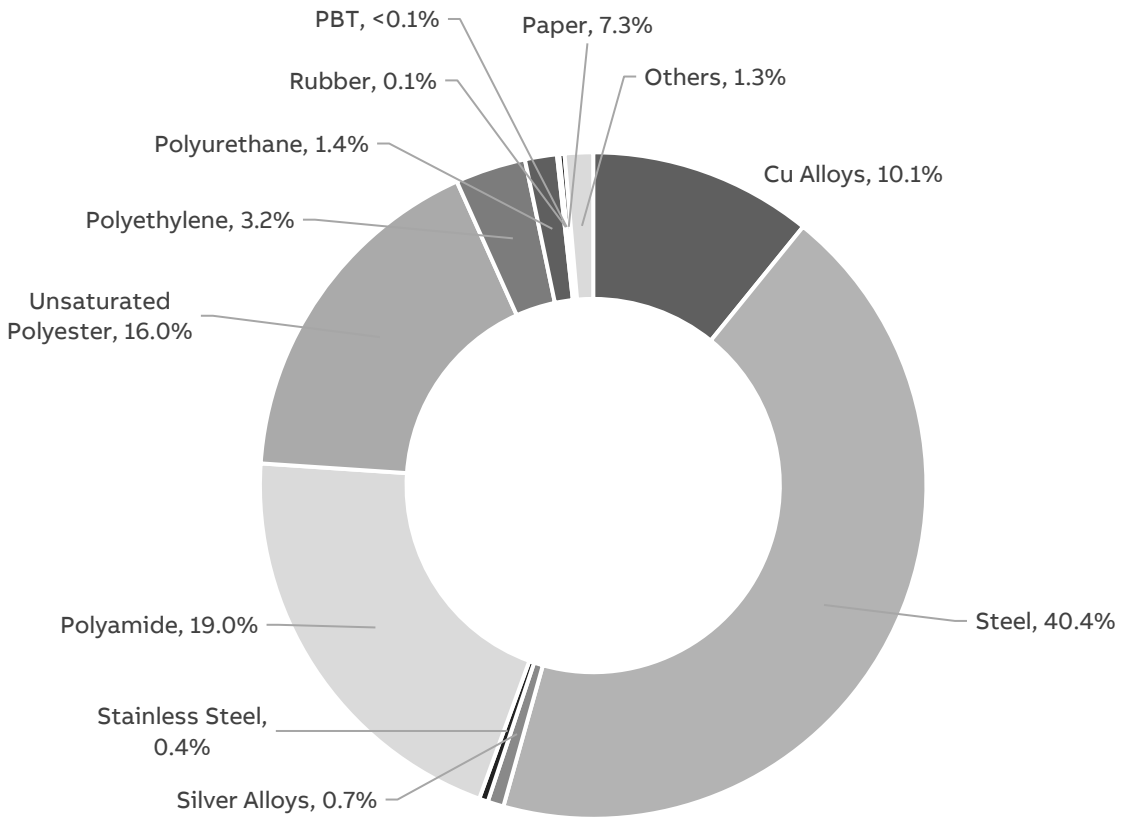


Figure 1: Composition of AF265-30

Along the whole AF265-370 product cluster, a set of different build configurations have been covered by this analysis.

Packaging weighs 372g, with the following substance composition:

Material	Unit	Total	%
Corrugated Cardboard	g	372	6.6%

Table 3: Weight of Packaging for AF265

Official declarations 2CMT2021-006277 [10] and 2CMT2021-006202 [11] states compliance of ABB AF Contactors respectively to RoHS and REACH regulations; 2CMT2021-006277 [10] provides exemptions considered for RoHS while 2CMT2021-006202 [11] lists REACH substances present in a concentration above 0.1% adding reference to products where involved parts are mounted.

© Copyright 2022 ABB. All rights reserved.

Functional unit and Reference Flow

The Functional Unit is to switch on and off during 20 years electrical power supply of a downstream installation with an electrical and/or mechanical control. The functional unit is characterized by type 3NO, a control circuit voltage U_c, a power circuit voltage U_p and a maximum allowed intensity by the power circuit I_p. (table 1)

The Reference Flow of the study is a single Contactor (including its packaging and accessories) with mass described, table 2.

System boundaries and life cycle stages

The life cycle of the Contactor, an EEPS (Electronic and Electrical Products and Systems), is a “from cradle to grave” analysis and covers the following main life cycle stages: manufacturing, including the relevant acquisition of raw material, preparation of semi-finished goods, etc. and processing steps; distribution; installation, including the relevant steps for the preparation of the product for use; use including the required maintenance steps within the RSL (reference service life of the product) associated to the reference product; end-of-life stage, including the necessary steps until final disposal or recovery of the product system.

The following table shows the stages of the product life cycle and the information stages according to EN 50693:2019 [3] for the evaluation of electronic and electrical products and systems.

Manufacturing	Distribution	Installation	Use	End-of-Life (EoL)
Acquisition of raw materials Transport to manufacturing site Components/parts manufacturing Assembly Packaging EoL treatment of generated waste	Transport to distributor/ logistic center Transport to place of use	Installation EoL treatment of generated waste (packaging)	Usage Maintenance	Deinstallation Collection and transport EoL treatment

Table 4: Phases for the evaluation of construction products according to EN50693:2019 [3].

© Copyright 2022 ABB. All rights reserved.

Temporal and geographical boundaries

The ABB component suppliers are sourced all over the world. All primary data collected are from 2021, which is a representative production year. Secondary data are also representative for this year, as provided by ecoinvent [6].

The selected ecoinvent [6] processes in the LCA model have a global representativeness, due to the unclear origin of each component. In this way, a conservative approach has been adopted.

Boundaries in the life cycle

As indicated in the PCR capital goods such as buildings, machinery, tools and infrastructure, the packaging for internal transport which cannot be allocated directly to the production of the reference product, may be excluded from the system boundary.

Infrastructures, when present, such as processes deriving from the ecoinvent [6] database have not been excluded.

Data quality

In this LCA, both primary and secondary data are used. Site specific foreground data have been provided by ABB. Main data sources are the bill of materials & drawings which are available on the ERP (SAP) & Windchill. For all processes for which primary are not available, generic data originating from the ecoinvent database [6], allocation cut-off by classification, are used. The ecoinvent database available in the SimaPro software [7] is used for the calculations.

The data quality characterized by quantitative and qualitative aspects, is presented in Ap-pendix 1. Each data quality parameter has been rated according to DQR tables from Chapter 7.19.2.2 of the Product Environmental Footprint Guide v.6.3 to give an indication of geography, technology and temporal representativeness.

Environmental impact indicators

The information obtained from the inventory analysis is aggregated according to the effects related to the various environmental issues. According to “PCR-ed4-EN-2021 09 06” and EN 50693 [3] the environmental impact indicators must be determined using the characterization factors and impact assessment methods specified in EN 15804:2012+A2:2019 [8].

PCR-ed4-EN-2021 09 06 and the EN 50693:2019 [3] standard establish four indicators for GWP: GWP (total) which includes all greenhouse gases; GWP (fossil fuels); GWP (biogenic) which includes the emissions and absorption of biogenic carbon dioxide and biogenic carbon stored in the product; GWP (land use) - land use and land use transformation. Other indicators as per the PCR[1].

Allocation rules

Allocation coefficients are based on the AF265 line’s occupancy area for electricity consumption since, apart from assembly processes, the whole production line is temperature-regulated throughout the year. The allocation of the total amount of waste generated by the production line and water consumption, has been based on this criterion.

© Copyright 2022 ABB. All rights reserved.

Limitations and simplifications

Raw materials life cycle stage includes the extraction of raw materials as well as the transport distances to the manufacturing suppliers. These distances are assumed to be 1000 km as per PCR. This distance has been added to the one already included in the market processes used for the model, as a result of a conservative choice made by the LCA operators.

Surface treatments like galvanizing, tin and silver plating as well as their related transport processes (back and forth from the finishing suppliers) have been considered in the LCA model.

Scraps for metal working and plastic processes are included when already defined in ecoinvent[6].

Energy Models

LCA Stage	EN 15804:2012 +A2:2019 module	Energy model	Notes
Raw material extraction and processing	A1-A2	Electricity, {RER}| market group for | Cut-off Electricity, {GLO}| market group for | Cut-off	Based on materials and supplier locations
Manufacturing	A3	Electricity, {SE}| market for | Cut-off	Specific Energy model for ABB Vasteras manufacturing plant, 100% renewable
Installation (Packaging EoL)	A5	Electricity, {GLO}| market group for | Cut-off	
Use Stage	B1	Electricity, [country]x | market for | Cut-off, S **	Low voltage, based on 2021 country sales mix
EoL	C1-C4	Electricity, {GLO}| market group for | Cut-off	

Table 5: Energy models used in each LCA stage

** Please refer the use phase page 11 for further description

Inventory analysis

In this LCA, both primary and secondary data are used. Site specific foreground data have been provided by ABB. For data collection, Bills of Material (BOM) extracted from ABB's internal SAP software were used. They are a list of all the components and assemblies that constitute the finished product, organized by level. Each item is matched with its code, quantity, weight and supplier. The BOMs were then processed, adding material, surface area and other weight data, taken from technical drawings. Finally, the manufacturing process and surface treatment were assigned, according to information provided by R&D personnel. Road distances between the suppliers and ABB were calculated using Google Maps, and marine distances using Distances & Time (Searates).

© Copyright 2022 ABB. All rights reserved.

All primary data collected from ABB are from 2021, which was a representative production year. The ecoinvent v3.8 cut-off by classification system processes [6] are used to model the background system of the processes.

Due to a lot of components in the Contactors, raw material inputs have been modelled with data from ecoinvent representing either a European [RER] or Global [RoW] market coverage based on the supplier's location. These datasets are assumed to be representative.

Manufacturing stage

The Contactors are composed of a multitude of components, all of which are made from of numerous materials. Most of the inputs to the products' manufacturing stage are already produced component parts.

The single use packaging as well as paper documentation are also included in the analysis in the manufacturing stage. ABB receives packaging components from outside suppliers and packages the Contactors before shipping them.

Most of the inputs to the products' manufacturing stage are already produced component parts from the supply chain. In the ABB manufacturing plant, the different components and subassemblies are assembled into the Contactor. All the semi-finished and ancillary products are produced by ABB's suppliers.

The entire AF Contactors suppliers' network has been modelled with the calculation of each transportation stage: from the first manufacturing supplier to the next. All the distances from the last subassembly suppliers' factories to the ABB manufacturing facility have been calculated.
All the distances from the last subassembly suppliers' factories to the ABB manufacturing facility have been calculated.
In the ABB factory, the different components and subassemblies are assembled into the Contactor. All the semi-finished and ancillary products are produced by ABB's suppliers.

The energy mix used for the production phase is representative for ABB Vasteras production site and includes renewable energy only (Hydroelectric).

The complete energy mix has been modeled considering the certificate on Guarantee of origins provided to ABB for the year 2021.

Distribution

The transport distances from ABB manufacturing plant to the distribution centers (regional distribution centers / local sales organizations) have been calculated considering the specific 2021 sales mix data for AF265-370 Contactors product cluster (SAP ERP sales data as a source).
Since no specific data is available for the transport distances from the Distribution Centre to place of actual use (Customer site), distances of 1000 km are assumed (local/domestic transport by lorry, according to PCR [1]).

© Copyright 2022 ABB. All rights reserved.

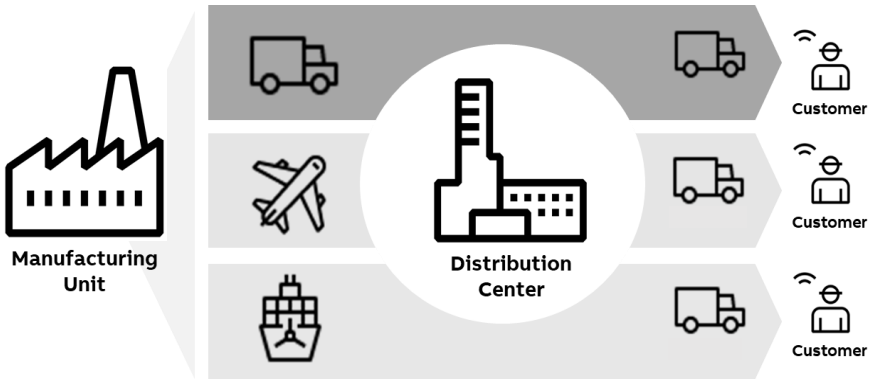


Figure 2: Distribution Methodology

Installation

The installation phase only implies manual activities, and no energy is consumed. This phase also includes the disposal of the packaging and paper technical documentation of the AF Contactor.

For the disposal of the packaging and documentation after installation of the AF Contactor at the end of its life, a transport distance of 1000 km (according to PCR[1]) was assumed. The chosen transportation datasets from Ecoinvent [6].

The actual disposal site is unknown and is managed by the customer.

Use

Use and maintenance are modelled according to the PCR [1].

During the use phase, AF Contactor, dissipates some electricity due to power losses. They are calculated according to the data provided in the catalogue of the AF Contactor and following the PCR [1] & PSR [2] rules:

Parameters		
Iu	[A]	265
Iu	[%]	50
h/year	[h]	8760
RSL	[years]	20
Time operating coefficient α	[%]	50

Table 6: Use phase parameters

The formula for the calculation of the electricity consumed is shown below and it is described as follows, where P_{use} is the power consumed by the Contactor at a given value of current:

$$E_{use}\ [kWh] = \frac{P_{use} * 8760 * RSL * \alpha}{1000}$$

© Copyright 2022 ABB. All rights reserved.

The above calculations have been performed according to the number of poles (3) on which relevant current flows during use phase.

The Energy model used for this phase has been modeled based on the 2021 actual sales mix data (SAP ERP sales data as a source). From the Ecoinvent [6] database, the low voltage electricity country mix for each country(x) has been selected with its respective percentage on the total sales mix (Electricity, low voltage [country]x | market for | Cut-off, S).

Since no maintenance happens during the use phase, the environmental impacts linked to this procedure have been considered as null in the analysis.

End of life

The end-of-life stage is modelled according to PCR [1] and IEC/TR 62635 [9]. The percentages for end-of-life treatments of materials are taken from IEC/TR 62635 [9].

Since no specific data is available, the transport distances from the place of use to the place of disposal are assumed to be 1000 km (local/domestic transport by lorry, according to PCR [1]).

Disassembly manuals can be provided to the customer to support product disposal.

© Copyright 2022 ABB. All rights reserved.

Environmental impacts

The following tables show the environmental impact indicators of the life cycle of a single Contactor, as indicated by PEP Ecopassport PCR and EN 50693:2019 [3]. The indicators are divided into the contribution of the processes to the different modules (upstream, core and downstream) and stages (manufacturing, distribution, installation, use and end-of-life).

AF265-30

Impact category	Unit	Total	Manufacturing	Distribution	Installation	Use	End of Life
GWP-total	kg CO2 eq	5.74E+02	4.40E+01	1.40E+00	2.46E-01	5.27E+02	1.32E+00
GWP-fossil	kg CO2 eq	5.57E+02	4.36E+01	1.40E+00	1.00E-01	5.11E+02	1.30E+00
GWP-biogenic	kg CO2 eq	1.48E+01	3.82E-01	1.07E-03	1.46E-01	1.43E+01	2.15E-02
GWP-luluc	kg CO2 eq	1.81E+00	5.34E-02	6.31E-04	3.56E-05	1.76E+00	7.18E-04
ODP	kg CFC11 eq	3.33E-05	3.14E-06	3.23E-07	2.28E-08	2.97E-05	1.30E-07
AP	mol H+ eq	3.03E+00	5.69E-01	1.63E-02	4.97E-04	2.44E+00	5.67E-03
EP-freshwater	kg P eq	3.85E-01	7.00E-02	7.80E-05	6.27E-06	3.15E-01	2.17E-04
EP-marine	kg N eq	5.30E-01	8.87E-02	4.53E-03	2.53E-04	4.33E-01	3.50E-03
EP-terrestrial	mol N eq	5.22E+00	9.78E-01	5.00E-02	1.86E-03	4.17E+00	1.48E-02
POCP	kg NMVOC eq	1.43E+00	2.60E-01	1.38E-02	5.73E-04	1.15E+00	4.33E-03
ADP-m&m	kg Sb eq	3.32E-02	2.96E-02	2.93E-06	2.23E-07	3.54E-03	1.39E-06
ADP-fossil	MJ	8.95E+03	6.25E+02	2.11E+01	1.49E+00	8.29E+03	1.29E+01
WDP	m3	1.51E+02	2.34E+01	6.57E-02	7.16E-03	1.27E+02	9.61E-02
PENRE	MJ	8.88E+03	5.63E+02	2.11E+01	1.49E+00	8.29E+03	1.29E+01
PENRM	MJ	6.19E+01	6.19E+01	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PENRT	MJ	8.95E+03	6.25E+02	2.11E+01	1.49E+00	8.29E+03	1.29E+01
PERE	MJ	1.52E+03	5.34E+01	2.43E-01	1.99E-02	1.47E+03	7.69E-01
PERM	MJ	7.03E+00	7.03E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PERT	MJ	1.53E+03	6.04E+01	2.43E-01	1.99E-02	1.47E+03	7.69E-01
SM	kg	1.57E+00	1.57E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
RSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
NRSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
FW	m3	6.99E+00	6.69E-01	2.25E-03	2.41E-04	6.32E+00	3.77E-03
HWD	kg	1.30E-02	6.77E-03	4.46E-05	3.60E-06	6.19E-03	2.00E-05
N-HWD	kg	4.24E+01	7.06E+00	1.58E+00	1.83E-01	3.17E+01	1.85E+00
RWD	kg	4.03E-02	1.88E-03	1.43E-04	1.01E-05	3.82E-02	6.50E-05
MfR	kg	4.95E+00	6.68E-01	0.00E+00	4.37E-01	0.00E+00	3.85E+00
MfER	kg	1.32E-01	0.00E+00	0.00E+00	3.87E-02	0.00E+00	9.30E-02
Efp	disease inc.	1.56E-05	2.93E-06	1.39E-07	1.15E-08	1.25E-05	1.02E-07
IrHH	kBq U-235 eq	1.58E+02	4.81E+00	1.04E-01	7.57E-03	1.53E+02	8.02E-02
ETX FW	CTUe	1.47E+04	7.23E+03	1.57E+01	1.41E+00	7.40E+03	1.92E+01
HTX CE	CTUh	2.96E-07	1.18E-07	5.64E-10	3.62E-11	1.77E-07	9.71E-10
HTX N-CE	CTUh	1.09E-05	5.47E-06	1.60E-08	1.55E-09	5.36E-06	5.83E-08
IrLS	Pt	1.98E+03	3.62E+02	1.98E+01	1.71E+00	1.59E+03	1.10E+01

Table 7: Impact indicators for AF265-30

© Copyright 2022 ABB. All rights reserved.

Impact category	Unit	Total
Biogenic Carbon content of the product	kg	2.9E-01
Biogenic Carbon content of the associated packaging	kg	7.9E-02

Table 8: Inventory Flow indicators of AF265-30.

Environmental impact indicators

GWP-total	Global Warming Potential total (Climate change)
GWP-fossil	Global Warming Potential fossil
GWP-biogenic	Global Warming Potential biogenic
GWP-luluc	Global Warming Potential land use and land use change
ODP	Depletion potential of the stratospheric ozone layer
AP	Acidification potential
EP-freshwater	Eutrophication potential - freshwater compartment
EP-marine	Eutrophication potential - fraction of nutrients reaching marine end compartment
EP-terrestrial	Eutrophication potential -Accumulated Exceedance
POCP	Formation potential of tropospheric ozone
ADP-m&m	Abiotic Depletion for non-fossil resources potential
ADP-fossil	Abiotic Depletion for fossil resources potential, WDP
WDP	Water deprivation potential.

Resource use indicators

PENRE	Use of non-renewable primary energy excluding renewable primary energy resources used as raw material
PENRM	Use of non-renewable primary energy resources used as raw material
PENRT	Total use of non-renewable primary energy resources (primary energy and primary energy resources used as raw materials)
PERE	Use of renewable primary energy excluding non-renewable primary energy resources used as raw material
PERM	Use of renewable primary energy resources used as raw material
PERT	Total use of renewable primary energy resources (primary energy and primary energy resources used as raw materi-als)

© Copyright 2022 ABB. All rights reserved.

Secondary materials, water and energy resources

SM	Use of secondary materials
RSF	Use of renewable secondary fuels
NRSF	Use of non-renewable secondary fuels
FW	FW: Net use of fresh water

Waste category indicators

HWD	Hazardous waste disposed
N-HWD	Non-hazardous waste disposed
RWD	Radioactive waste disposed

Output flow indicators

MfR	Materials for recycling
MfER	Materials for energy recovery

Others indicators

Efp	Emissions of Fine particles
IrHH	Ionizing radiation, human health
ETX FW	Ecotoxicity, freshwater
HTX CE	Human toxicity, carcinogenic effects
HTX N-CE	Human toxicity, non-carcinogenic effects

© Copyright 2022 ABB. All rights reserved.

Extrapolation for Homogeneous environmental family

This LCA covers different build configurations than the representative product from the IEC and UL types. All the analyzed configurations have the same main functionality, product standards and manufacturing technology

The different life cycle stages can be extrapolated to other products of the same homogeneous environmental family by applying a rule of proportionality to the parameters in the following tables, divided by different life cycle stages.

Contactors	LCA Stage	GWP-total	GWP-fossil	GWP-biogenic	GWP-luluc	ODP	AP	EP-freshwater	EP-marine	EP-terrestrial	POCP	ADP-minerals & metals	ADP-fossil	WDP
AF265-AF370	Manu.	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	EoL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
AF265-AF370 (PLC)	Manu.	1.08	1.08	0.79	1.12	1.07	1.06	1.08	1.07	1.07	1.07	1.06	1.09	1.07
	EoL	1.03	1.04	1.00	1.02	1.01	1.02	1.02	0.98	1.02	1.01	1.01	1.01	1.01
AF265-AF370	Manu.	1.26	1.27	0.94	1.34	1.29	1.52	1.42	1.35	1.37	1.36	1.40	1.25	1.32
	EoL	1.45	1.41	3.86	1.58	1.31	1.52	1.65	1.30	1.42	1.41	1.34	1.45	1.49

Table 9: Extrapolation factors for AF265
Reference product: AF265-30 -Manufacturing / End of Life

Contactor	LCA Stage	Factor
AF265-AF370	Distribution	1.00
AF265-AF370 (PLC)		1.01
AF265-AF370		1.23

Table10: Extrapolation factors for AF265.
Reference product: AF265-30 -Distribution

Contactor	LCA Stage	Factor
AF265 (With & Without PLC)	Use Phase	1.00
AF305 (With & Without PLC)		1.23
AF370 (With & Without PLC)		1.74
AF265		1.20
AF305		1.63
AF370		2.32

Table 11: Extrapolation factors for AF265
Reference product: AF265-30-Use Phase

© Copyright 2022 ABB. All rights reserved.

Additional environmental information

According to the waste treatment scenario calculation in Simapro, based on the recycling rate in the technical report IEC/TR 62635 Edition 1.0 [9] Table D.6, the following recyclability potentials were calculated. The recyclability potential is calculated based on the product weight (excluding packaging).

	AF265-30
Recyclability potential	76.88%

Table 12: Recyclability potential of AF265-30.

© Copyright 2022 ABB. All rights reserved.

References

[1] PEP Ecopassport PCR-ed4-EN-2021 09 06 “Product Category Rules for Electrical, Electronic and HVAC-R Products”
(published: 6th September 2021)

[2] PEP Ecopassport PSR-0005-ed2-EN-2016 03 29 “Product Specific Rules for Electrical Switchgear and Control gear Solutions” (published: May 2016)

[3] EN 50693:2019 - Product category rules for life cycle assessments of electronic and electrical products and systems

[4] ISO 14040:2006 - Environmental management -Life cycle assessment - Principles and framework

[5] ISO 14044:2006 - Environmental management - Life cycle assessment - Requirements and guidelines

[6] ecoinvent v3.8 (2021). ecoinvent version 3.8. Swiss, Centre for Life Cycle Inventories, Dübendorf, Switzerland

[7] PRé Consultants, 2021. Software SimaPro versione 9.3.1 (www.pre.nl).

[8] UNI EN 15804:2012+A2:2019: Sustainability of constructions - Environmental product declarations (September 2019).

[9] IEC/TR 62635 - Guidelines for end-of-life information provided by manufacturers and recyclers and for recyclability rate calculation of electrical and electronic equipment - Edition 1.0 2012-10

[10] 2CMT2021-006277- RoHS

[11] 2CMT2021-006202- REACh

[12] 1SFC100101D0201-LCA Report

© Copyright 2022 ABB. All rights reserved.